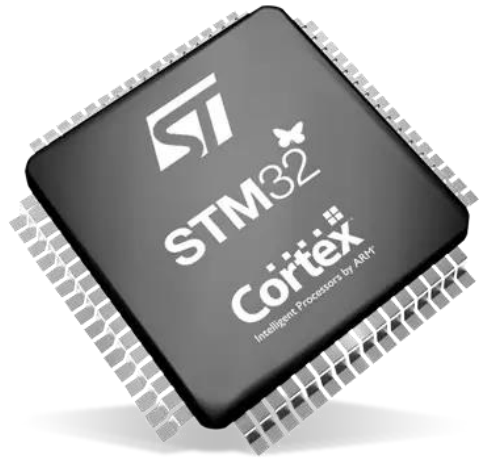




STM32  
CubeIDE

Introducción a STM32

Agenda

- Arquitectura ARM
- Características de la plataforma STM32
- Formas de programar STM32
- GPIOs
- Interrupciones
- Instalación de software
- Descarga de documentos necesarios

ARM

Arquitectura

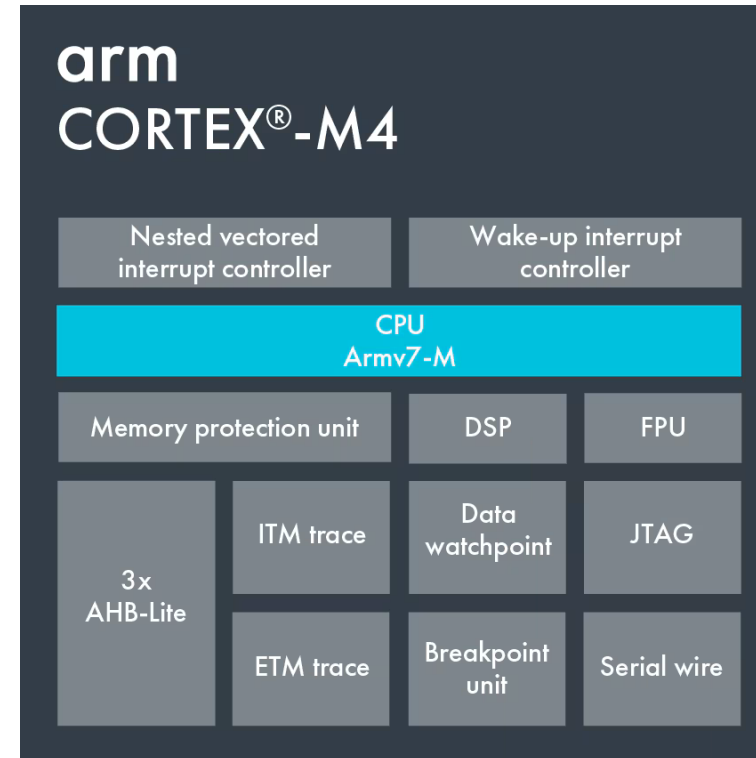
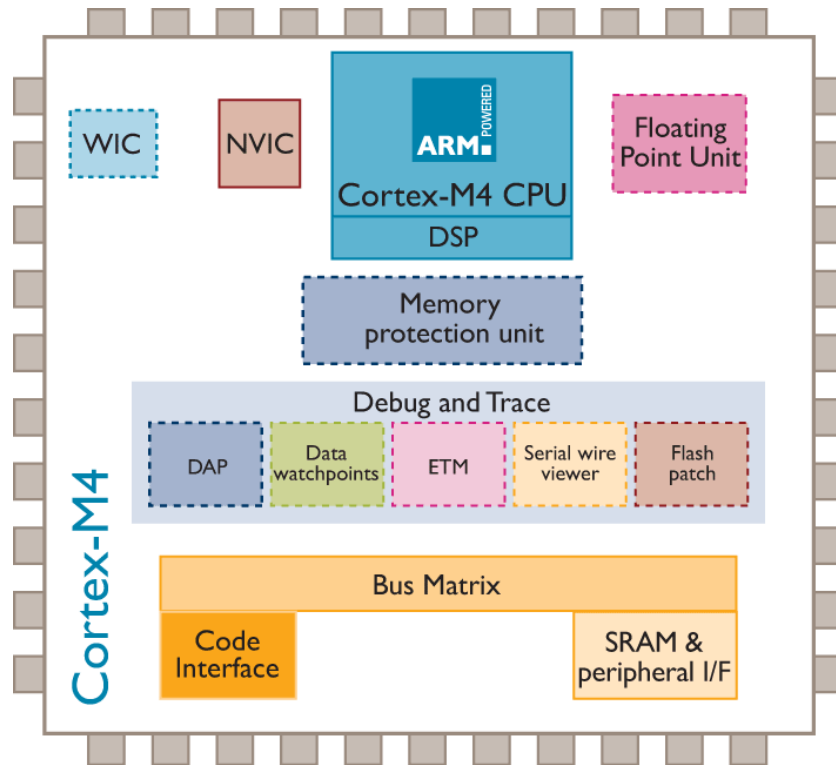
Arquitectura

95% de los dispositivos móviles utilizan procesadores ARM

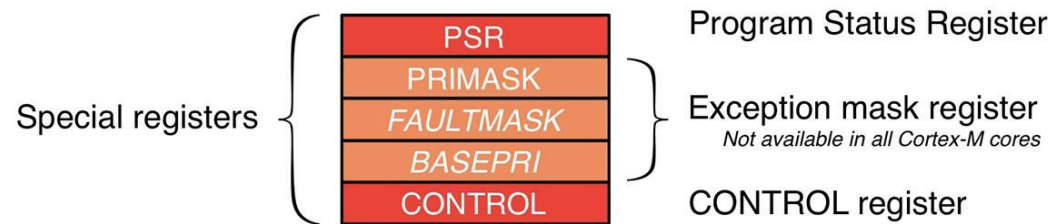
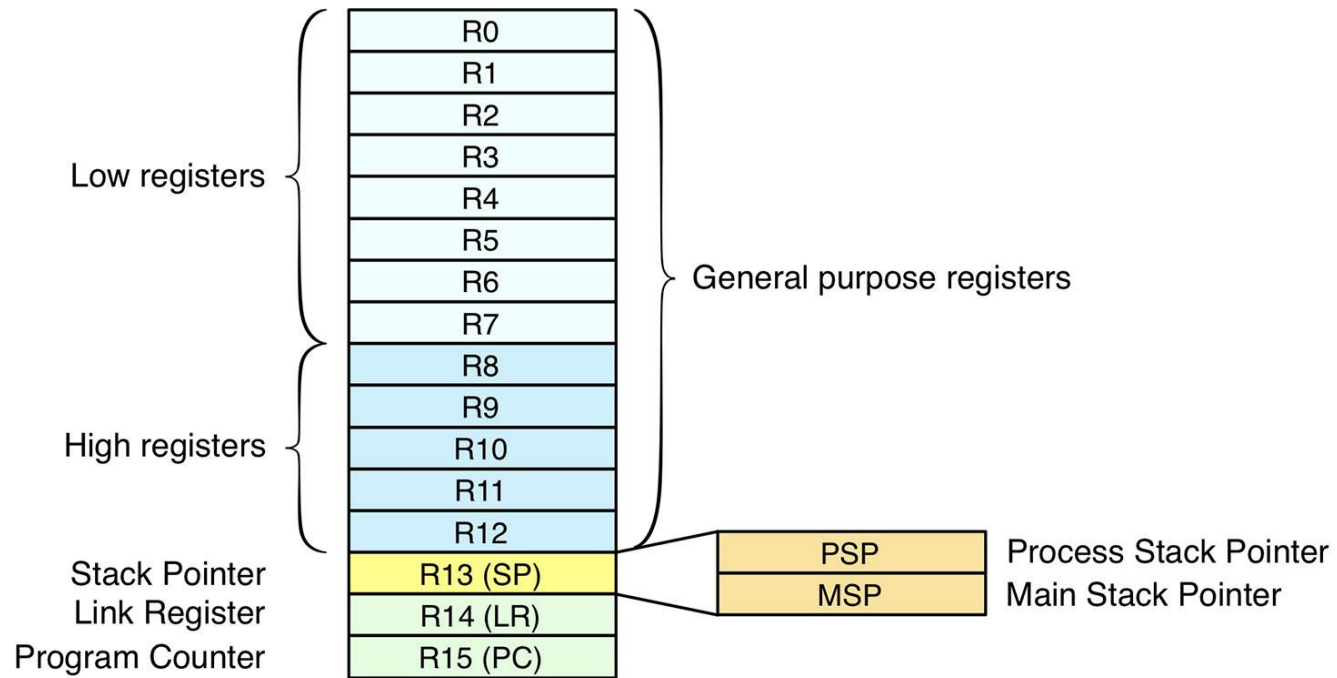
Al 2020 se habían fabricado 160 billones de procesadores ARM

Los procesadores Cortex-M son específicos para soluciones embebidas.

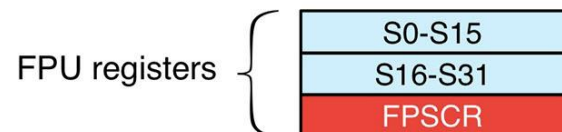




Arquitectura

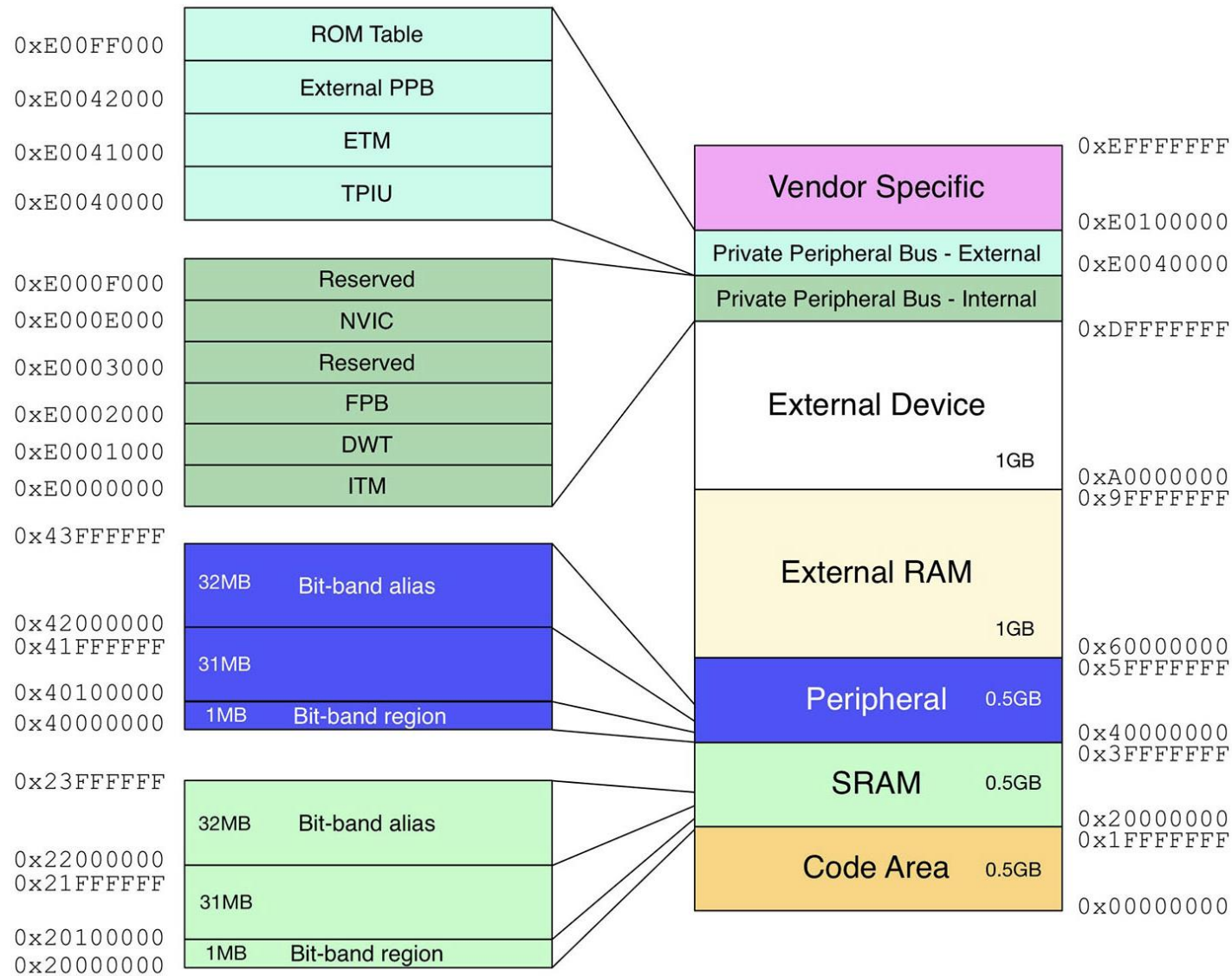


Floating point registers - Available only in Cortex-M4F/M7 cores

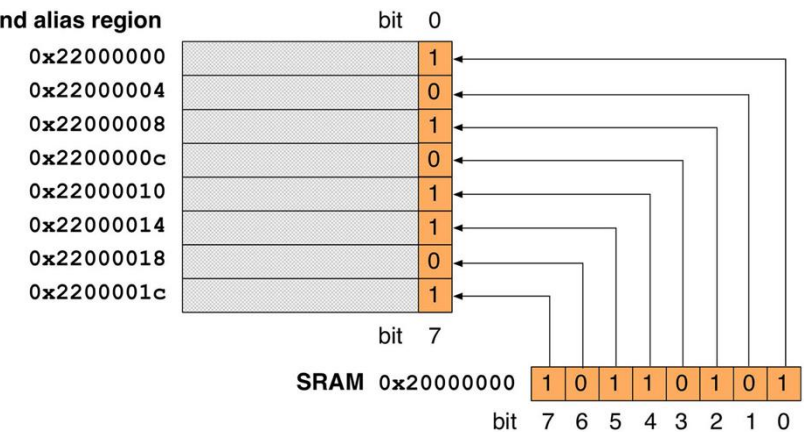


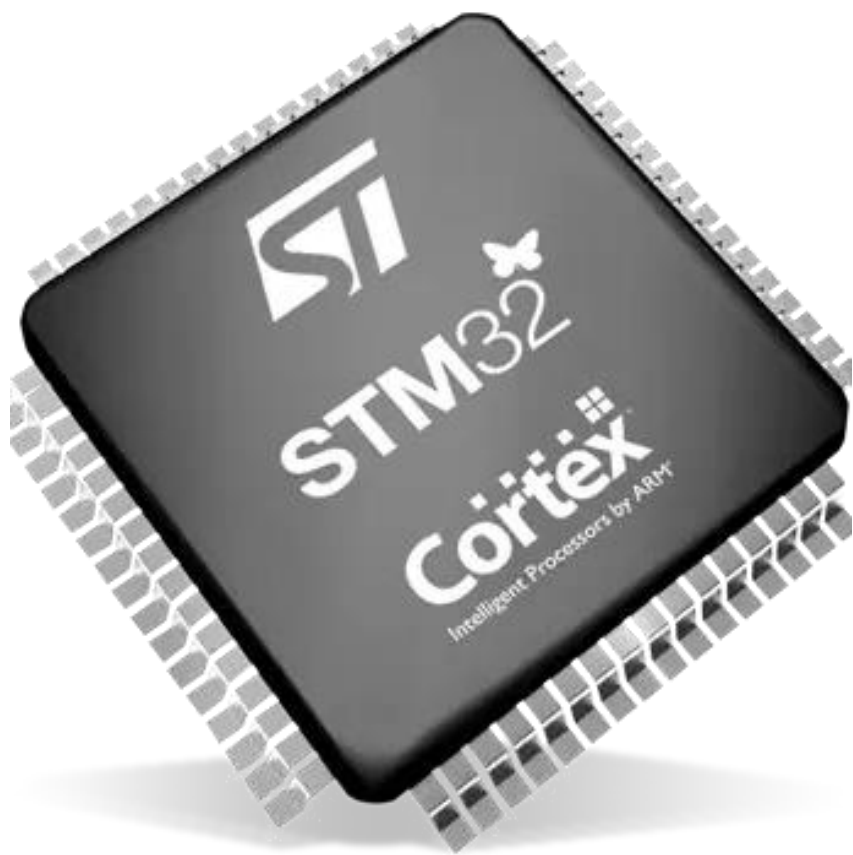
Registros

Memoria



Bit-band alias region





STM32



STM32 MCUs
32-bit Arm® Cortex®-M



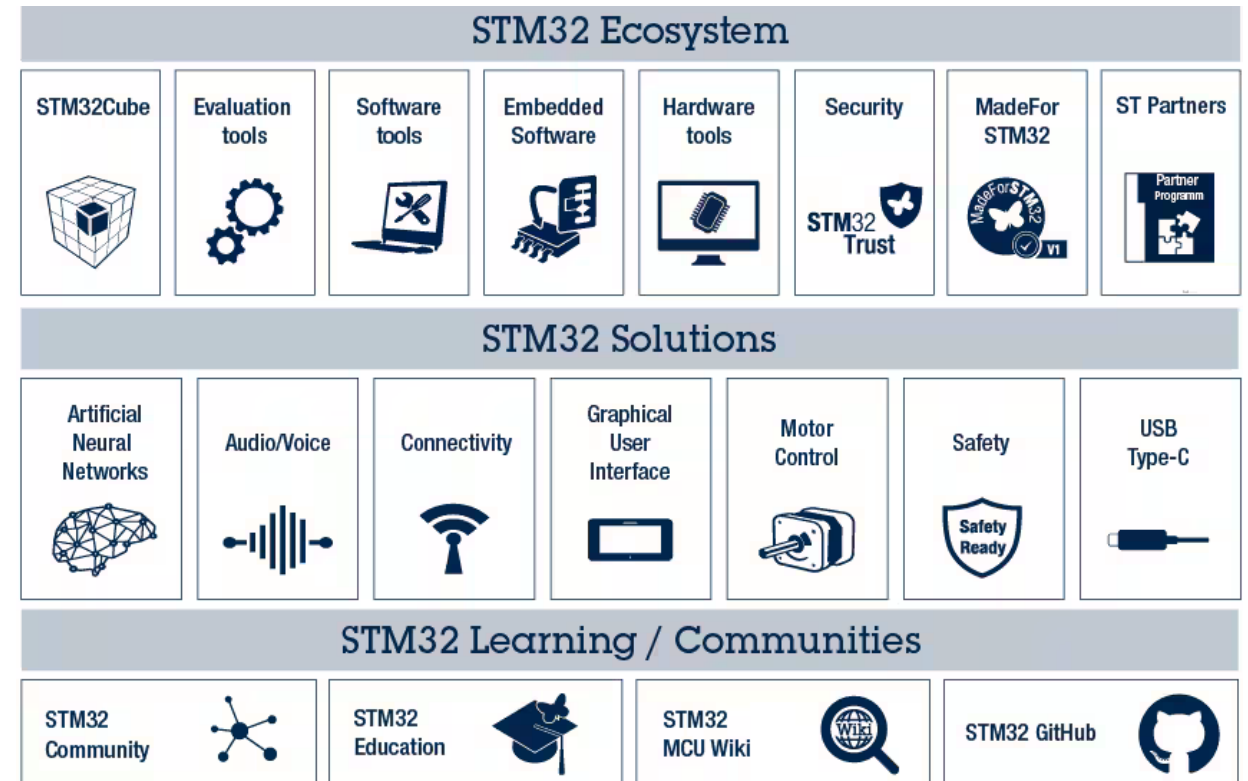
★ High Performance				STM32F7 1082 CoreMark 216 MHz Cortex-M7	STM32H7 Up to 3224 CoreMark Up to 600 MHz Cortex-M7 240 MHz Cortex-M4
	STM32F2 398 CoreMark 120 MHz Cortex-M3	STM32F4 608 CoreMark 180 MHz Cortex-M4	STM32H5 Up to 1023 CoreMark 250 MHz Cortex-M33		
» Mainstream		STM32G0 142 CoreMark 64 MHz Cortex-M0+	STM32G4 569 CoreMark 170 MHz Cortex-M4		● Optimized for mixed-signal applications
	STM32C0 114 CoreMark 48 MHz Cortex-M0+	STM32F0 106 CoreMark 48 MHz Cortex-M0	STM32F1 177 CoreMark 72 MHz Cortex-M3	STM32F3 245 CoreMark 72 MHz Cortex-M4	
🔋 Ultra-low-power			STM32L4+ 409 CoreMark 120 MHz Cortex-M4	STM32U5 651 CoreMark 160 MHz Cortex-M33	
	STM32L0 75 CoreMark 32 MHz Cortex-M0+	STM32U0 140 CoreMark 56 MHz Cortex-M0+	STM32L4 273 CoreMark 80 MHz Cortex-M4	STM32L5 443 CoreMark 110 MHz Cortex-M33	
📶 Wireless			STM32WL 162 CoreMark 48 MHz Cortex-M4 48 MHz Cortex-M0+	STM32WB0 64 MHz Cortex-M0+	STM32WB 216 CoreMark 64 MHz Cortex-M4 32 MHz Cortex-M0+
					STM32WBA 407 CoreMark 100 MHz Cortex-M33
● Cortex-M0+ Radio co-processor					

● Cortex-M0+
Radio co-processor

Series STM32F4



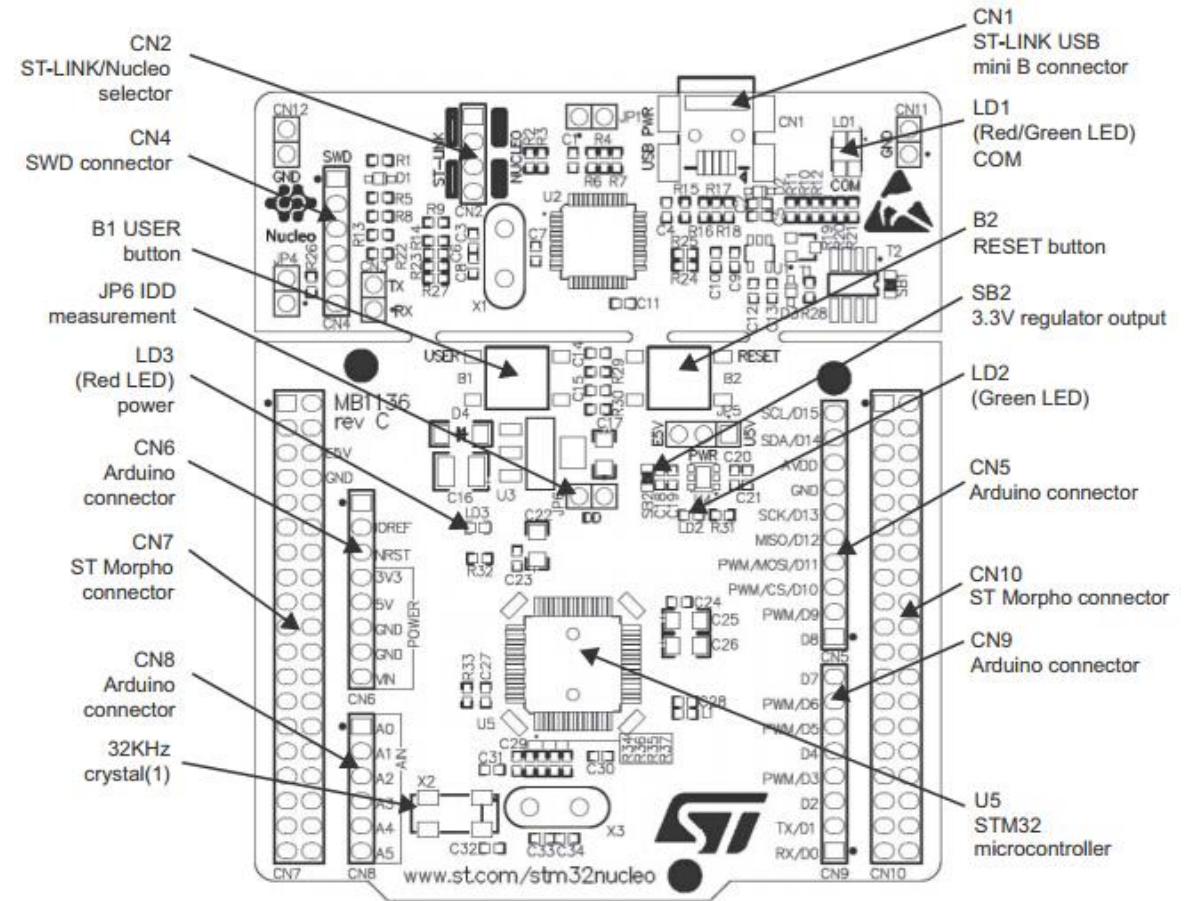
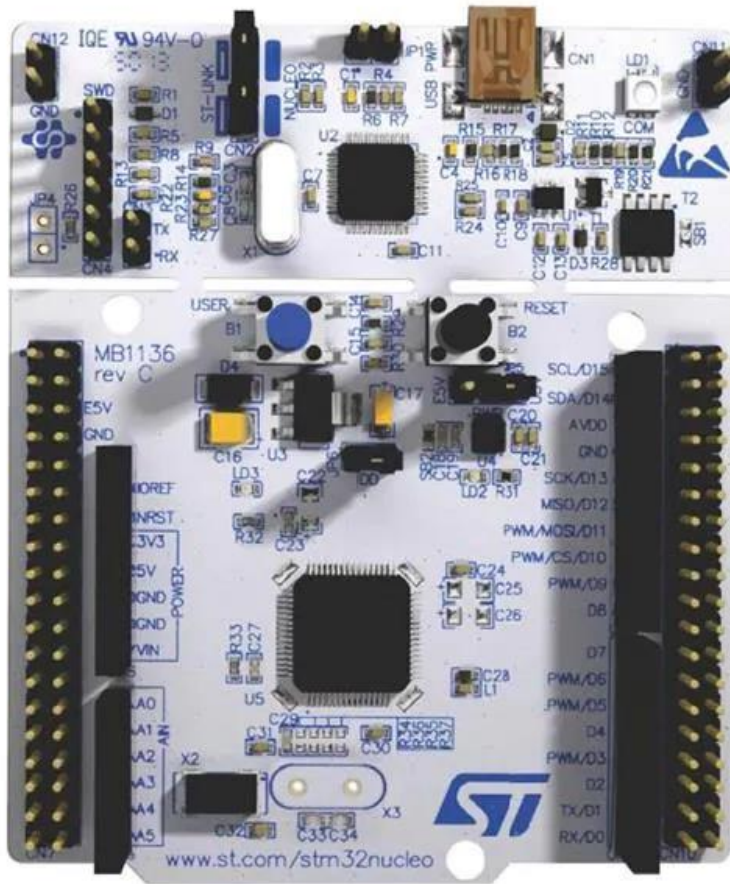
- Arm® Cortex®-M4 32-bit RISC
- F_{cpu} 180 MHz
- Flash 256 KB a 512KB
- RAM 128 KB
- USART, SPI, I²C
- Timers de 16 y 32 bits
- ADC de 12 bits
- Generador de números aleatorios
- Bajo voltaje de 1.7 a 3.6V
- Temperatura 40°C a 125°C

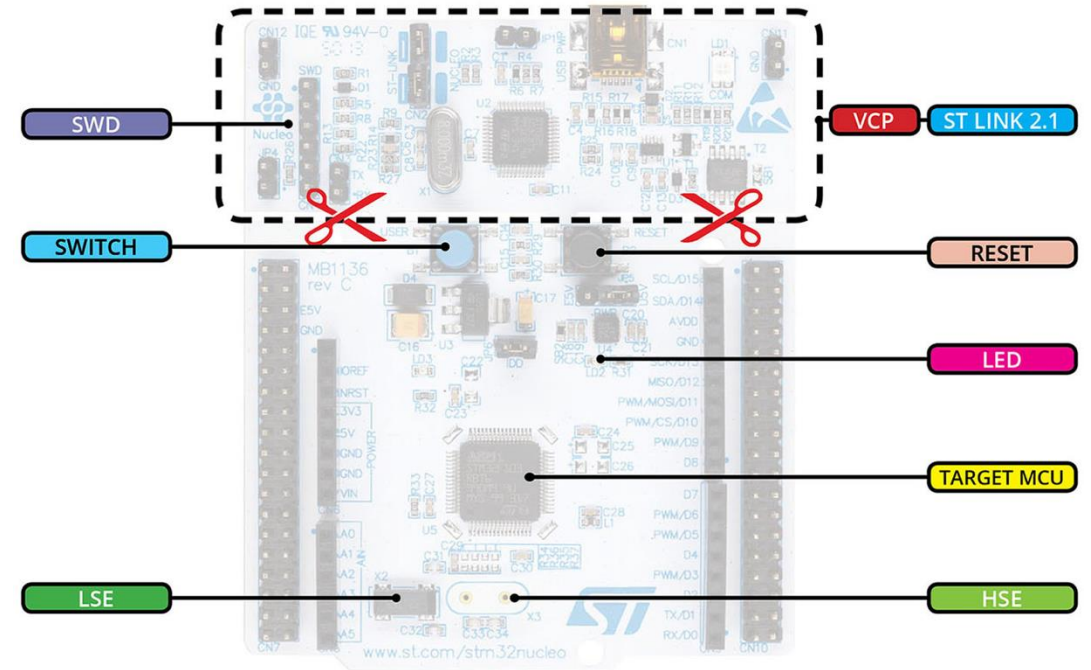
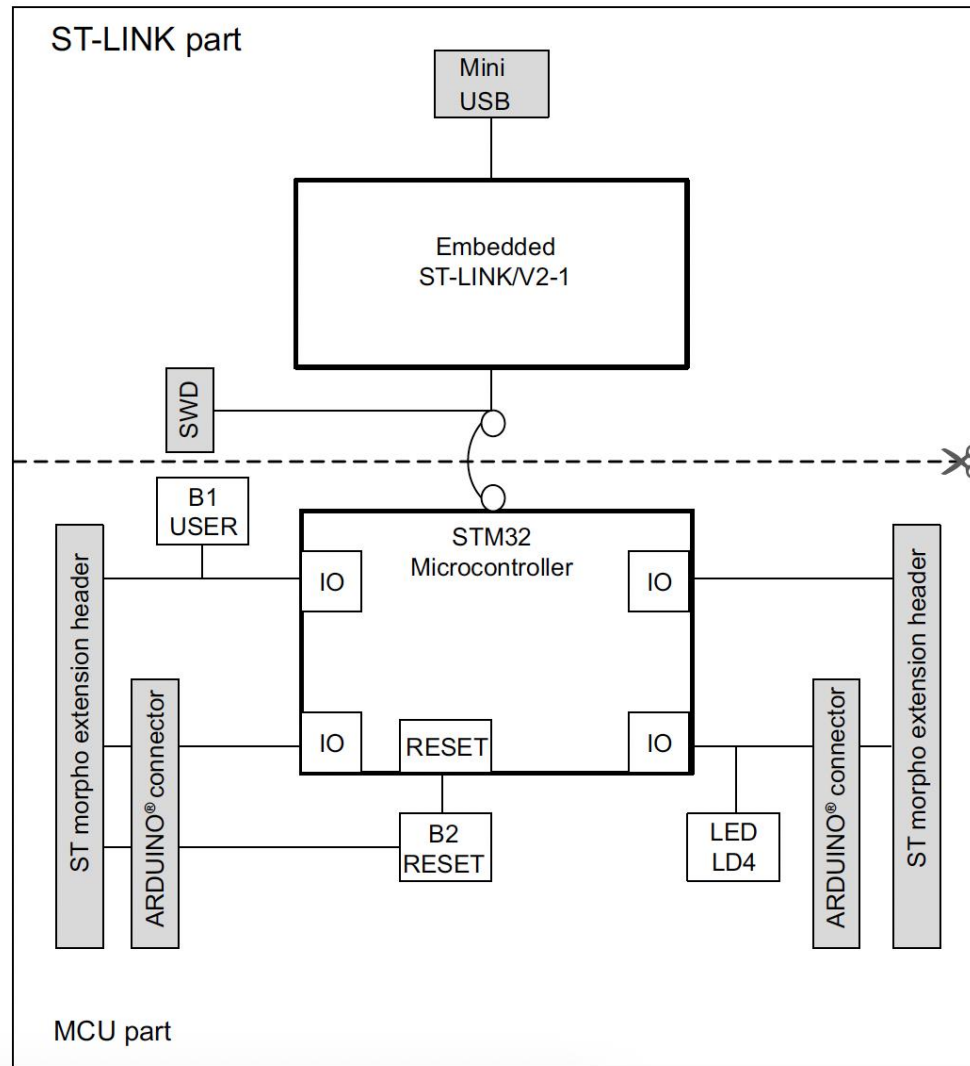


STM32F446

System	Chrom-ART Accelerator™	
Power supply 1.2 V internal regulator POR/PDR/PVD	ART Accelerator™	Connectivity
Xtal oscillators 32 kHz + 4 ~26 MHz	180 MHz Arm® Cortex®-M4 CPU	4x SPI (3x with I²S)
Internal RC oscillators 32 kHz + 16 MHz	Floating Point Unit (FPU)	Camera interface
PLL	Nested Vector Interrupt Controller (NVIC)	4x I²C
Clock control	JTAG/SW debug	2x CAN 2.0B
RTC/AWU	Embedded Trace Macrocell (ETM)	1x USB 2.0 OTG FS/HS
1x SysTick timer	Memory Protection Unit (MPU)	1x USB 2.0 OTG FS
2x watchdogs (independent and window)	Mutli-AHB bus matrix	1x SDMMC
50/63/81/114 I/Os	16-channel DMA	4x USART + 2 UART LIN, smartcard, IrDA, modem control
Cyclic Redundancy Check (CRC)		2x SAI (Serial Audio Interface)
96-bit unique ID		HDMI CEC
Voltage scaling		SPDIF input x4
Control	True random number generator (RNG)	Analog
2x 16-bit motor-control PWM synchronized AC timer	Up to 512-Kbyte Flash memory	2-channel 2x 12-bit DAC
10x 16-bit timers 2x 32-bit timers	128-Kbyte SRAM	Up to 3x 12-bit ADC 2.4 MSPS Up to 24 channels 7.2 MSPS
	External memory interface W/SDRAM support	Temperature sensor
	80-byte + 4-Kbyte backup data	
	512 OTP bytes	
	Dual Quad SPI	

Nucleo64 – STM32F446RE

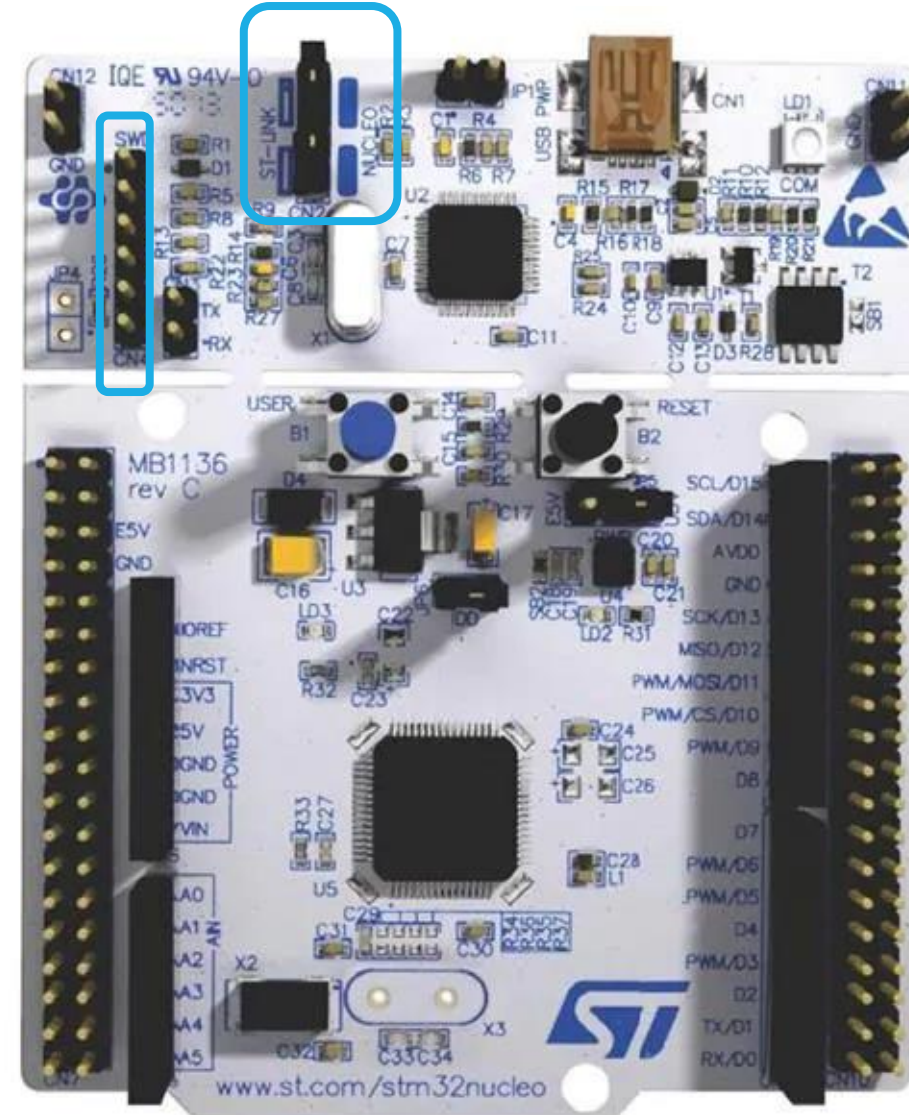


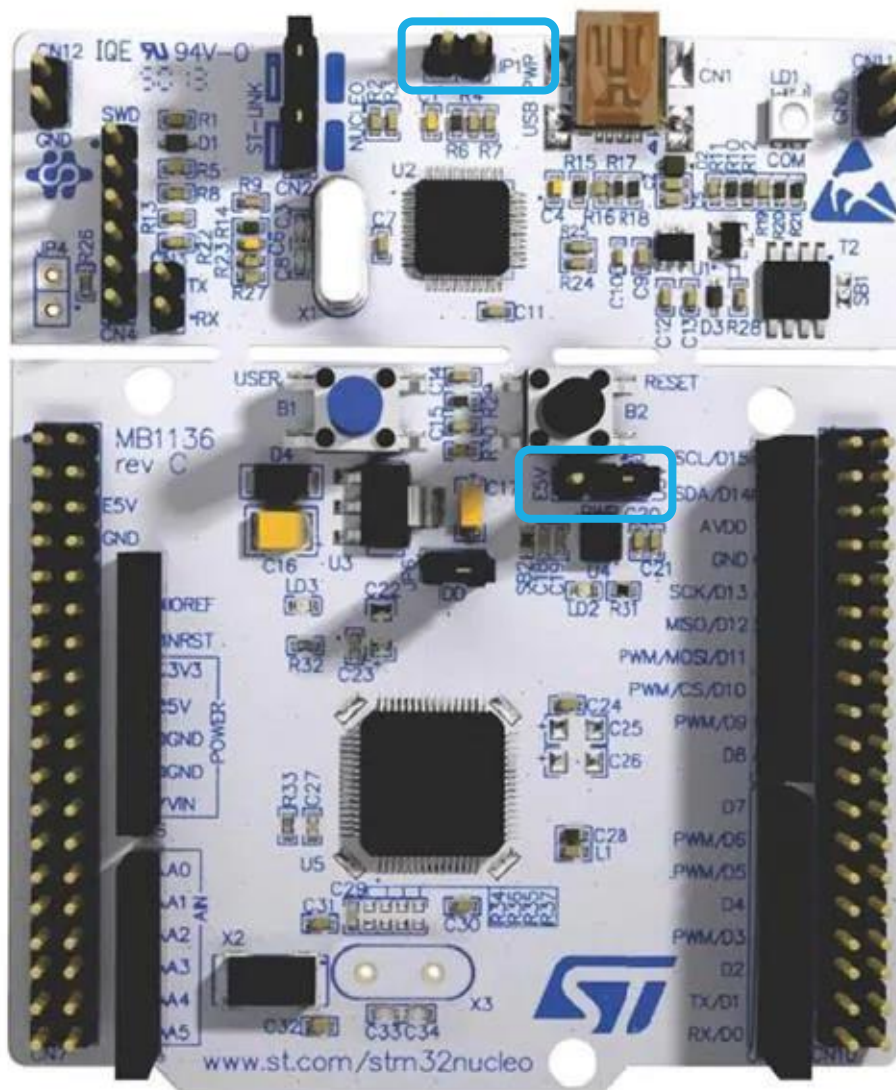


Sin los jumpers

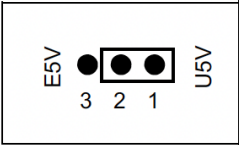
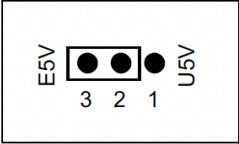
Pin	CN4	Designation
1	VDD_TARGET	VDD from application
2	SWCLK	SWD clock
3	GND	ground
4	SWDIO	SWD data input/output
5	NRST	RESET of target STM32
6	SWO	Reserved

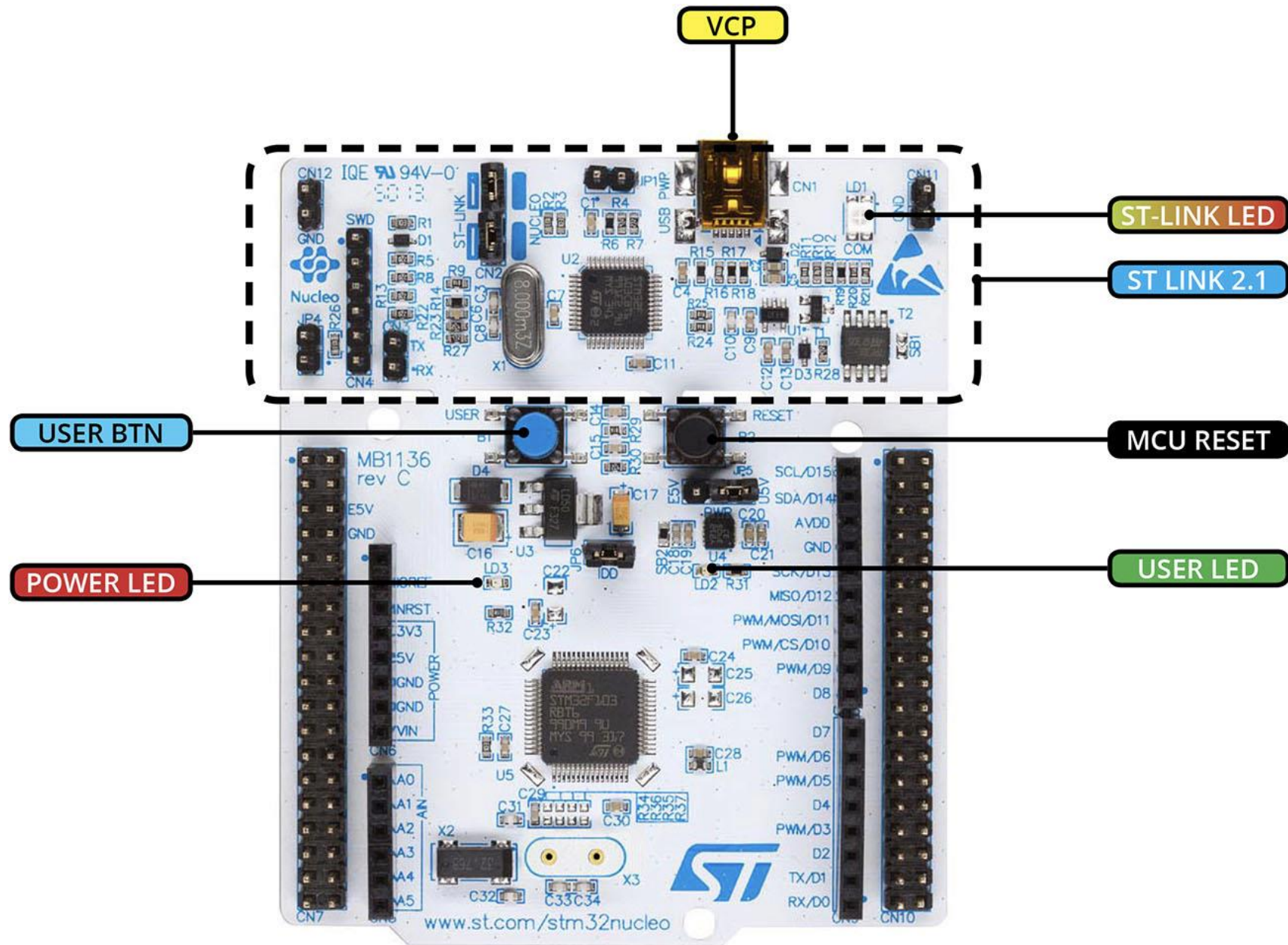
Programador Externo

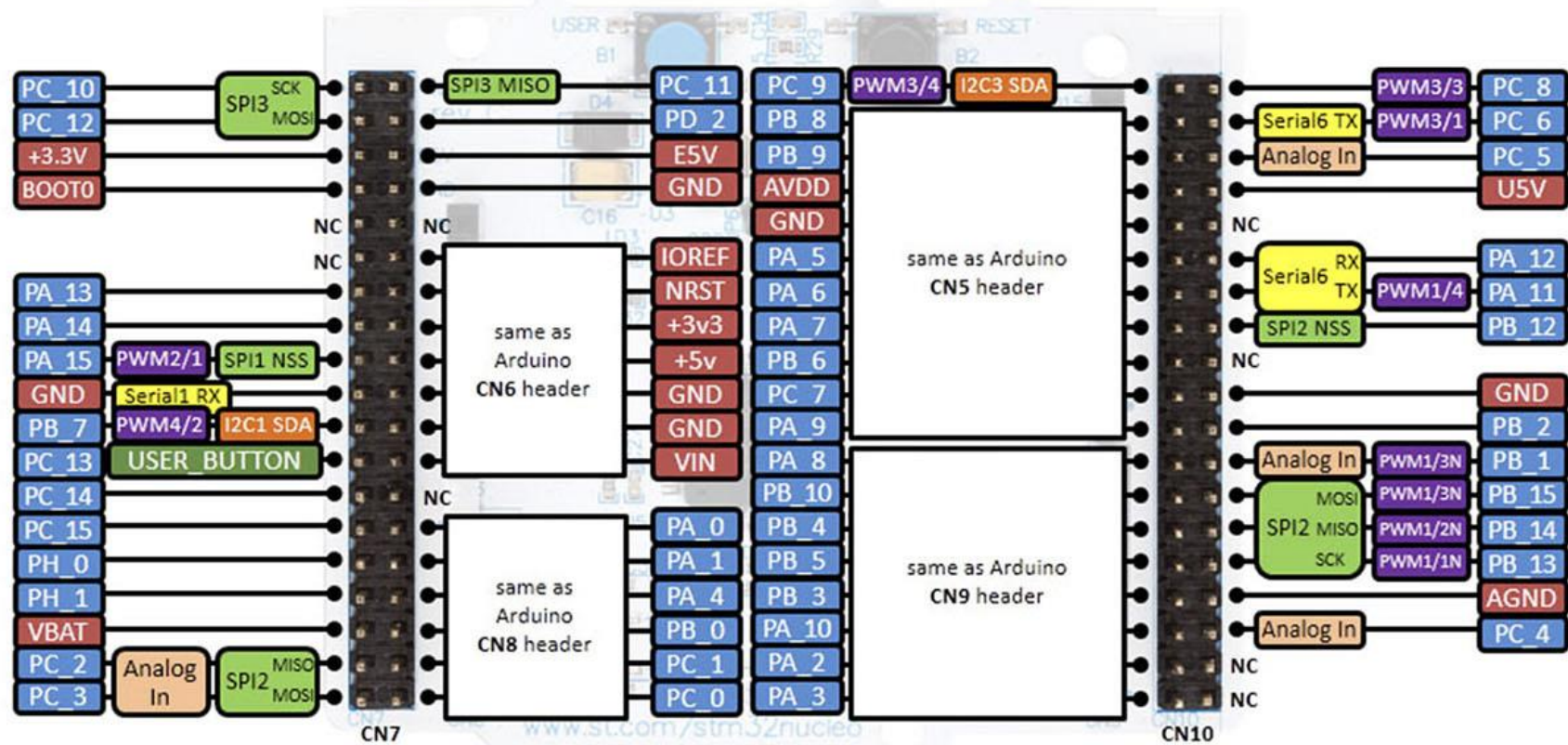
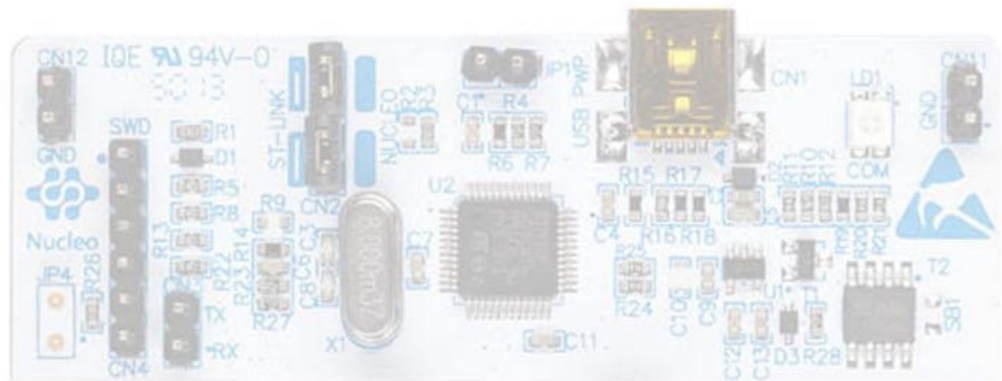




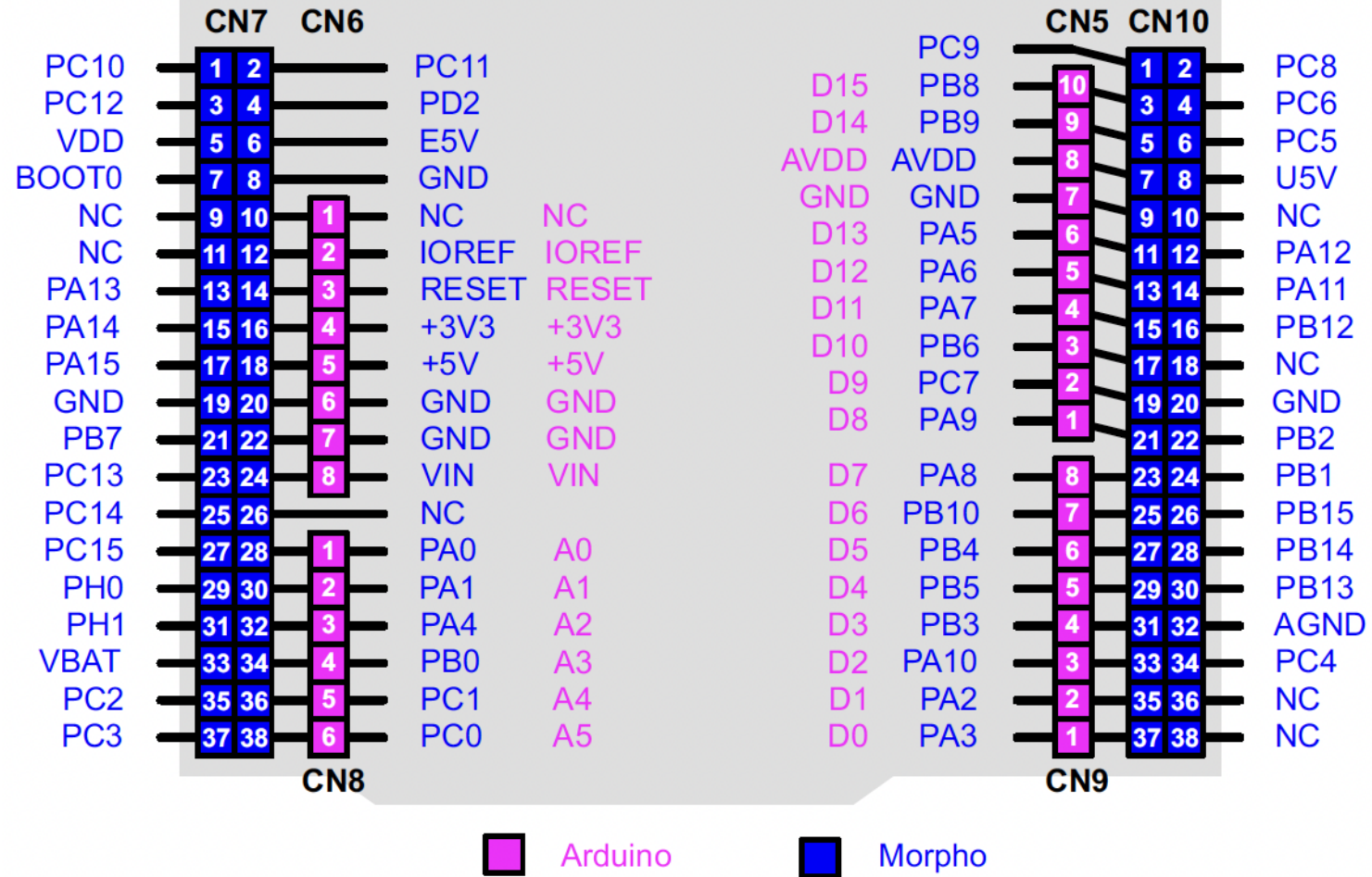
Jumper state	Power supply	Allowed current
JP1 jumper OFF	USB power through CN1	300 mA max
JP1 jumper ON		100 mA max

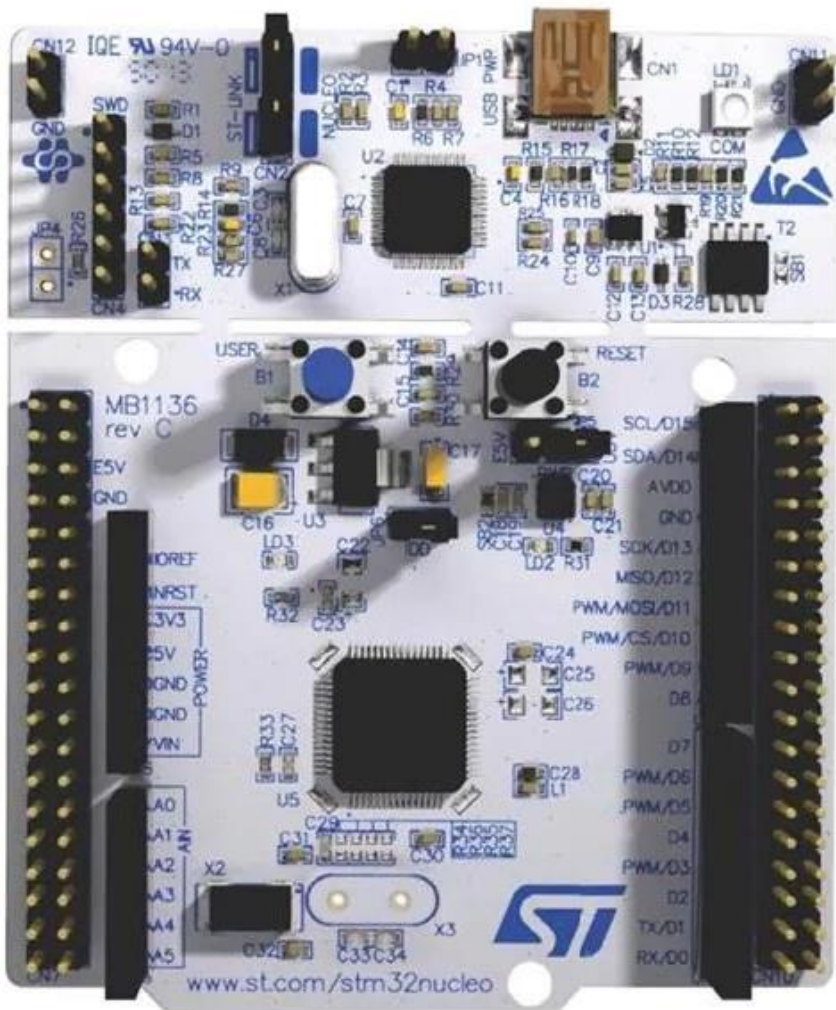
Jumper	Description
JP5	U5V (ST-LINK VBUS) is used as a power source when JP5 is set as shown below (Default setting) 
	VIN or E5V is used as a power source when JP5 is set as shown below. 



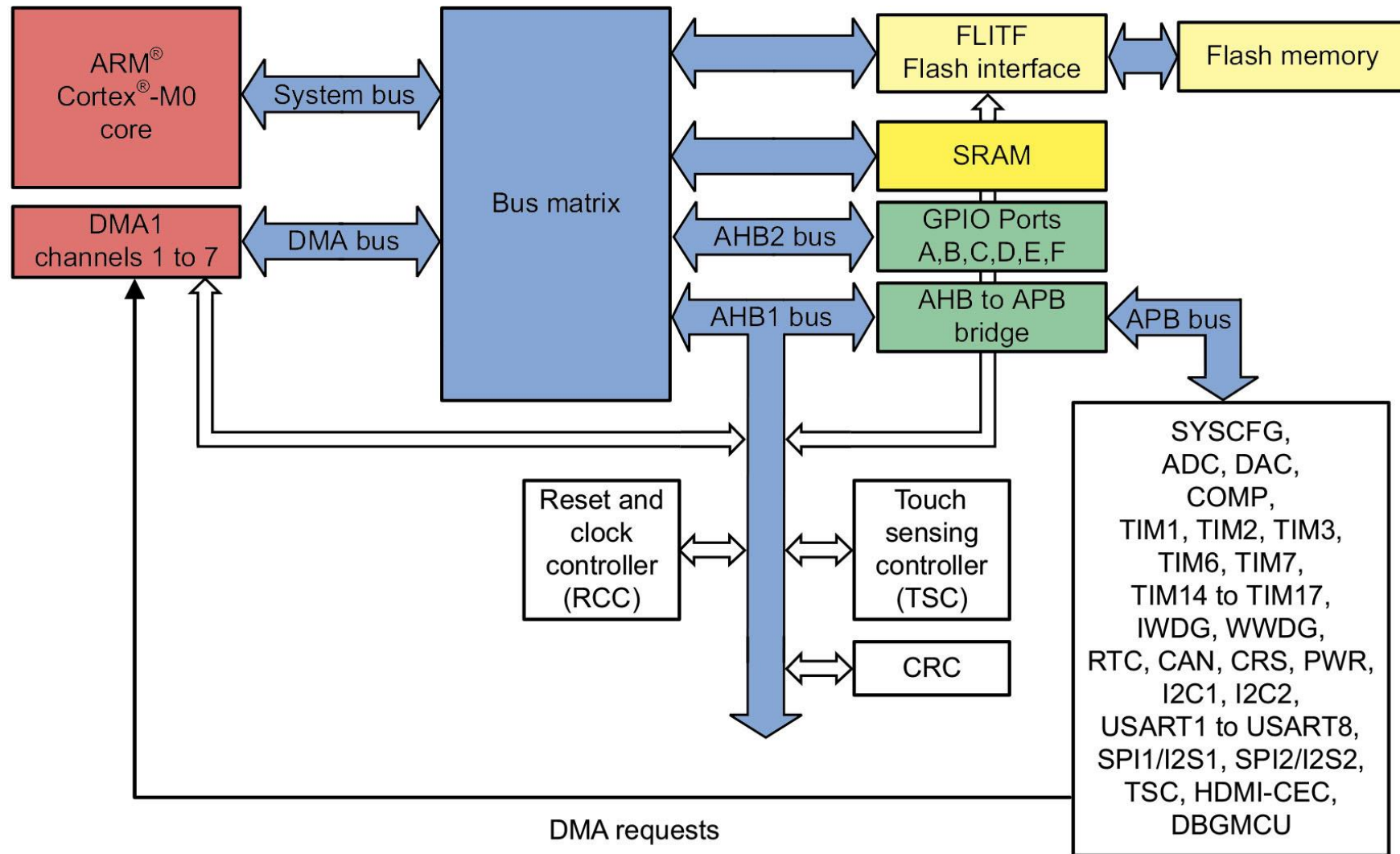


NUCLEO-F446RE

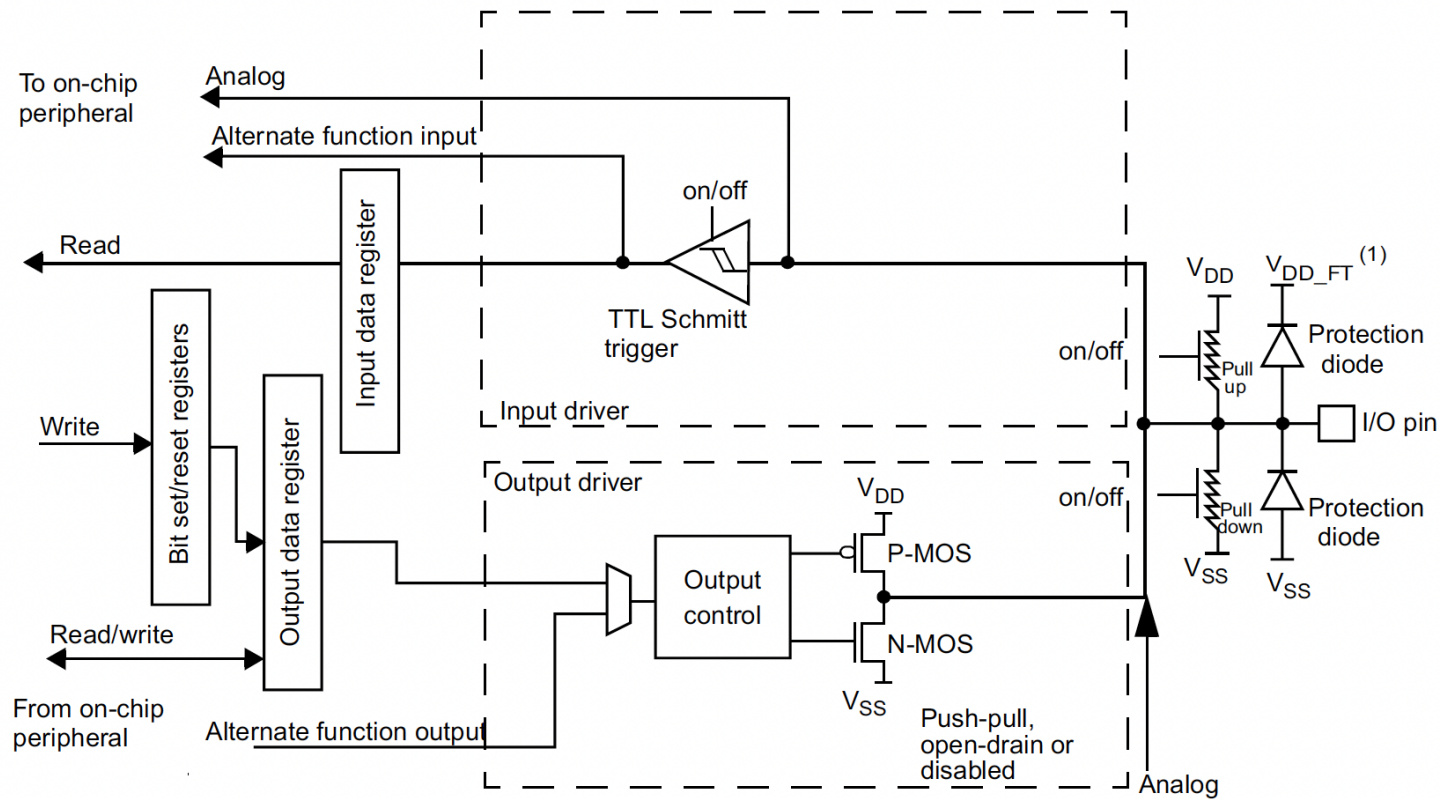




GPIOs



Configuración de GPIOs



Pueden funcionar en los siguientes modos:

Modo Entrada

- Pull-up
- Pull-down
- Velocidad

Modo Salida

- Push-Pull
- Open Collector
- Velocidad
- Estado inicial

Función alterna

Modo Analógico

Configuración de GPIOs

```
typedef struct {  
    uint32_t Pin;  
    uint32_t Mode;  
    uint32_t Pull;  
    uint32_t Speed;  
    uint32_t Alternate;  
} GPIO_InitTypeDef;
```

Pin Mode	Description
GPIO_MODE_INPUT	Input Floating Mode ⁶
GPIO_MODE_OUTPUT_PP	Output Push Pull Mode
GPIO_MODE_OUTPUT_OD	Output Open Drain Mode
GPIO_MODE_AF_PP	Alternate Function Push Pull Mode
GPIO_MODE_AF_OD	Alternate Function Open Drain Mode
GPIO_MODE_ANALOG	Analog Mode
GPIO_MODE_IT_RISING	External Interrupt Mode with Rising edge trigger detection
GPIO_MODE_IT_FALLING	External Interrupt Mode with Falling edge trigger detection
GPIO_MODE_IT_RISING_FALLING	External Interrupt Mode with Rising/Falling edge trigger detection
GPIO_MODE_EVT_RISING	External Event Mode with Rising edge trigger detection
GPIO_MODE_EVT_FALLING	External Event Mode with Falling edge trigger detection
GPIO_MODE_EVT_RISING_FALLING	External Event Mode with Rising/Falling edge trigger detection

Pin Mode	Description
GPIO_NOPULL	No Pull-up or Pull-down activation
GPIO_PULLUP	Pull-up activation
GPIO_PULLDOWN	Pull-down activation

Registros (pg. 187)

GPIOx_ODR: Output Data Register

GPIOx_IDR: Input Data Register

GPIOx_MODER: Port Mode Register

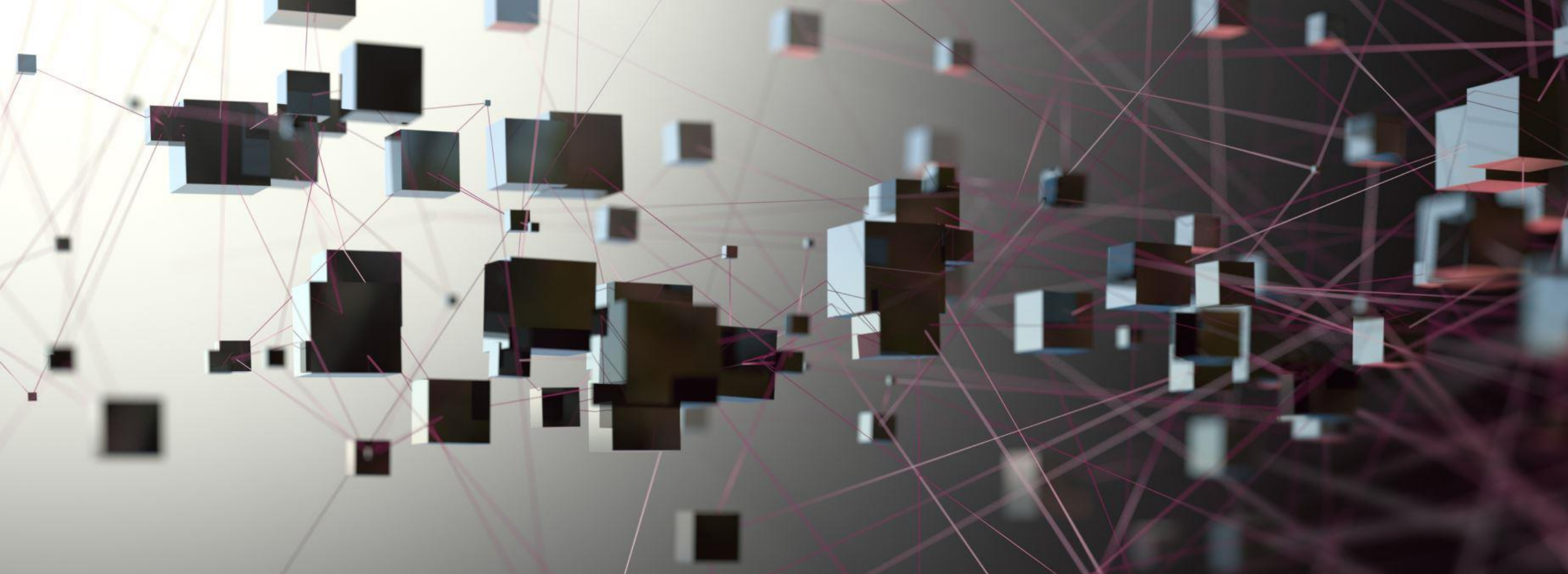
GPIOx_OTYPER: Output type Register

GPIOx_OSPEEDR: Output speed Register

GPIOx_PUPDR: Pull-up/Pull-down Register

GPIOx_BSRR: bit set/reset Register

GPIOx_LCKR: Configuration lock Register

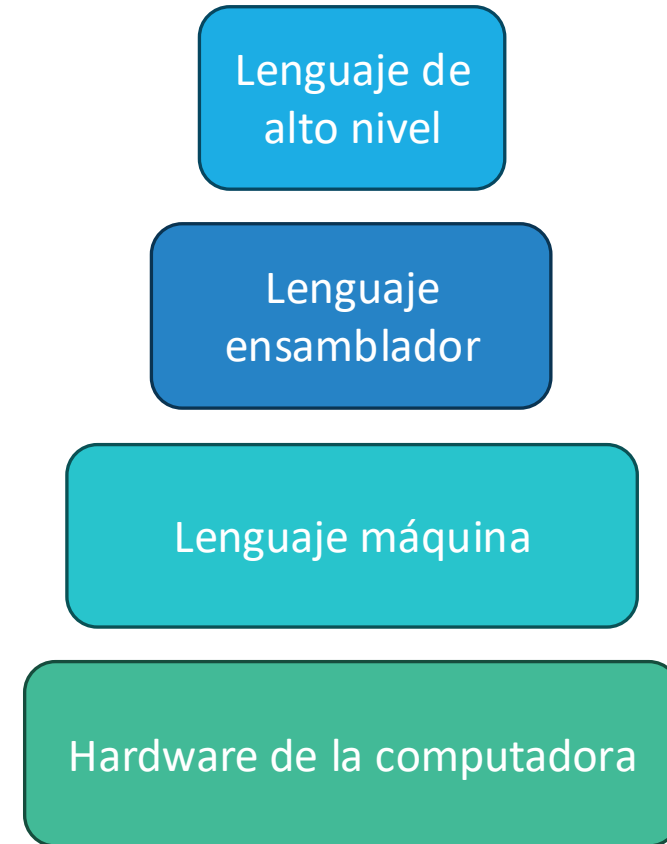
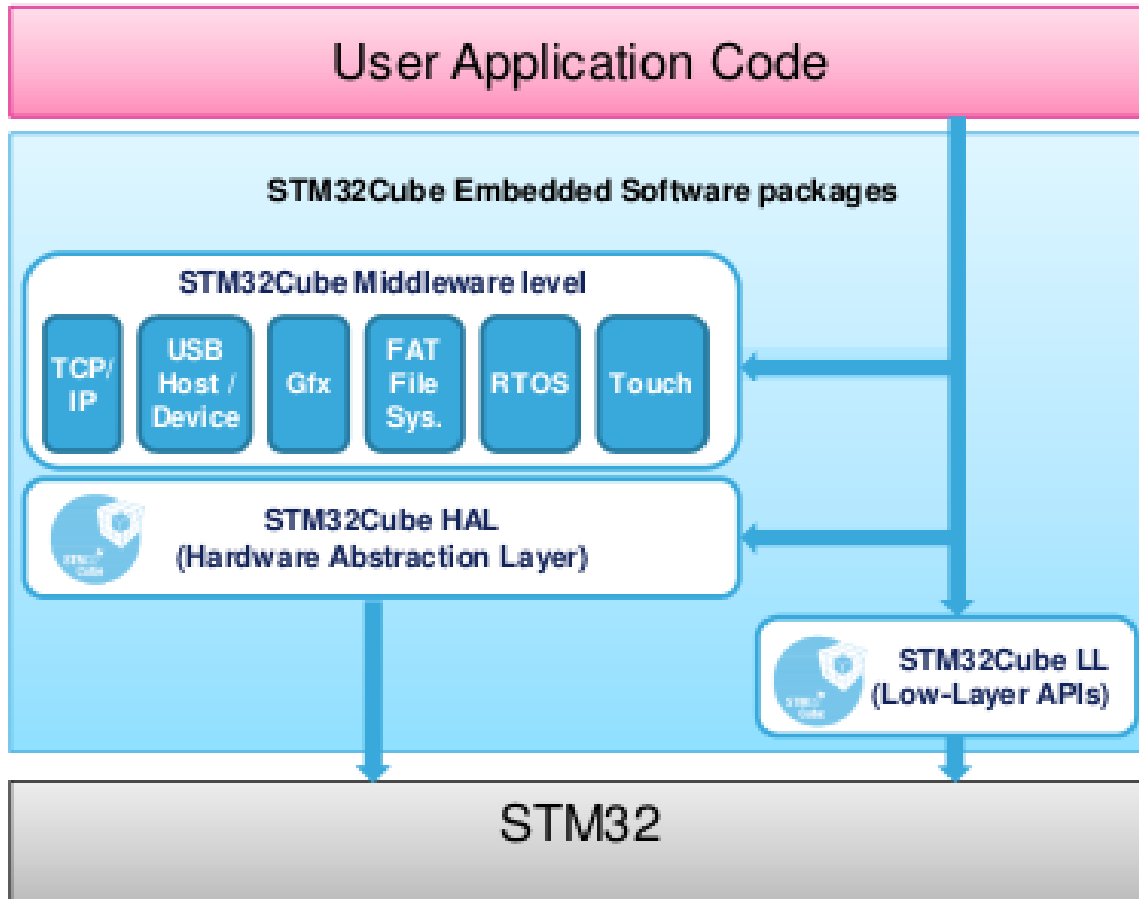


Formas de programar STM32

Formas de programar

Offer		Portability	Optimization (Memory & Mips)	Easy	Readiness	Hardware coverage
 Standard Peripheral Library		++	++	+	++	+++
STM32Cube 	HAL APIs	+++	+	++	+++	+++
	LL APIs	+	+++	+	++	++





STM32 Cube HAL

Podemos tener 4 tipos de acciones

Tipo	Función
Inicialización y desinicialización	HAL_Periférico_Init (), HAL_Periférico_DeInit ()
Entrada o Salida de datos	HAL_Periférico_Read (), HAL_Periférico_Write (), HAL_Periférico_Transmit (), HAL_Periférico_Receive ()
Control	HAL_Periférico_Set (), HAL_Periférico_Get ()
Estado y Error	HAL_Periférico_GetState (), HAL_Periférico_GetError ()

STM32 Cube HAL

Las funciones tienen una estructura estándar

Sintaxis	Función
HAL_Periférico_Función ()	Realiza una acción sobre el periférico
HAL_PeriféricoEx_Función ()	Realiza una acción extendida sobre el periférico
_HAL_Periférico_Función ()	Macro para acceder a un registro particular del periférico
_Periférico_Función ()	Macro para acceder directamente al periférico

STM32 Cube HAL GPIOs

Tiene 5 funciones principales

Obtener el estado del Pin

```
HAL_GPIO_ReadPin(GPIO_TypeDef* GPIOx, uint16_t GPIO_Pin)
```

Escribir en el Pin

```
HAL_GPIO_WritePin(GPIO_TypeDef* GPIOx, uint16_t GPIO_Pin, GPIO_PinState PinState)
```

Realizar toggle

```
| HAL_GPIO_TogglePin(GPIO_TypeDef* GPIOx, uint16_t GPIO_Pin)
```

Poner en lock el Pin

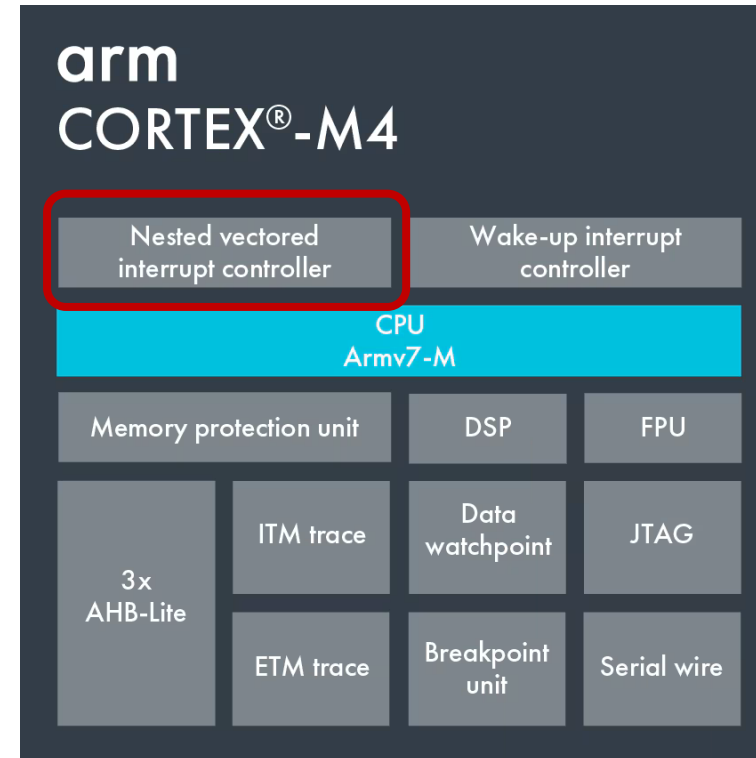
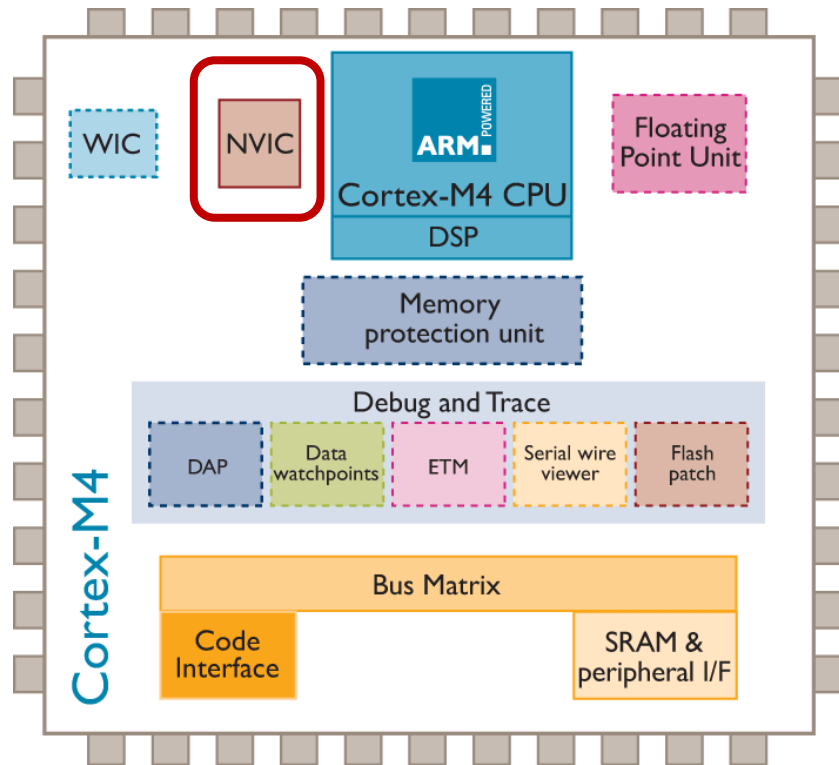
```
HAL_GPIO_LockPin(GPIO_TypeDef* GPIOx, uint16_t GPIO_Pin)
```

Desconfigurar un Pin

```
HAL_GPIO_DeInit(GPIO_TypeDef *GPIOx, uint32_t GPIO_Pin)
```

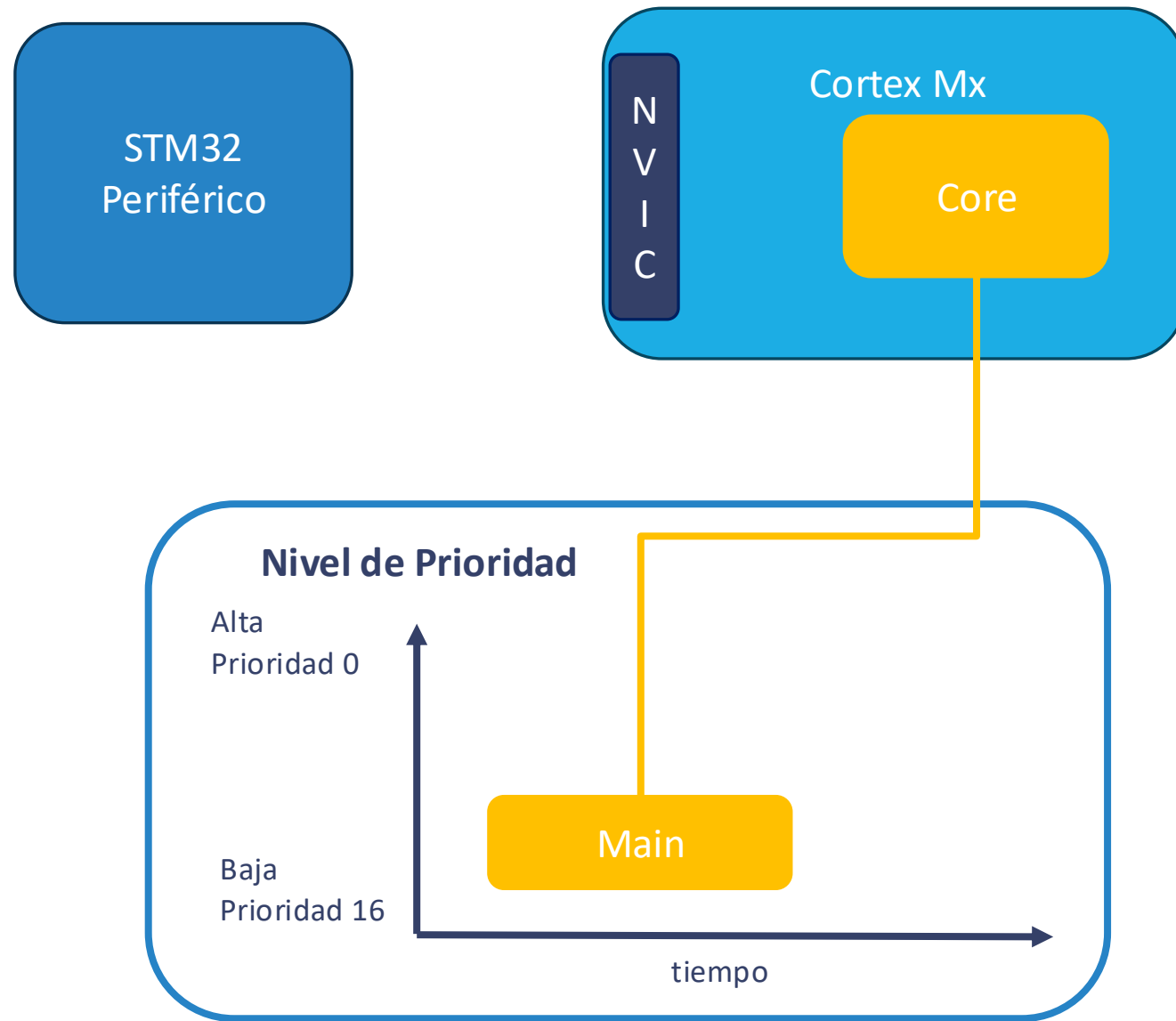
ARM

Interrupciones - Arquitectura

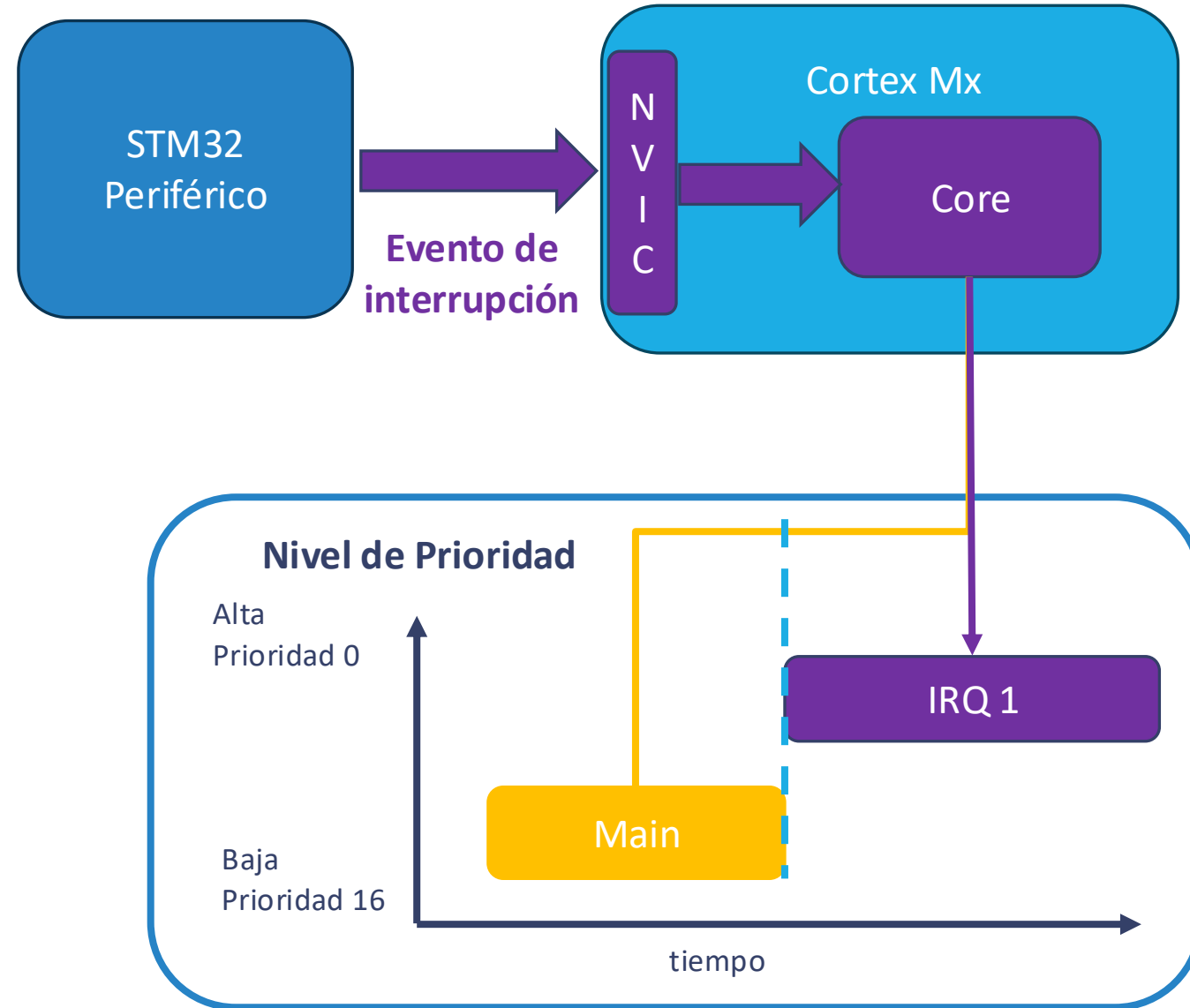


Arquitectura

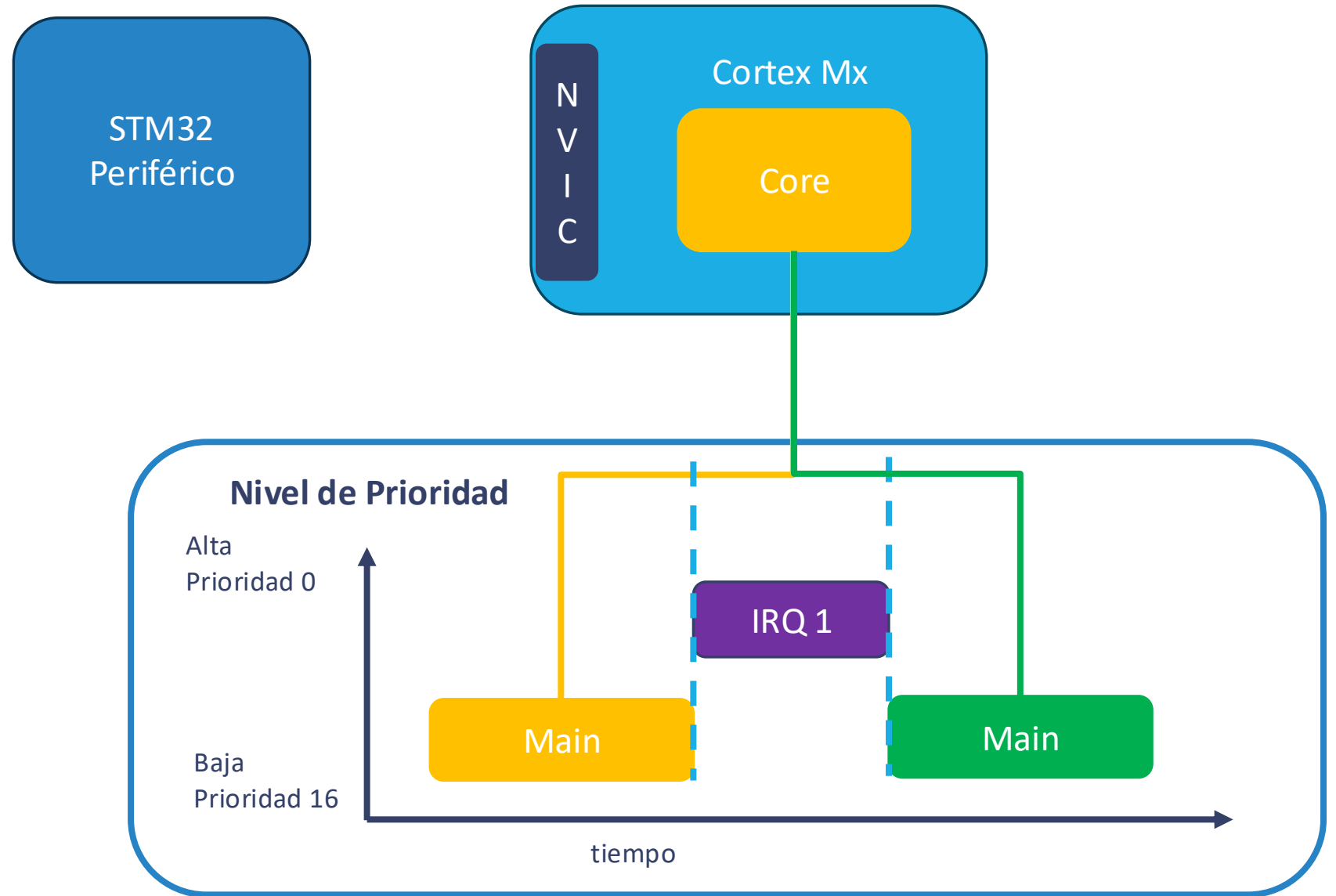
NVIC Nested Vector Interrupt Controller



NVIC Nested Vector Interrupt Controller



NVIC Nested Vector Interrupt Controller



Nested Vectored Interrupt Controller (NVIC) (RM pg. 239)

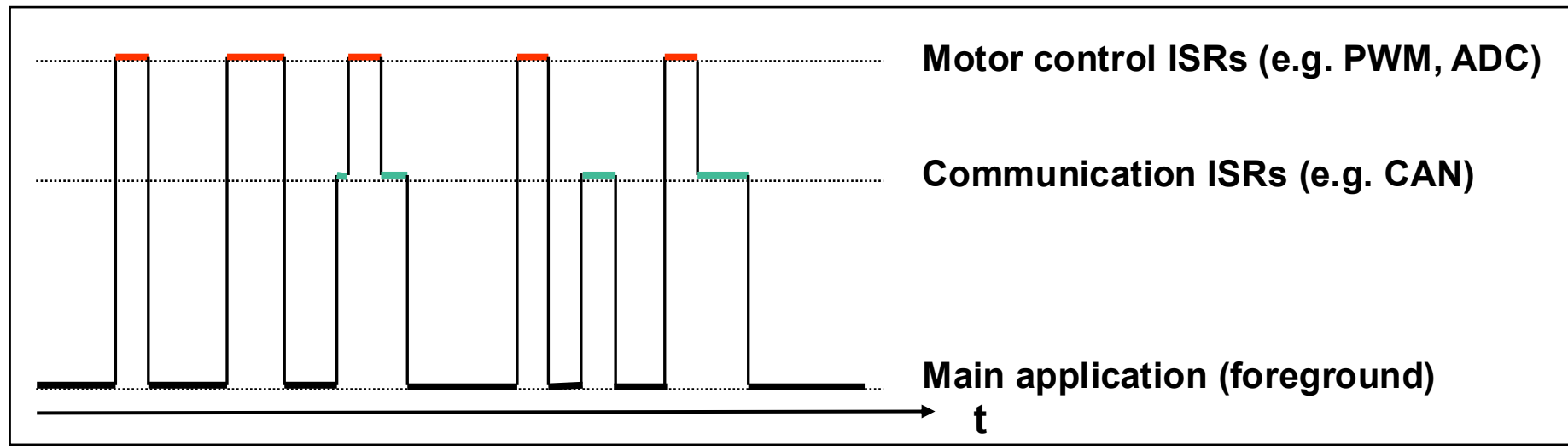
Maneja interrupciones y excepciones

16 niveles programables de prioridades (4 bits) y subprioridades

96 maskable interrupt channels

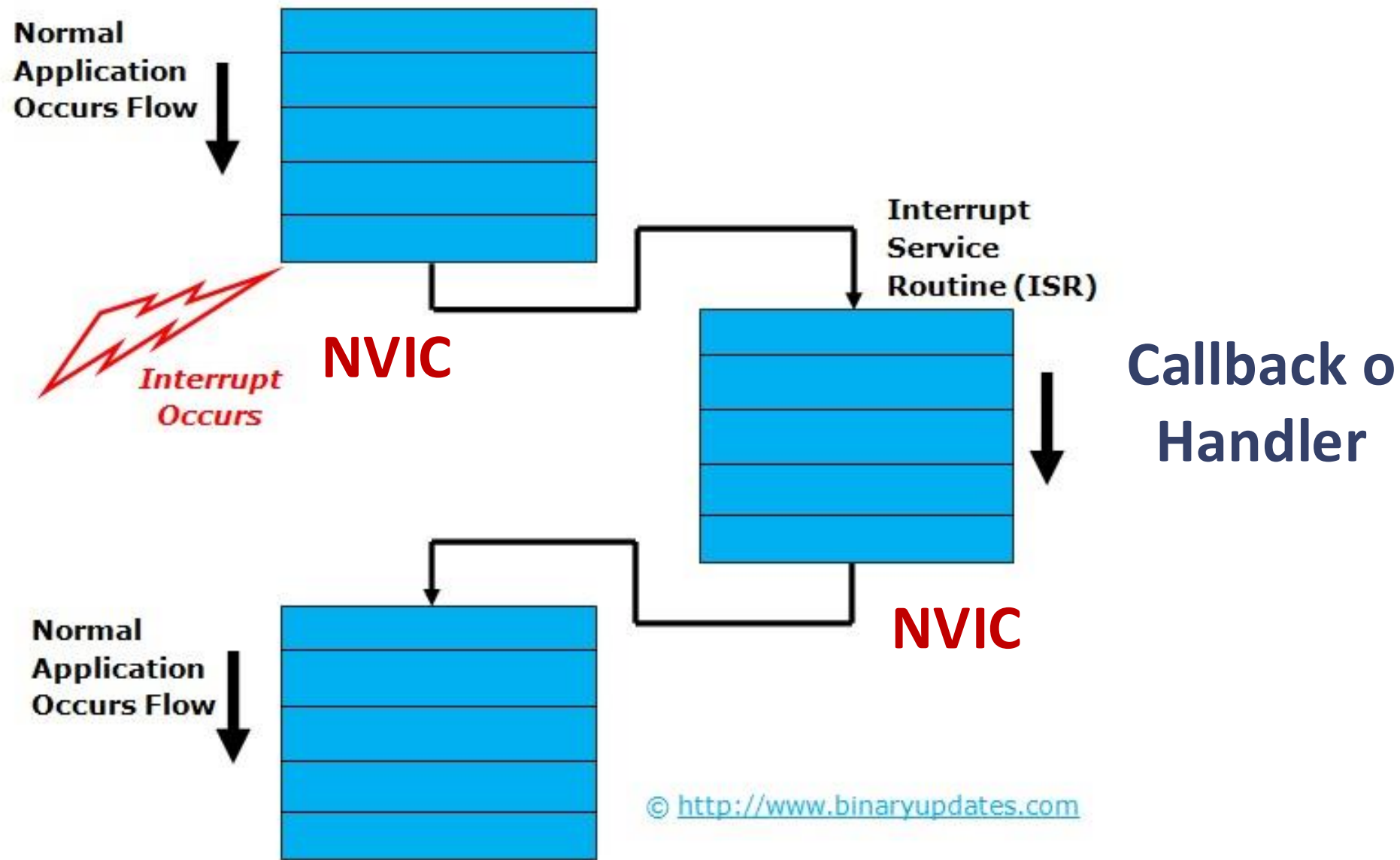
Automatic state saving and restoring

Automatic reading of the vector table entry

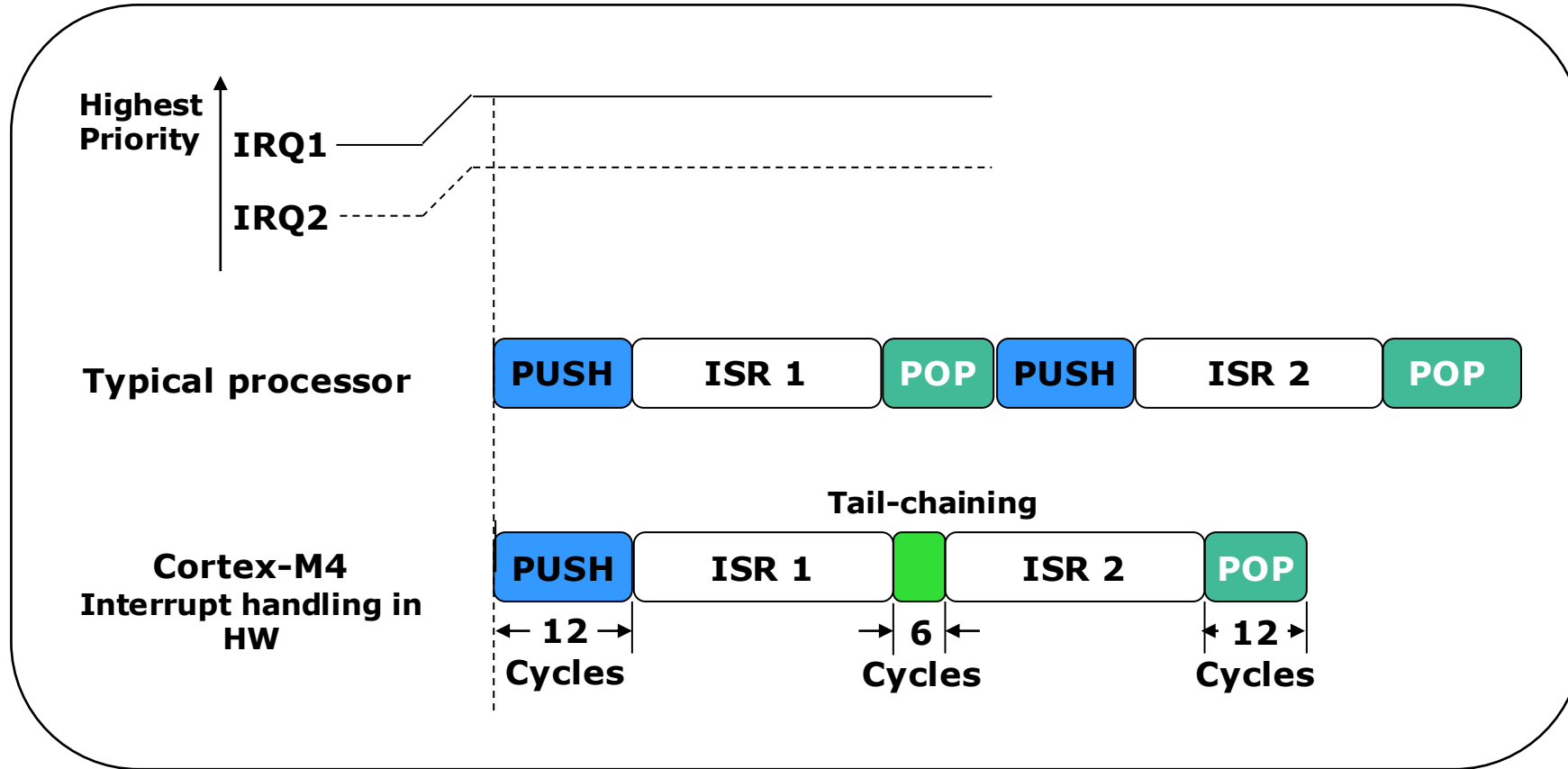


Modo de operación

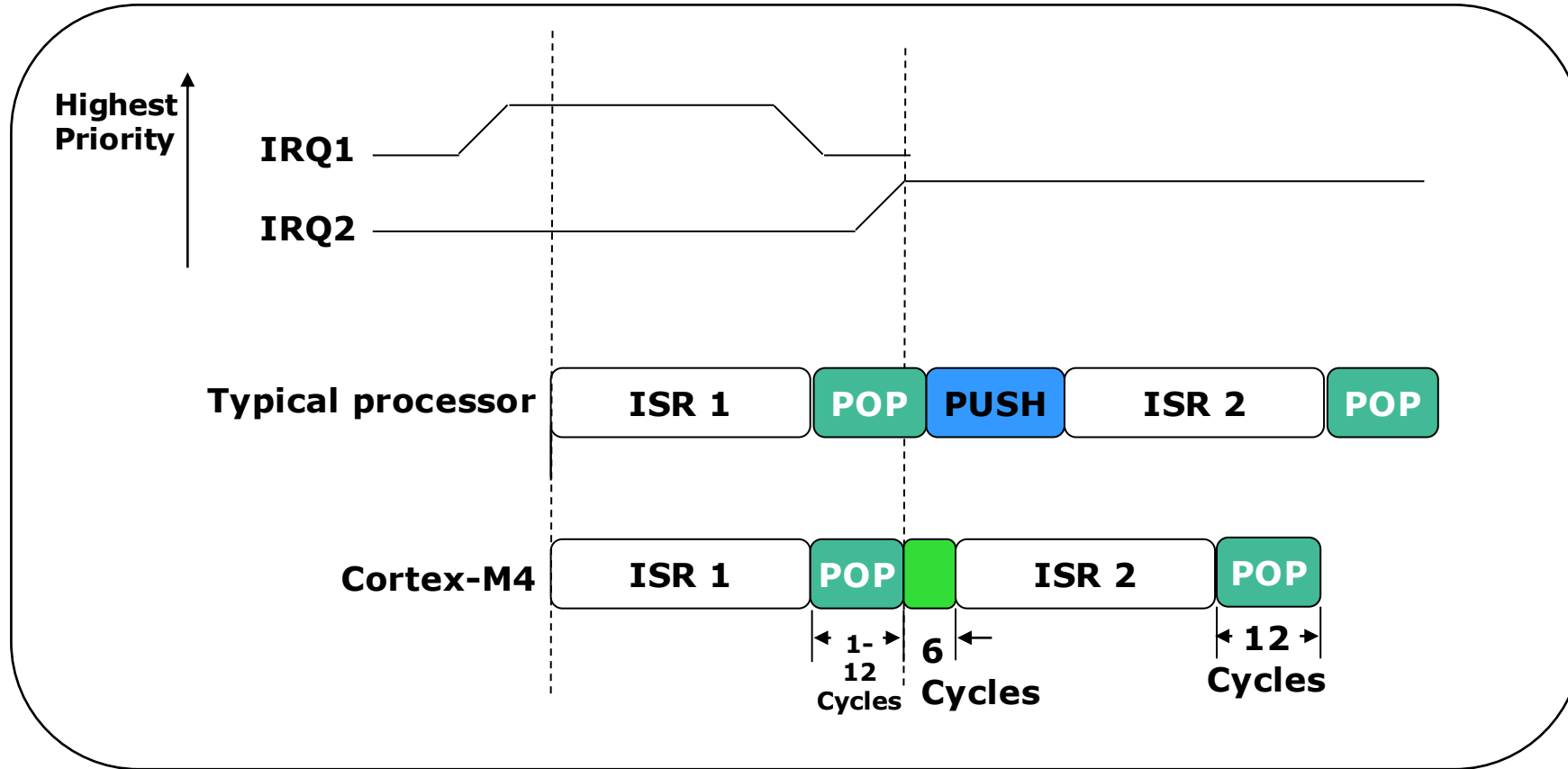
Interrupción



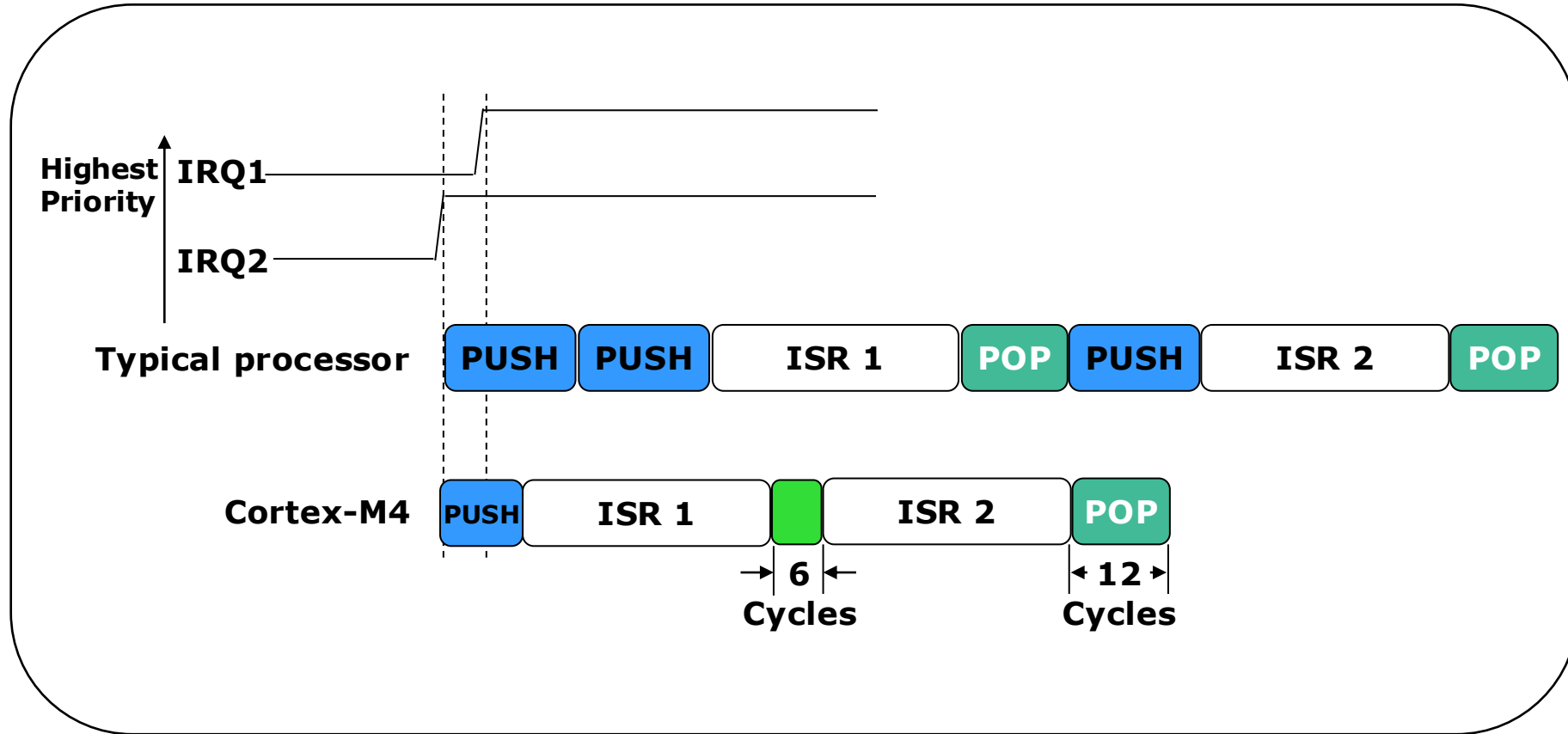
Interrupt Latency - Tail Chaining



Interrupt Latency – Pre-emption



Interrupt Latency – Late Arrival



Cortex-M4[®] Interrupt Handling

Interrupt handling is automatic. No instruction overhead.

Entry

- Automatically pushes registers R0–R3, R12, LR, PSR, and PC onto the stack
- In parallel, ISR is pre-fetched on the instruction bus. ISR ready to start executing as soon as stack PUSH complete

Exit

- Processor state is automatically restored from the stack
- In parallel, interrupted instruction is pre-fetched ready for execution upon completion of stack POP

Tabla 38. Tabla de vector del STM32F446 (RM pg. 239)

Position	Priority	Type of priority	Acronym	Description	Address
-	-	-	-	Reserved	0x0000 0000
-	-3	fixed	Reset	Reset	0x0000 0004
-	-2	fixed	NMI	Non maskable interrupt, Clock Security System	0x0000 0008
-	-1	fixed	HardFault	All class of fault	0x0000 000C
-	0	settable	MemManage	Memory management	0x0000 0010

Position	Priority	Type of priority	Acronym	Description	Address
-	1	settable	BusFault	Pre-fetch fault, memory access fault	0x0000 0014
-	2	settable	UsageFault	Undefined instruction or illegal state	0x0000 0018
-	-	-	-	Reserved	0x0000 001C - 0x0000 002B
-	3	settable	SVCall	System Service call via SWI instruction	0x0000 002C
-	4	settable	Debug Monitor	Debug Monitor	0x0000 0030
-	-	-	-	Reserved	0x0000 0034
-	5	settable	PendSV	Pendable request for system service	0x0000 0038
-	6	settable	SysTick	System tick timer	0x0000 003C
0	7	settable	WWDG	Window Watchdog interrupt	0x0000 0040
1	8	settable	PVD	PVD through EXTI line detection interrupt	0x0000 0044
2	9	settable	TAMP_STAMP	Tamper and TimeStamp interrupts through the EXTI line	0x0000 0048
3	10	settable	RTC_WKUP	RTC Wakeup interrupt through the EXTI line	0x0000 004C
4	11	settable	FLASH	Flash global interrupt	0x0000 0050
5	12	settable	RCC	RCC global interrupt	0x0000 0054
6	13	settable	EXTI0	EXTI Line0 interrupt	0x0000 0058
7	14	settable	EXTI1	EXTI Line1 interrupt	0x0000 005C
8	15	settable	EXTI2	EXTI Line2 interrupt	0x0000 0060
9	16	settable	EXTI3	EXTI Line3 interrupt	0x0000 0064
10	17	settable	EXTI4	EXTI Line4 interrupt	0x0000 0068
11	18	settable	DMA1_Stream0	DMA1 Stream0 global interrupt	0x0000 006C
12	19	settable	DMA1_Stream1	DMA1 Stream1 global interrupt	0x0000 0070
13	20	settable	DMA1_Stream2	DMA1 Stream2 global interrupt	0x0000 0074
14	21	settable	DMA1_Stream3	DMA1 Stream3 global interrupt	0x0000 0078
15	22	settable	DMA1_Stream4	DMA1 Stream4 global interrupt	0x0000 007C
16	23	settable	DMA1_Stream5	DMA1 Stream5 global interrupt	0x0000 0080
17	24	settable	DMA1_Stream6	DMA1 Stream6 global interrupt	0x0000 0084
18	25	settable	ADC	ADC1, ADC2 and ADC3 global interrupts	0x0000 0088
19	26	settable	CAN1_TX	CAN1 TX interrupts	0x0000 008C
20	27	settable	CAN1_RX0	CAN1 RX0 interrupts	0x0000 0090
21	28	settable	CAN1_RX1	CAN1 RX1 interrupt	0x0000 0094
22	29	settable	CAN1_SCE	CAN1 SCE interrupt	0x0000 0098
23	30	settable	EXTI9_5	EXTI Line[9:5] interrupts	0x0000 009C

Position	Priority	Type of priority	Acronym	Description	Address
24	31	settable	TIM1_BRK_TIM9	TIM1 Break interrupt and TIM9 global interrupt	0x0000 00A0
25	32	settable	TIM1_UP_TIM10	TIM1 Update interrupt and TIM10 global interrupt	0x0000 00A4
26	33	settable	TIM1_TRG_COM_TIM11	TIM1 Trigger and Commutation interrupts and TIM11 global interrupt	0x0000 00A8
27	34	settable	TIM1_CC	TIM1 Capture compare interrupt	0x0000 00AC
28	35	settable	TIM2	TIM2 global interrupt	0x0000 00B0
29	36	settable	TIM3	TIM3 global interrupt	0x0000 00B4
30	37	settable	TIM4	TIM4 global interrupt	0x0000 00B8
31	38	settable	I2C1_EV	I ² C1 event interrupt	0x0000 00BC
32	39	settable	I2C1_ER	I ² C1 error interrupt	0x0000 00C0
33	40	settable	I2C2_EV	I ² C2 event interrupt	0x0000 00C4
34	41	settable	I2C2_ER	I ² C2 error interrupt	0x0000 00C8
35	42	settable	SPI1	SPI1 global interrupt	0x0000 00CC
36	43	settable	SPI2	SPI2 global interrupt	0x0000 00D0
37	44	settable	USART1	USART1 global interrupt	0x0000 00D4
38	45	settable	USART2	USART2 global interrupt	0x0000 00D8
39	46	settable	USART3	USART3 global interrupt	0x0000 00DC
40	47	settable	EXTI15_10	EXTI Line[15:10] interrupts	0x0000 00E0
41	48	settable	RTC_Alarm	RTC Alarms (A and B) through EXTI line interrupt	0x0000 00E4
42	49	settable	OTG_FS_WKUP	USB On-The-Go FS Wakeup through EXTI line interrupt	0x0000 00E8
43	50	settable	TIM8_BRK_TIM12	TIM8 break interrupt and TIM12 global interrupt	0x0000 00EC
44	51	settable	TIM8_UP_TIM13	TIM8 Update interrupt and TIM13 global interrupt	0x0000 00F0
45	52	settable	TIM8_TRG_COM_TIM14	TIM8 Trigger and Commutation interrupts and TIM14 global interrupt	0x0000 00F4
46	53	settable	TIM8_CC	TIM8 Capture compare interrupt	0x0000 00F8
47	54	settable	DMA1_Stream7	DMA1 Stream7 global interrupt	0x0000 00FC
48	55	settable	FMC	FMC global interrupt	0x0000 0100
49	56	settable	SDIO	SDIO global interrupt	0x0000 0104
50	57	settable	TIM5	TIM5 global interrupt	0x0000 0108
51	58	settable	SPI3	SPI3 global interrupt	0x0000 010C
52	59	settable	UART4	UART4 global interrupt	0x0000 0110

Position	Priority	Type of priority	Acronym	Description	Address
53	60	settable	UART5	UART5 global interrupt	0x0000 0114
54	61	settable	TIM6_DAC	TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	0x0000 0118
55	62	settable	TIM7	TIM7 global interrupt	0x0000 011C
56	63	settable	DMA2_Stream0	DMA2 Stream0 global interrupt	0x0000 0120
57	64	settable	DMA2_Stream1	DMA2 Stream1 global interrupt	0x0000 0124
58	65	settable	DMA2_Stream2	DMA2 Stream2 global interrupt	0x0000 0128
59	66	settable	DMA2_Stream3	DMA2 Stream3 global interrupt	0x0000 012C
60	67	settable	DMA2_Stream4	DMA2 Stream4 global interrupt	0x0000 0130
61	68	-	-	Reserved	0x0000 0134
62	69	-	-	Reserved	0x0000 0138
63	70	settable	CAN2_TX	CAN2 TX interrupts	0x0000 013C
64	71	settable	CAN2_RX0	CAN2 RX0 interrupts	0x0000 0140
65	72	settable	CAN2_RX1	CAN2 RX1 interrupt	0x0000 0144
66	73	settable	CAN2_SCE	CAN2 SCE interrupt	0x0000 0148
67	74	settable	OTG_FS	USB On The Go FS global interrupt	0x0000 014C
68	75	settable	DMA2_Stream5	DMA2 Stream5 global interrupt	0x0000 0150
69	76	settable	DMA2_Stream6	DMA2 Stream6 global interrupt	0x0000 0154
70	77	settable	DMA2_Stream7	DMA2 Stream7 global interrupt	0x0000 0158
71	78	settable	USART6	USART6 global interrupt	0x0000 015C
72	79	settable	I2C3_EV	I ² C3 event interrupt	0x0000 0160
73	80	settable	I2C3_ER	I ² C3 error interrupt	0x0000 0164
74	81	settable	OTG_HS_EP1_OUT	USB On The Go HS End Point 1 Out global interrupt	0x0000 0168
75	82	settable	OTG_HS_EP1_IN	USB On The Go HS End Point 1 In global interrupt	0x0000 016C
76	83	settable	OTG_HS_WKUP	USB On The Go HS Wakeup through EXTI interrupt	0x0000 0170
77	84	settable	OTG_HS	USB On The Go HS global interrupt	0x0000 0174
78	85	settable	DCMI	DCMI global interrupt	0x0000 0178
79	86	-	-	Reserved	0x0000 017C
80	87	-	-	Reserved	0x0000 0180
81	88	settable	FPU	FPU global interrupt	0x0000 0184
82	89	-	-	Reserved	0x0000 0188
83	90	-	-	Reserved	0x0000 018C

Position	Priority	Type of priority	Acronym	Description	Address
84	91	settable	SPI4	SPI 4 global interrupt	0x0000 0190
85	92	-	-	Reserved	0x0000 0194
86	93	-	-	Reserved	0x0000 0198
87	94	settable	SAI1	SAI1 global interrupt	0x0000 019C
88	95	-	-	Reserved	0x0000 01A0
89	96	-	-	Reserved	0x0000 01A4
90	97	-	-	Reserved	0x0000 01A8
91	98	settable	SAI2	SAI2 global interrupt	0x0000 01AC
92	99	settable	QuadSPI	QuadSPI global interrupt	0x0000 01B0
93	100	settable	HDMI-CEC	HDMI-CEC global interrupt	0x0000 01B4
94	101	settable	SPDIF-Rx	SPDIF-Rx global interrupt	0x0000 01B8
95	102	settable	FMPI2C1	FMPI2C1 event interrupt	0x0000 01BC
96	103	settable	FMPI2C1 error	FMPI2C1 error interrupt	0x0000 01C0

HAL service peripherals NVIC interrupt settings

Se utilizan las siguientes funciones

- *HAL_NVIC_SetPriority*
- *HAL_NVIC_EnableIRQ / HAL_NVIC_DisableIRQ*
- *HAL_XXX_Proceso_IT()*

```
void HAL_NVIC_EnableIRQ(IRQn_Type IRQn);  
void HAL_NVIC_DisableIRQ(IRQn_Type IRQn);
```

Y la subrutina se denomina, genéricamente:

- *HAL_XXX_ProcesoCptlCallback()*

Las funciones de las interrupciones se pueden programar en los archivos ***stm32f4xxit.c*** y ***stm32f4xxit.h***

Interrupciones externas EXTI

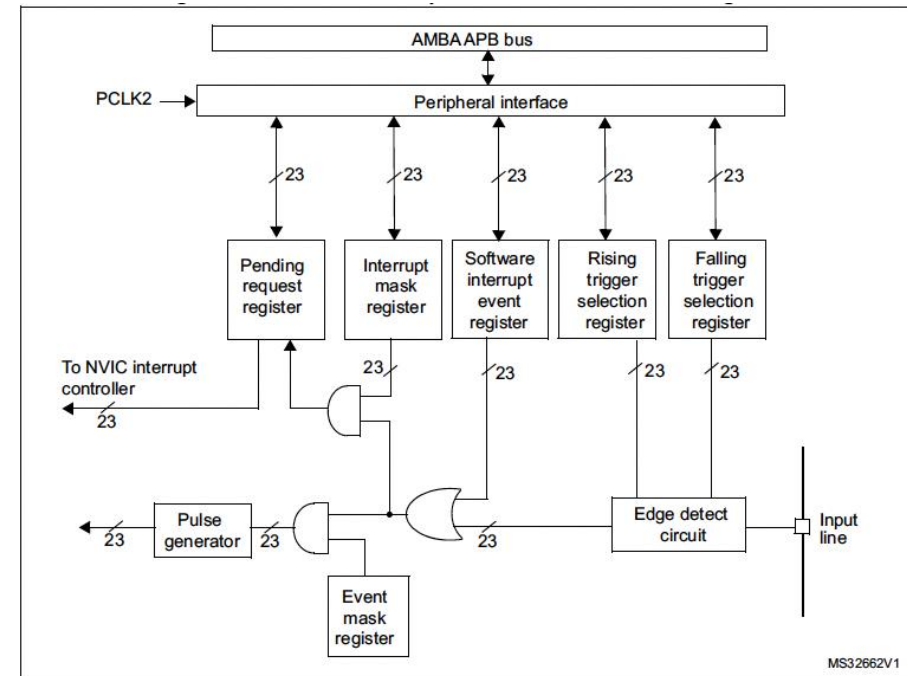
16 canales EXTI0 – EXTI15

Pueden ser activadas por

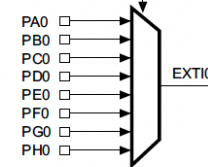
- Flanco positivo
- Flanco negativo
- Cambio de estado

Callback:

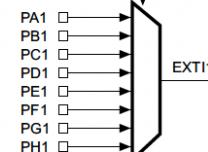
```
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin) {  
}
```



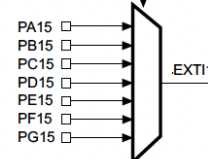
EXTI0[3:0] bits in the SYSCFG_EXTICR1 register



EXTI1[3:0] bits in the SYSCFG_EXTICR1 register

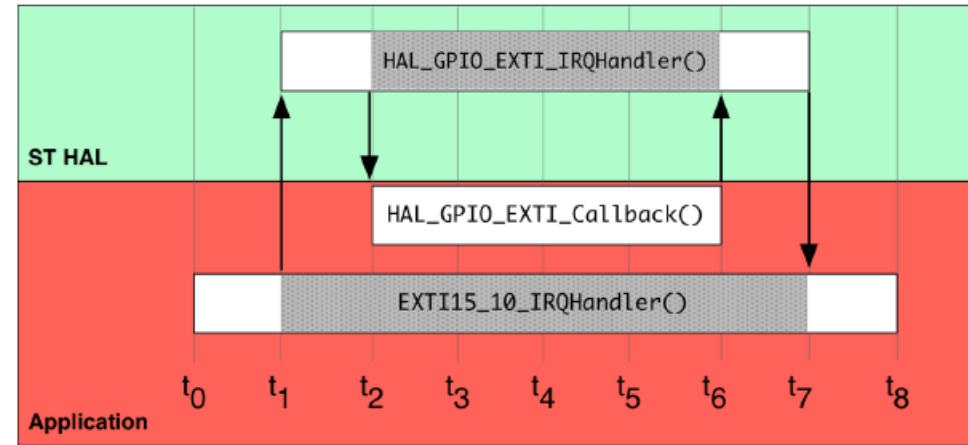


EXTI15[3:0] bits in the SYSCFG_EXTICR4 register



EXTI 0
EXTI 1
EXTI 2
EXTI 3
EXTI 4
EXTI 5:9
EXTI 10:15

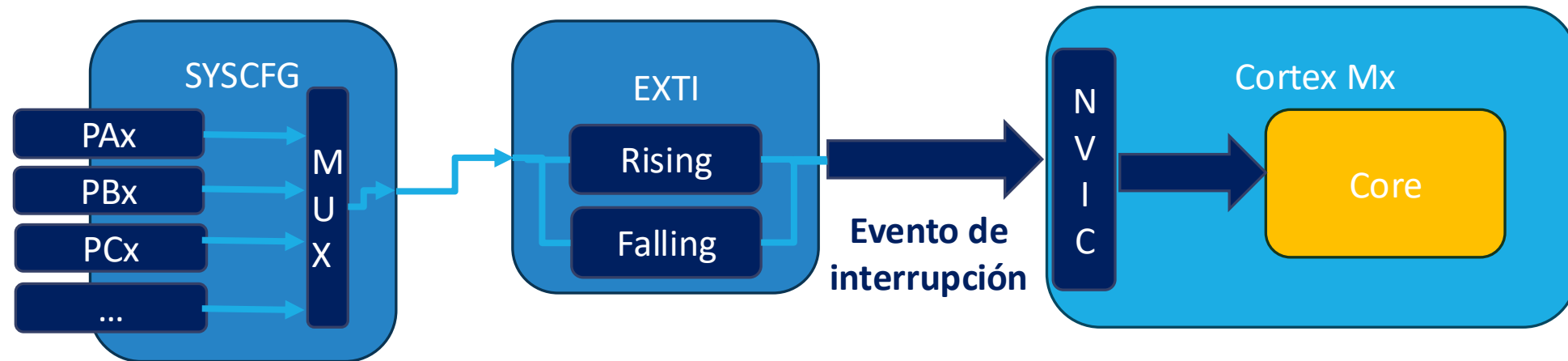
Ejemplo:



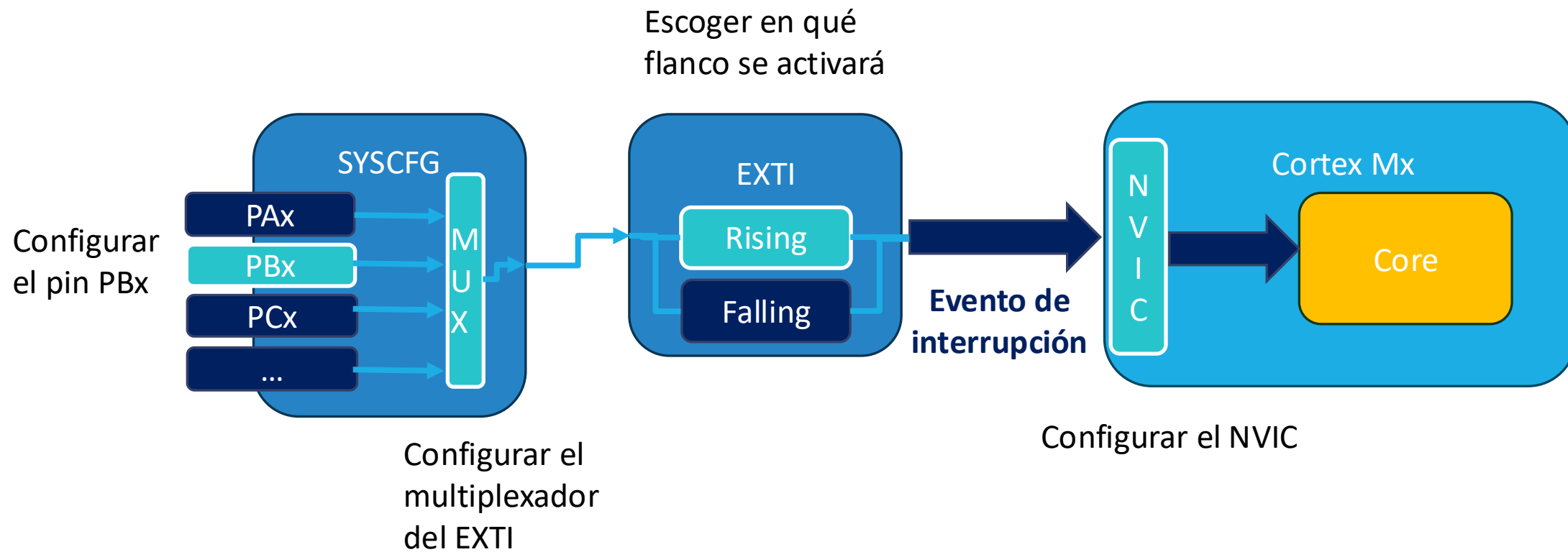
```
void EXTI15_10_IRQHandler(void) {  
    HAL_GPIO_EXTI_IRQHandler(GPIO_PIN_12);  
    HAL_GPIO_EXTI_IRQHandler(GPIO_PIN_13);  
}
```

```
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin) {  
    if(GPIO_Pin == GPIO_PIN_13)  
        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, SET);  
    else if(GPIO_Pin == GPIO_PIN_12)  
        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, RESET);  
}
```

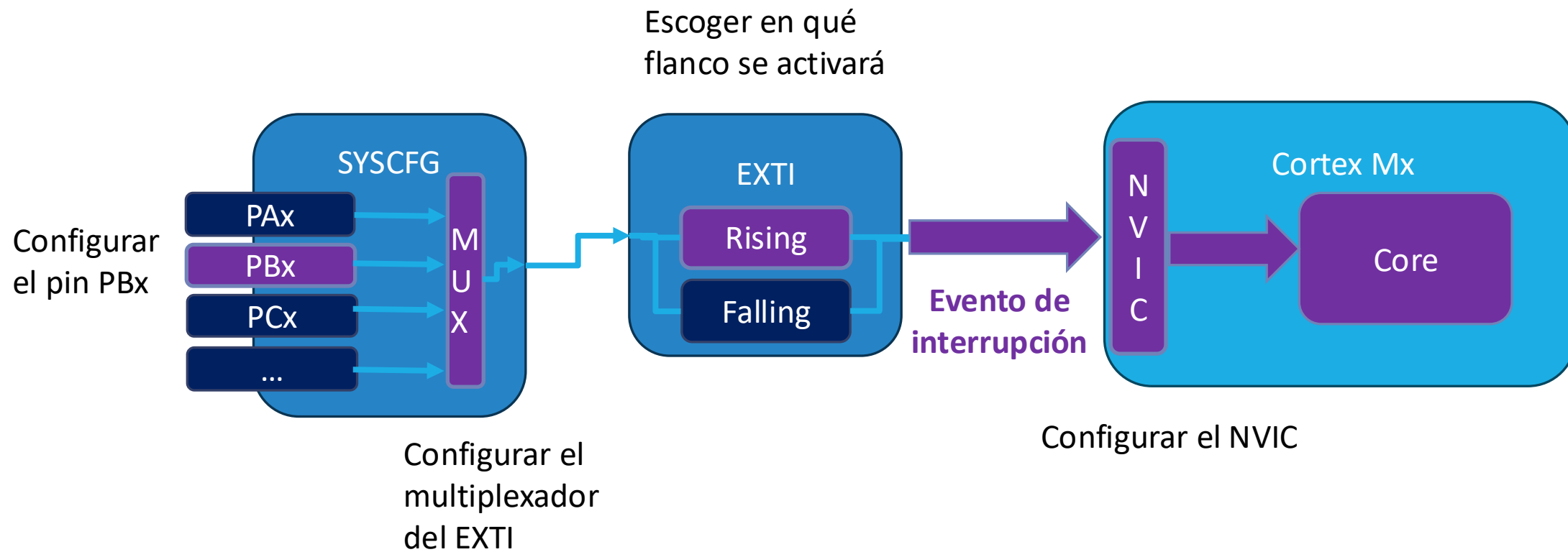
Workflow



Workflow

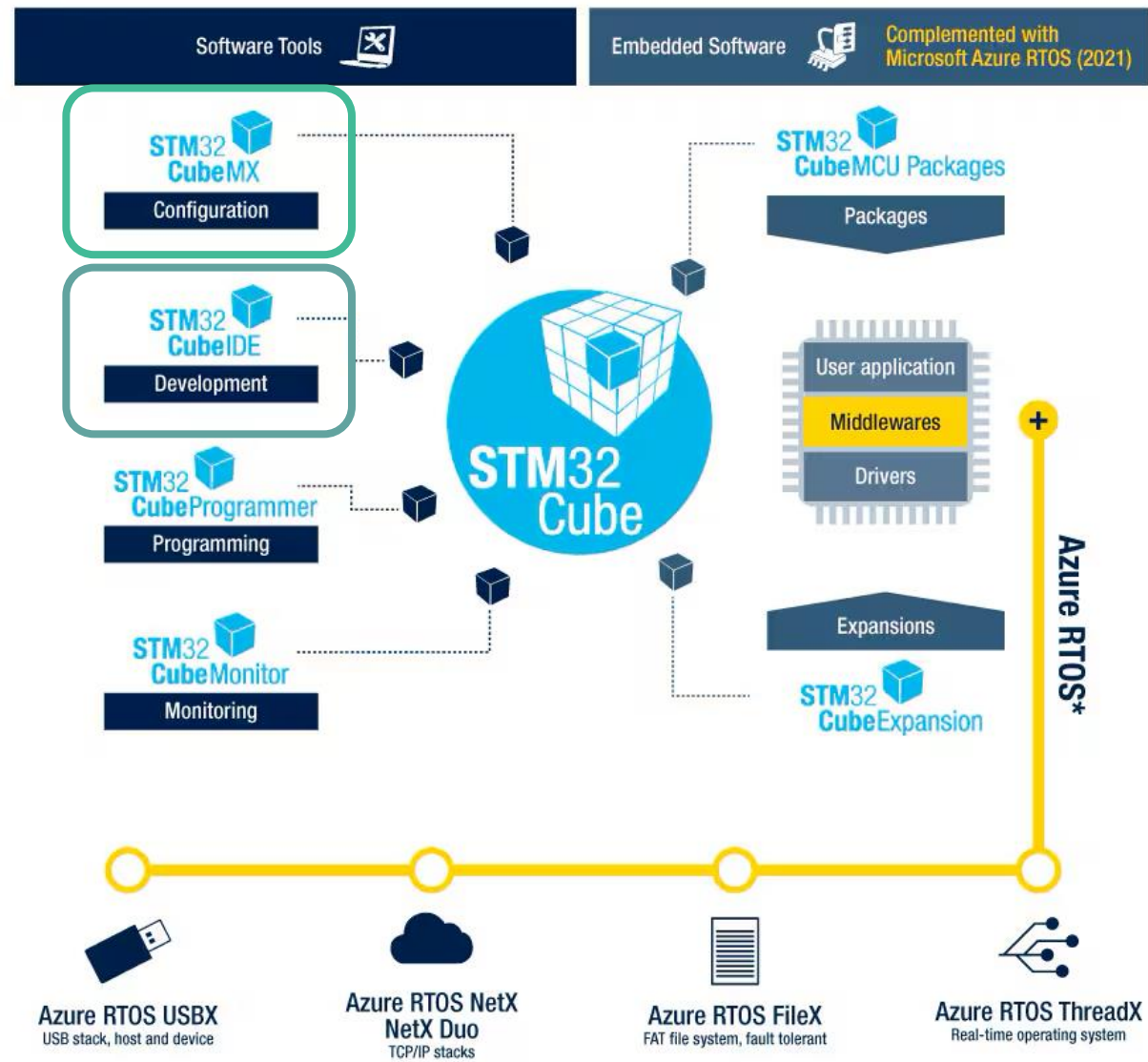


Workflow

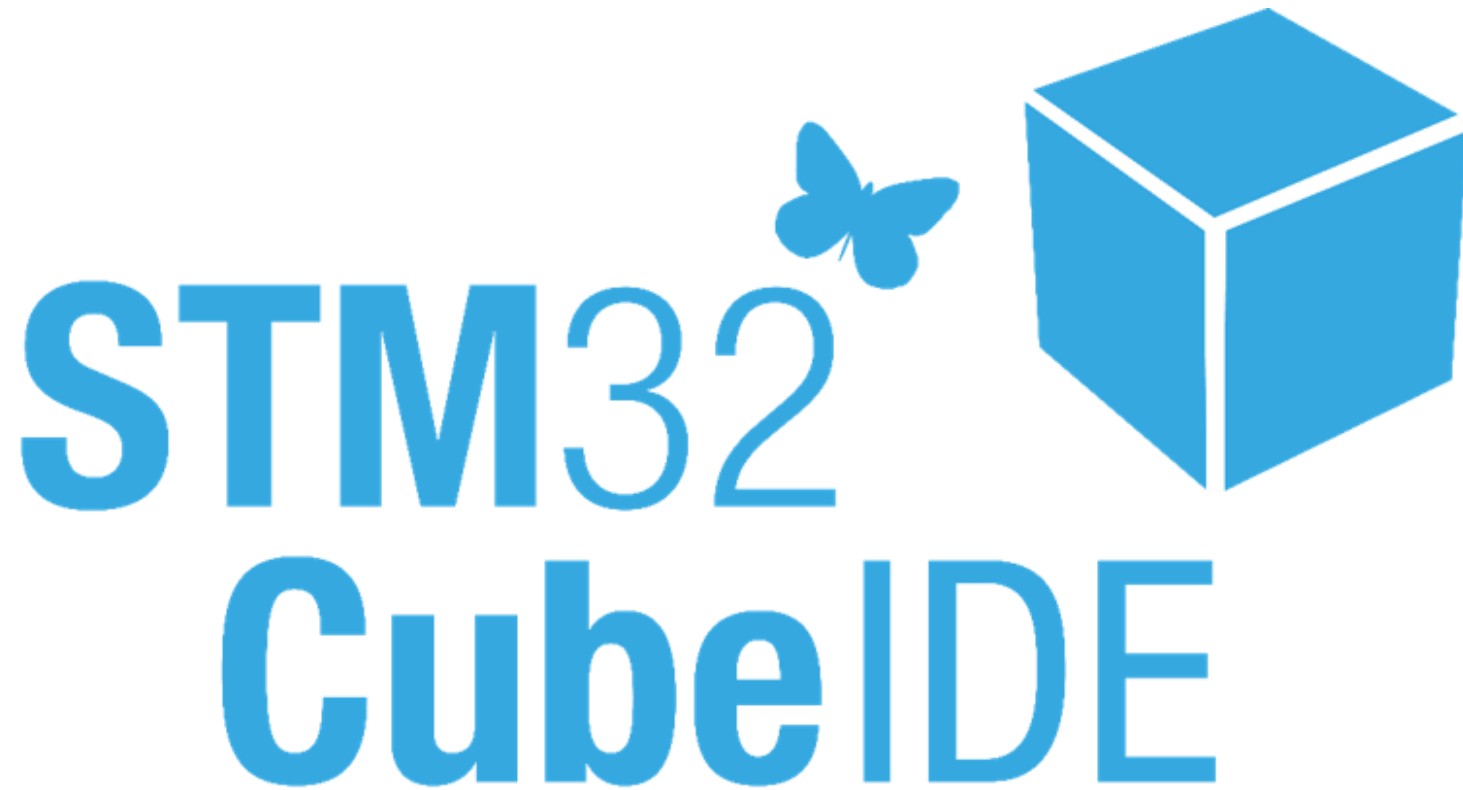




Instalación de entorno de programación



*Production License for any STM32



Descarga
de STM32
Cube IDE

STM32CubeIDE - Integrated De x +

st.com/en/development-tools/stm32cubeide.html

ST

Products ▾

Development tools > Software development tools > STM32 Software Development Tools > STM32 IDEs > STM32CubeIDE >

STM32CubeIDE ACTIVE

Integrated Development Environment for ST

 Download databrief

Overview Documentation Tools & Software

Product overview

Description All features **Get Software** Featured Videos Recommended for you

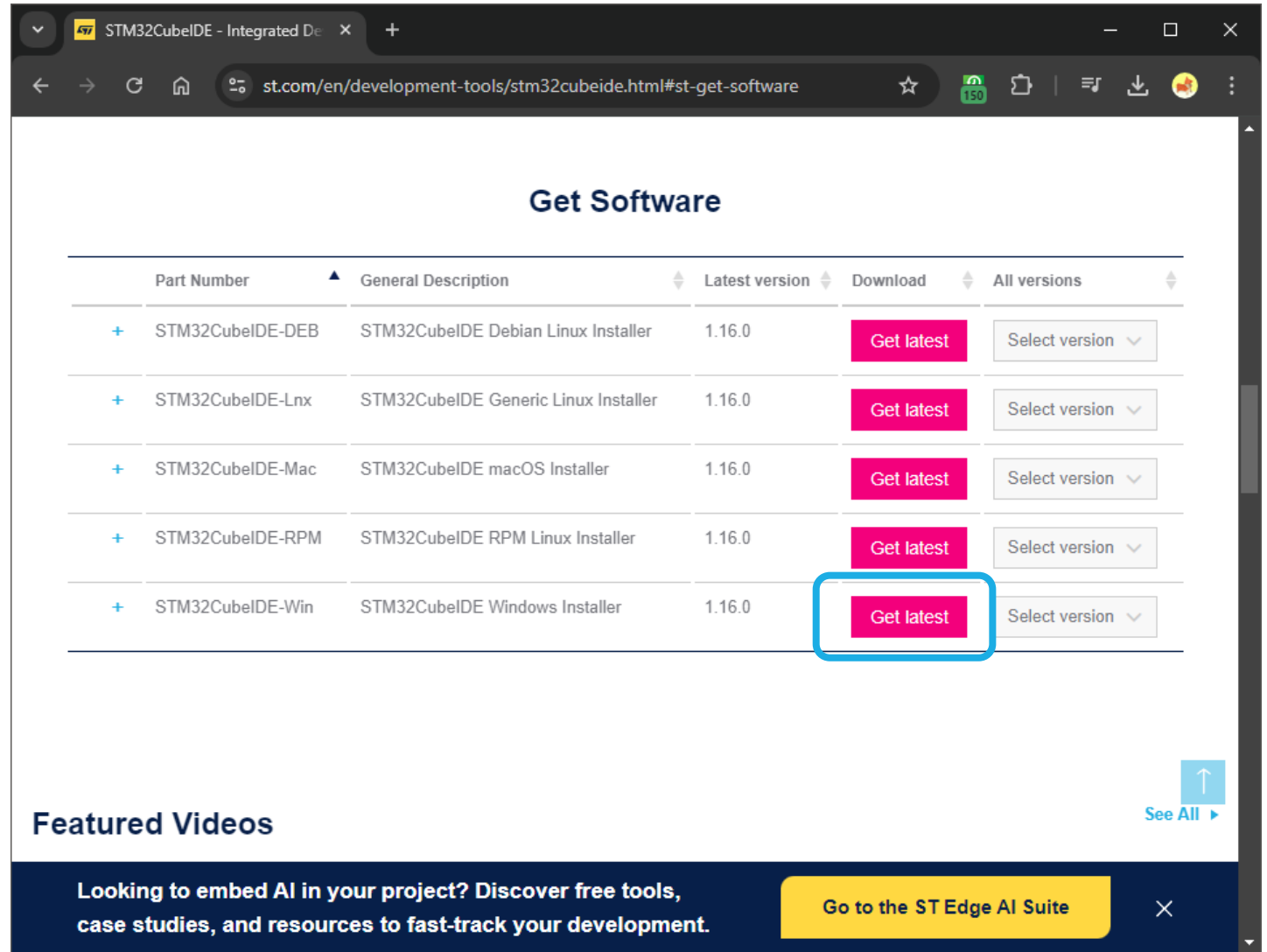
Looking to embed AI in your project? Discover free tools, case studies, and resources to fast-track your development.

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Ingresa al siguiente enlace:

<https://www.st.com/en/development-tools/stm32cubeide.html>

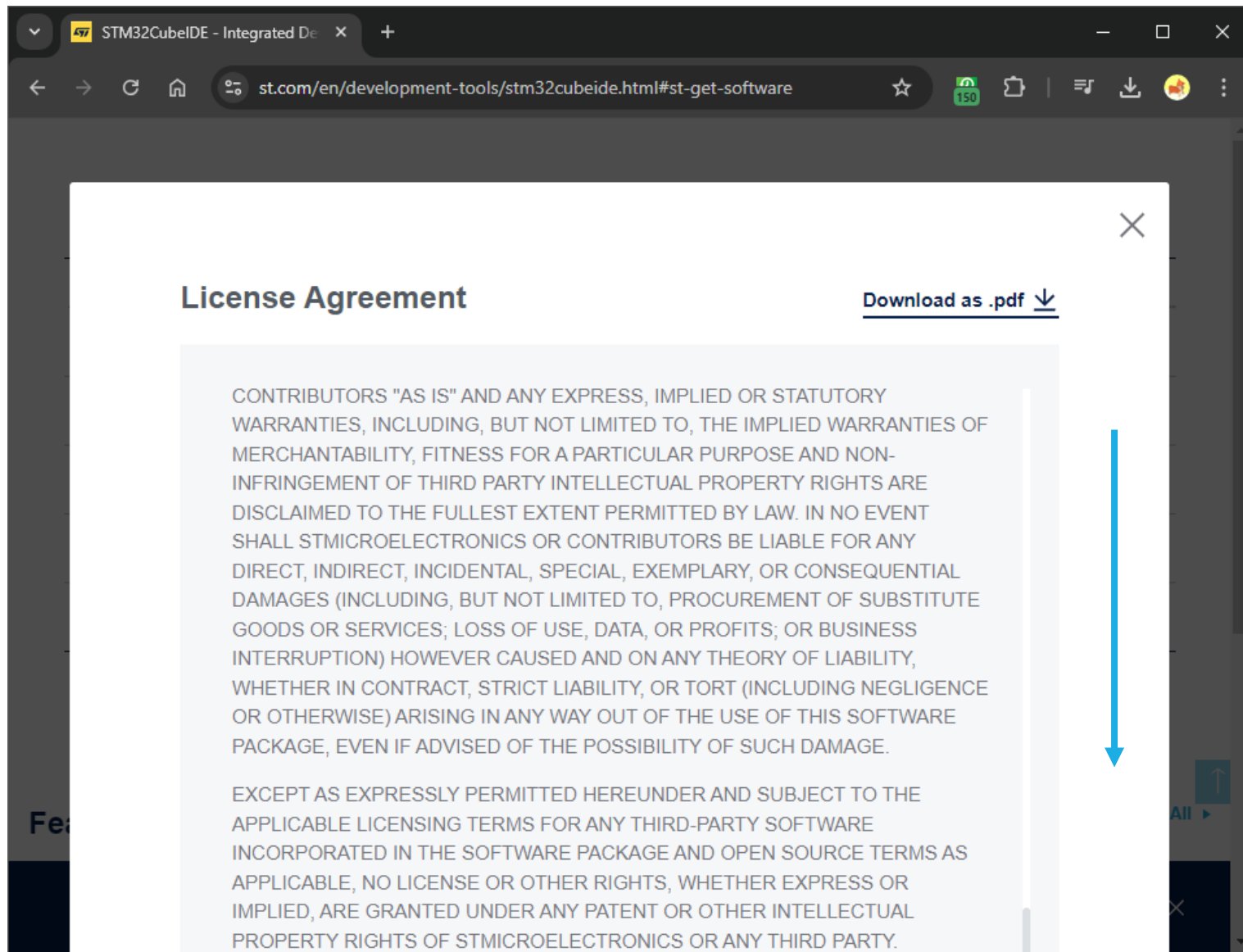
Seleccione la
versión
correspondiente a
su sistema
operativo



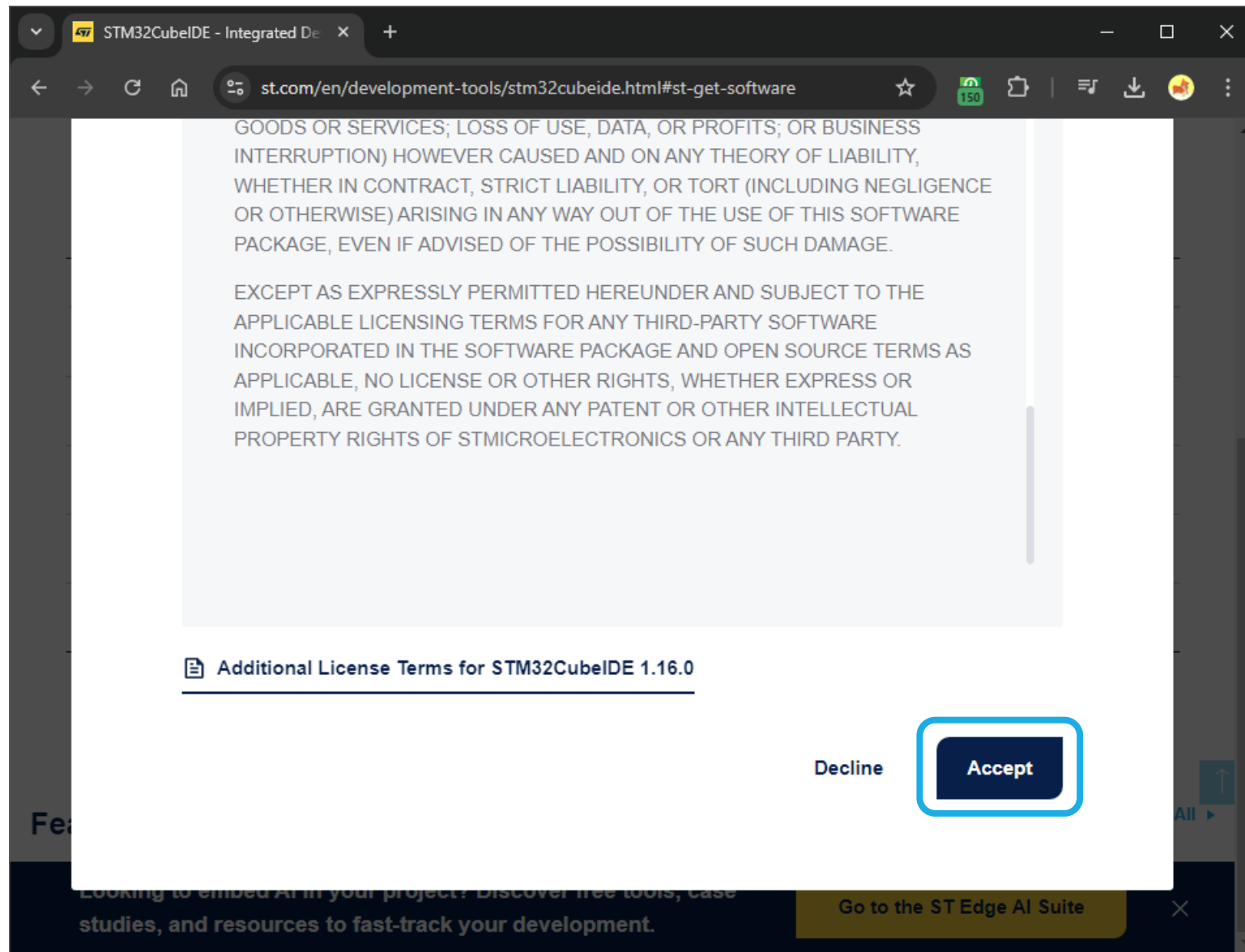
The screenshot shows the 'Get Software' page for STM32CubeIDE. The browser address bar indicates the URL: `st.com/en/development-tools/stm32cubeide.html#st-get-software`. The page features a table with the following columns: Part Number, General Description, Latest version, Download, and All versions.

Part Number	General Description	Latest version	Download	All versions
STM32CubeIDE-DEB	STM32CubeIDE Debian Linux Installer	1.16.0	Get latest	Select version
STM32CubeIDE-Lnx	STM32CubeIDE Generic Linux Installer	1.16.0	Get latest	Select version
STM32CubeIDE-Mac	STM32CubeIDE macOS Installer	1.16.0	Get latest	Select version
STM32CubeIDE-RPM	STM32CubeIDE RPM Linux Installer	1.16.0	Get latest	Select version
STM32CubeIDE-Win	STM32CubeIDE Windows Installer	1.16.0	Get latest	Select version

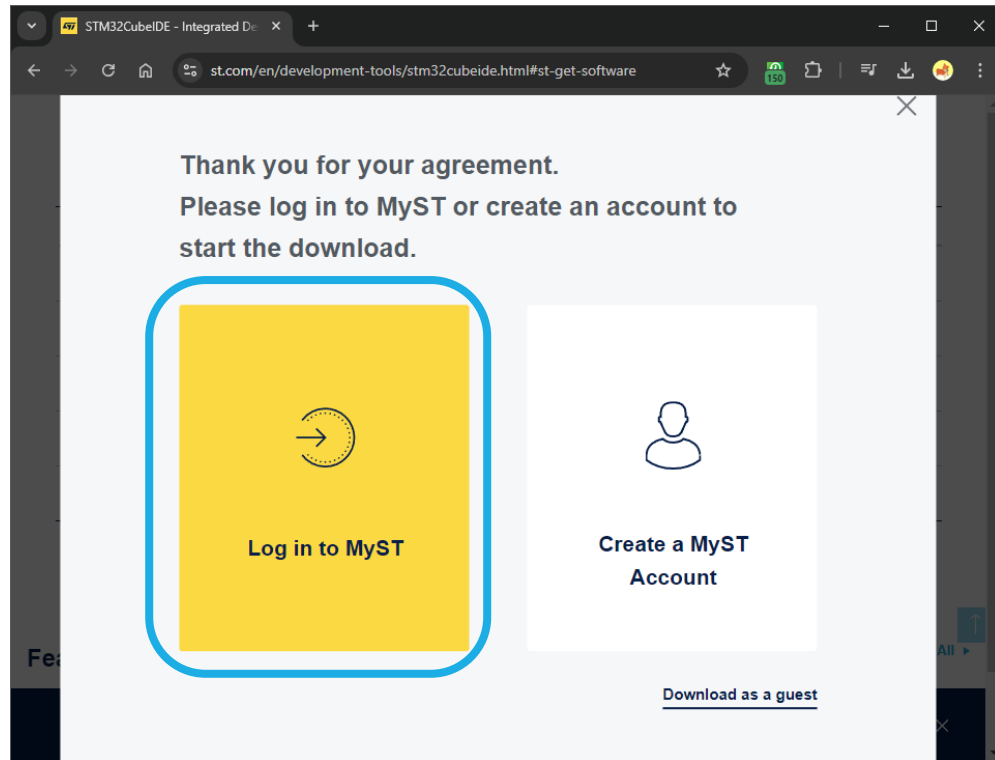
The 'Get latest' button for the Windows installer is highlighted with a red box. Below the table, there is a 'Featured Videos' section with a 'See All' link. At the bottom, a dark blue banner contains the text: 'Looking to embed AI in your project? Discover free tools, case studies, and resources to fast-track your development.' and a yellow button labeled 'Go to the ST Edge AI Suite'.



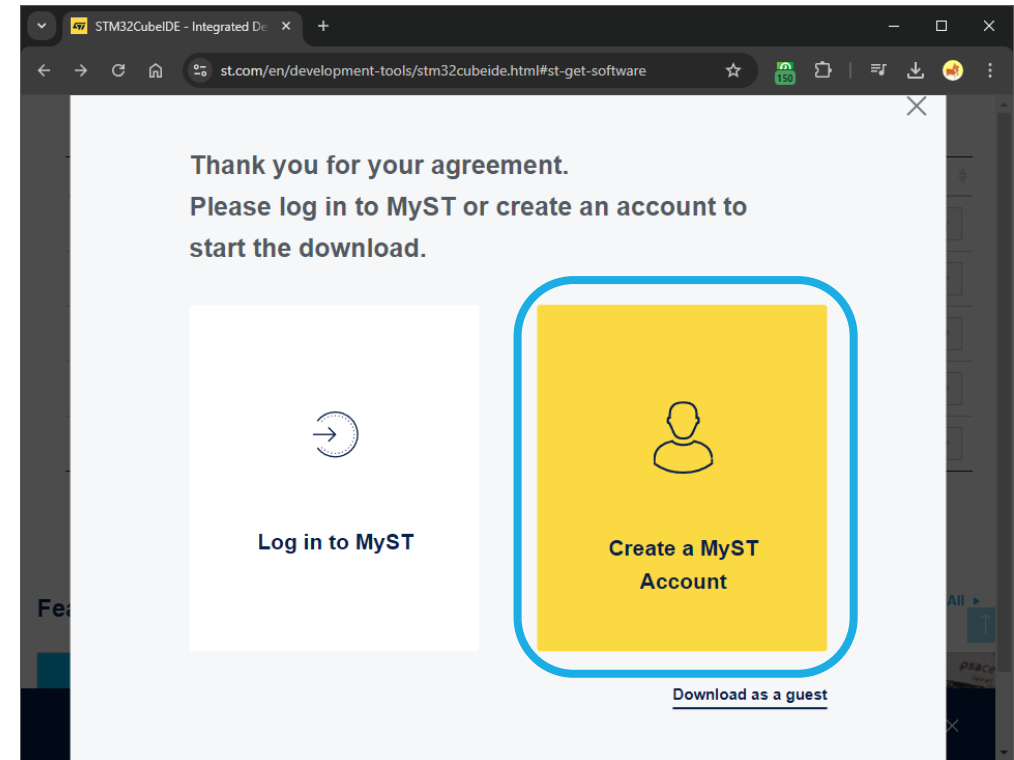
Nos aparecerá la siguiente ventana



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Nos pedirá ingresar sesión



Si no tenemos usuario
podemos crear una cuenta

Creando una cuenta en ST

Llenamos los datos que se nos solicitan

my.st.com account creation

Your Profile

Salutation:	Select one
First name*:	
Last name*:	
Email*:	
Email Confirmation*:	

Function*:	Select one
Company/University*:	
Industry*:	Select one
Country/Region*:	Select one
State/Province:	Select one
Zip/Postcode*:	
Phone Number:	

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E-mail address

pdmazariegos@uvg.edu.gt

Password

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☐ Remember me on this computer. ⓘ

Login

New user?

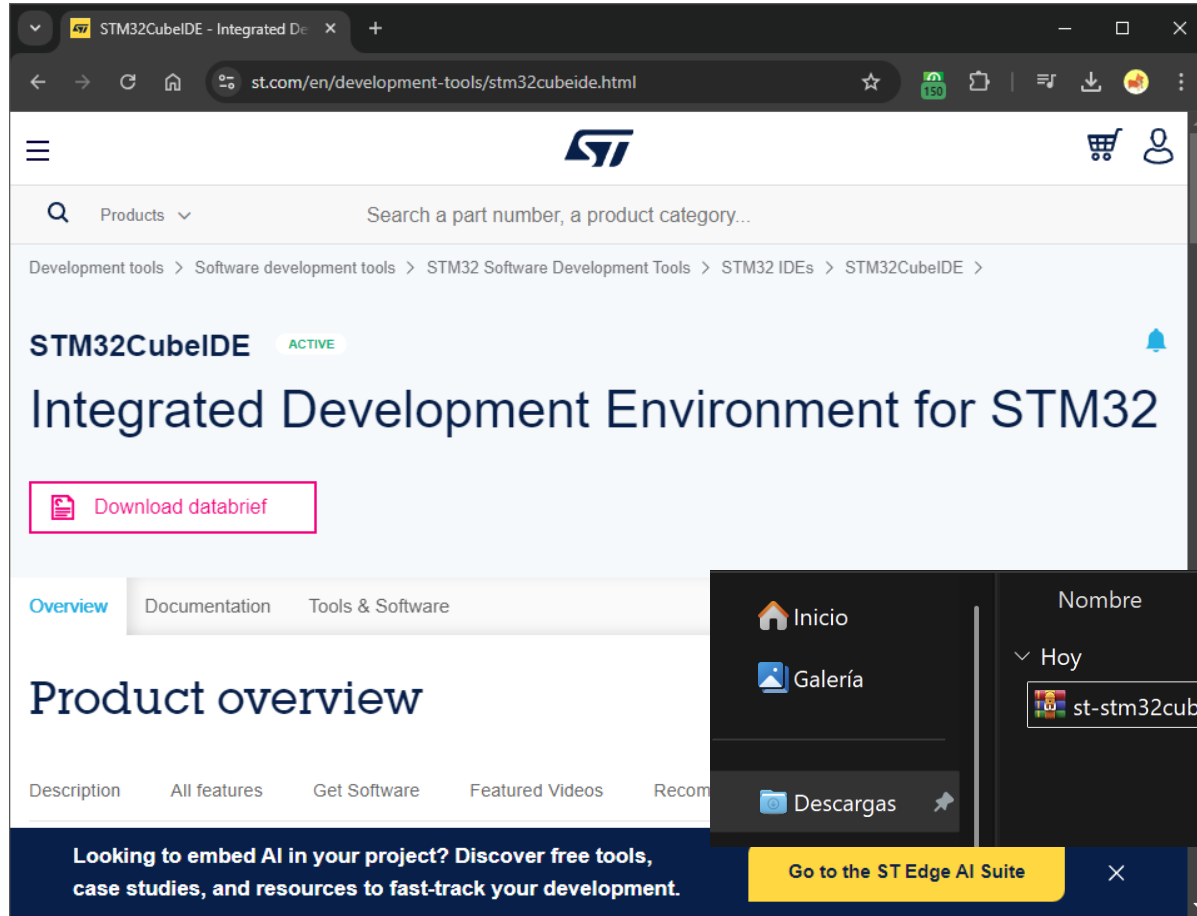
myST brings you a set of personalized features:

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Feedback

Iniciará la descarga



STM32CubeIDE - Integrated Development Environment for STM32

Download databrief

Product overview

Looking to embed AI in your project? Discover free tools, case studies, and resources to fast-track your development.

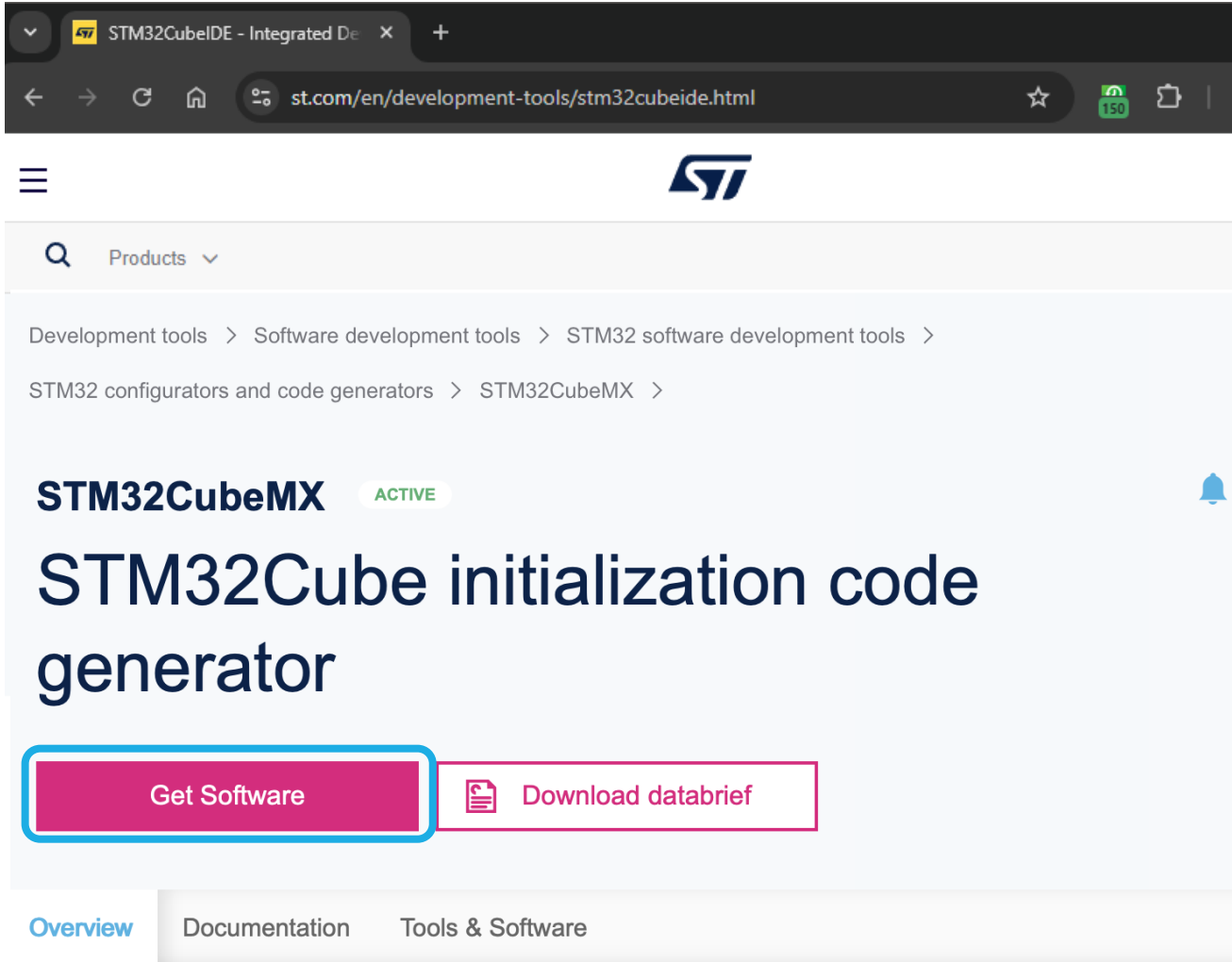
Go to the ST Edge AI Suite

Esperamos a que se descargue el siguiente zip

Nombre	Fecha de modificación	Tipo	Tamaño
Hoy			
st-stm32cubeide_2.0.0_26820_20251114_1348_...	17/02/2026 22:09	Archivo WinRAR ZIP	813,541 KB



Descarga
instalador
de STM32
Cube MX



Descargamos ahora el
STM32CubeMX
Ingresando al siguiente
enlace:

<https://www.st.com/en/development-tools/stm32cubemx.html>

Product overview

Seleccione la
versión
correspondiente a
su sistema
operativo

STM32CubeIDE - Integrated De x +

st.com/en/development-tools/stm32cubeide.html#st-get-software

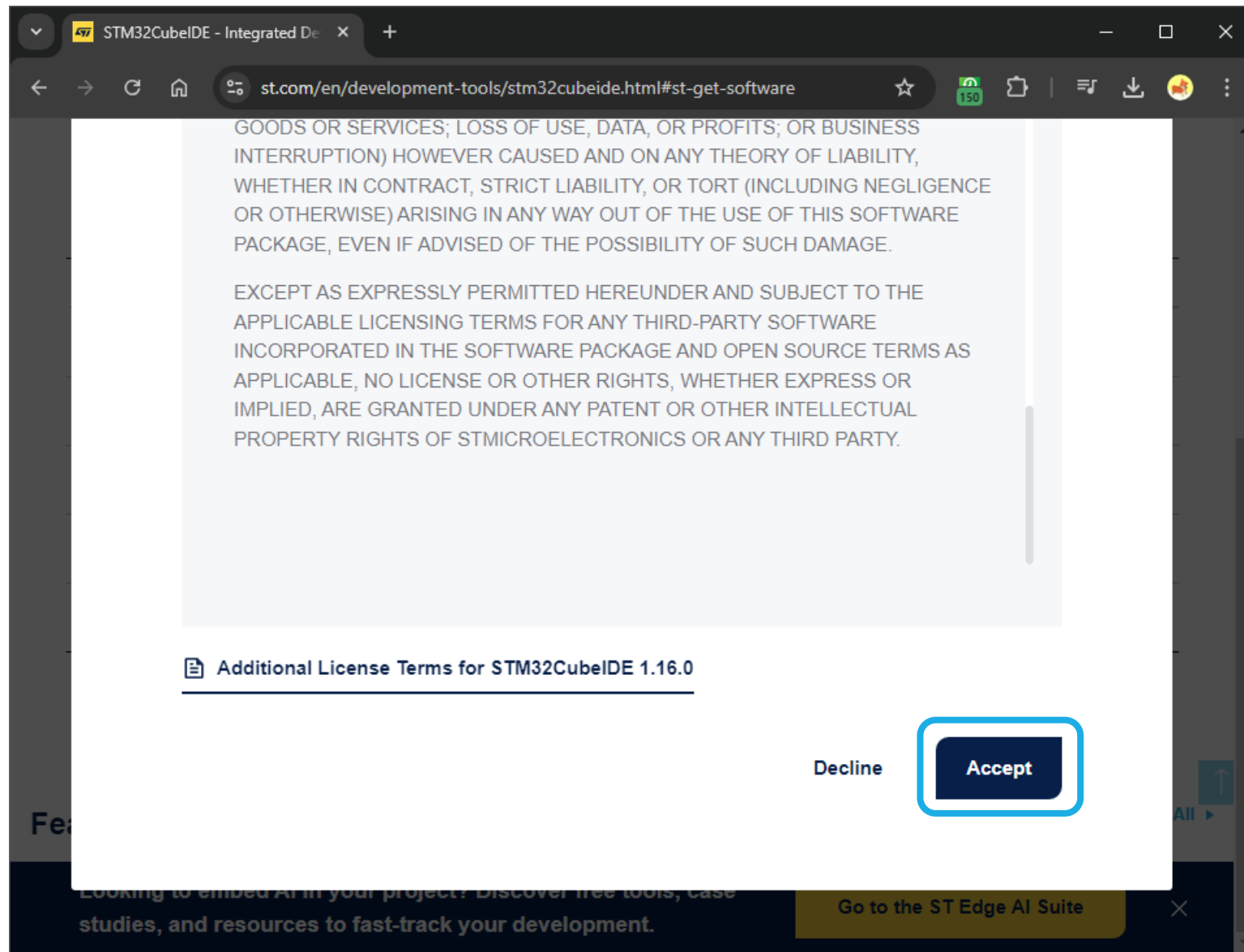
Get Software

Part Number	Download	All versions
+ STM32CubeMX-Lin	Get latest	Select version ▾
+ STM32CubeMX-Mac	Get latest	Select version ▾
+ STM32CubeMX-MacA	Get latest	Select version ▾
+ STM32CubeMX-Win	Get latest	Select version ▾

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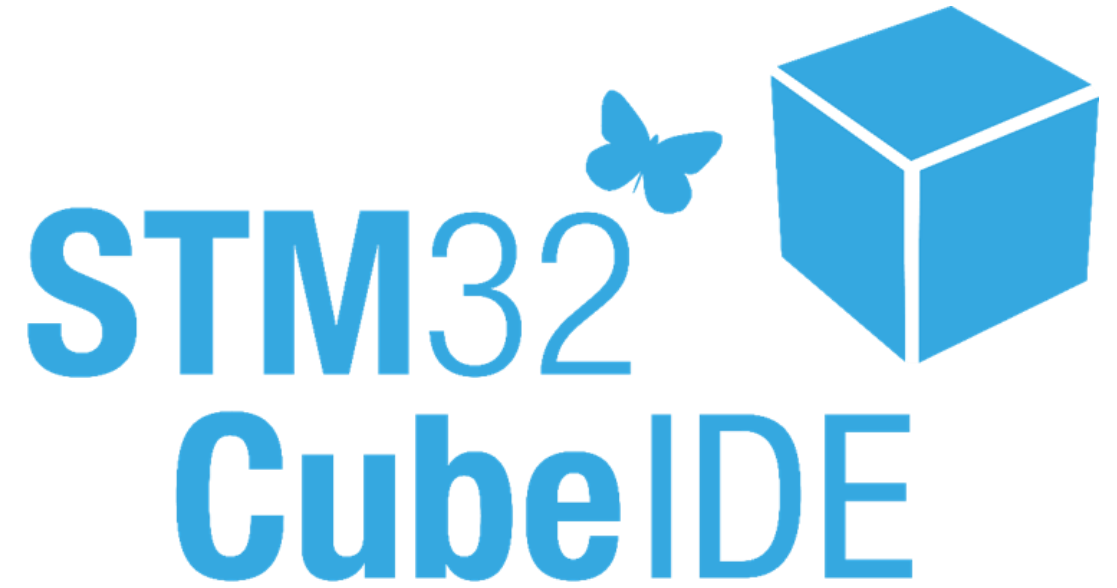
Nos volverá aparecer la ventana de aceptar términos,
Aceptamos los términos

Iniciará la descarga

The screenshot shows the STM32CubeMX website. The browser address bar displays 'st.com/en/development-tools/stm32cubeide.html'. The page features the STM logo, a search bar, and a breadcrumb trail: 'Development tools > Software development tools > STM32 software development tools > STM32 configurators and code generators > STM32CubeMX >'. The main heading reads 'STM32CubeMX' with a green 'ACTIVE' tag, followed by 'STM32Cube inicializa generator'. At the bottom, there are two buttons: 'Get Software' and 'Download databrief'.

Esperamos a que se descargue el siguiente zip

	Nombre	Fecha de modificación	Tipo	Tamaño
	Hoy			
	stm32cubemx-win-v6-16-1	17/02/2026 22:12	Archivo WinRAR ZIP	656,213 KB
	st-stm32cubeide_2.0.0_20020_20251114_1348_...	17/02/2026 22:09	Archivo WinRAR ZIP	813,541 KB



Instalación

Descomprimir ambos zips

 Inicio	Nombre	Fecha de modificación	Tipo	Tamaño
	▼ Hoy			
	 stm32cubemx-win-v6-16-1	17/02/2026 22:12	Archivo WinRAR ZIP	656,213 KB
	 st-stm32cubeide_2.0.0_26820_20251114_1348_...	17/02/2026 22:09	Archivo WinRAR ZIP	813,541 KB

 Inicio	Nombre	Fecha de modificación	Tipo	Tamaño
	▼ Hoy			
	 stm32cubemx-win-v6-16-1	17/02/2026 22:12	Archivo WinRAR ZIP	656,213 KB
	 st-stm32cubeide_2.0.0_26820_20251114_1348_...	17/02/2026 22:09	Archivo WinRAR ZIP	813,541 KB
	▼ Hace mucho tiempo			

 Galería	Nombre	Fecha de modificación	Tipo	Tamaño

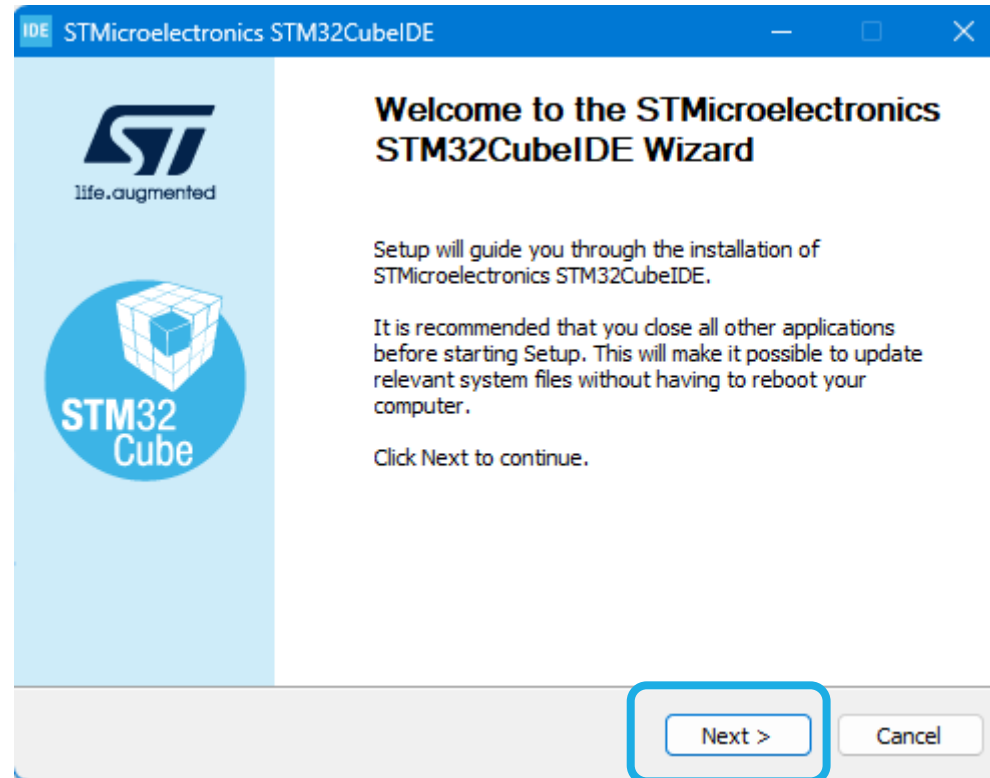
 Descargas	Nombre	Fecha de modificación	Tipo	Tamaño

 Documentos	Nombre	Fecha de modificación	Tipo	Tamaño
	 SetupSTM32CubeMX-6.16.1-Win	10/12/2025 07:13	Aplicación	656,273 KB
	 st-stm32cubeide_2.0.0_26820_20251114_1348_...	17/11/2025 02:50	Aplicación	823,999 KB

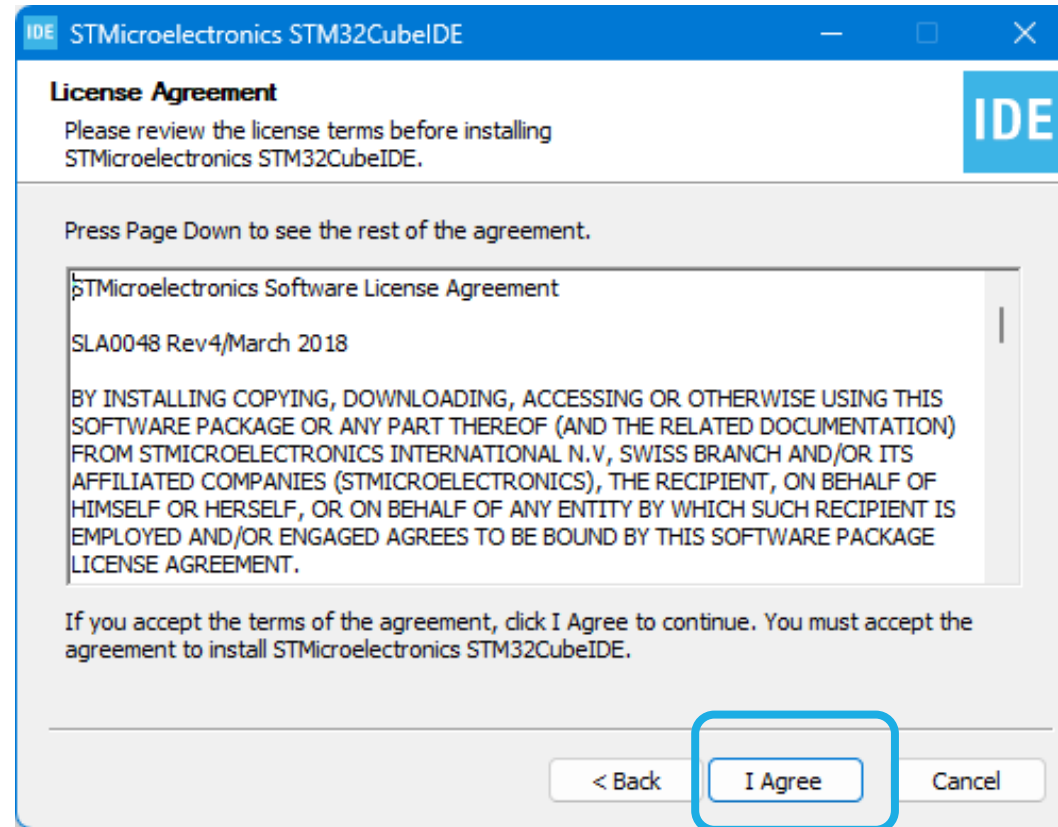
 Imágenes	Nombre	Fecha de modificación	Tipo	Tamaño

 Música	Nombre	Fecha de modificación	Tipo	Tamaño

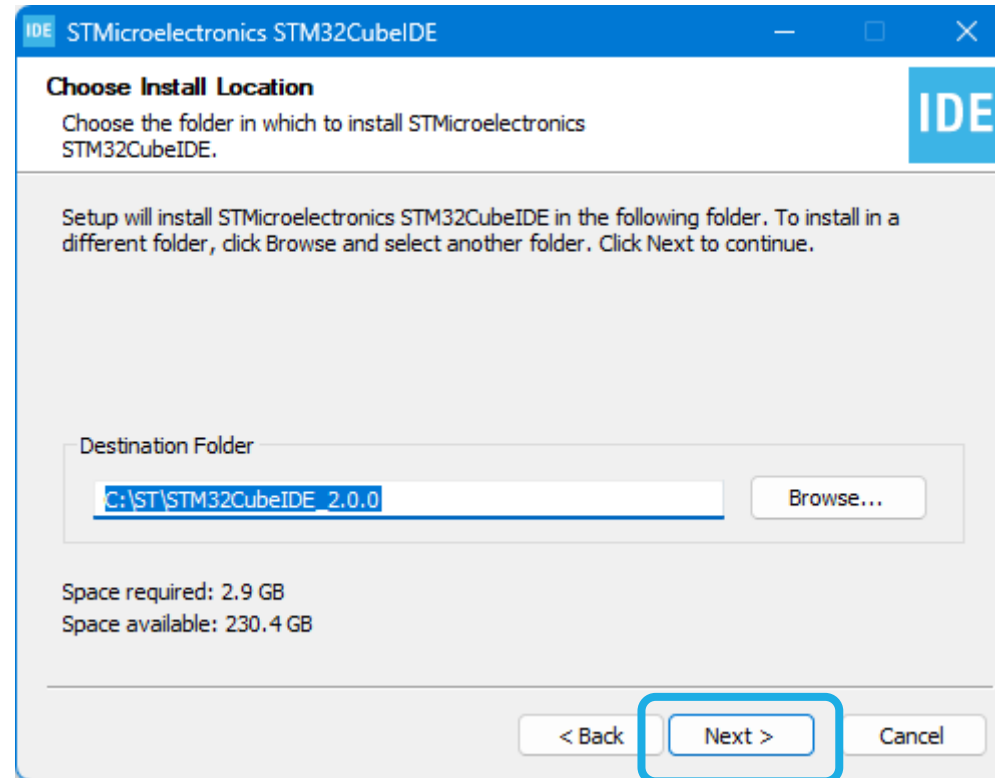
Iniciamos el instalador del STMCubeIDE



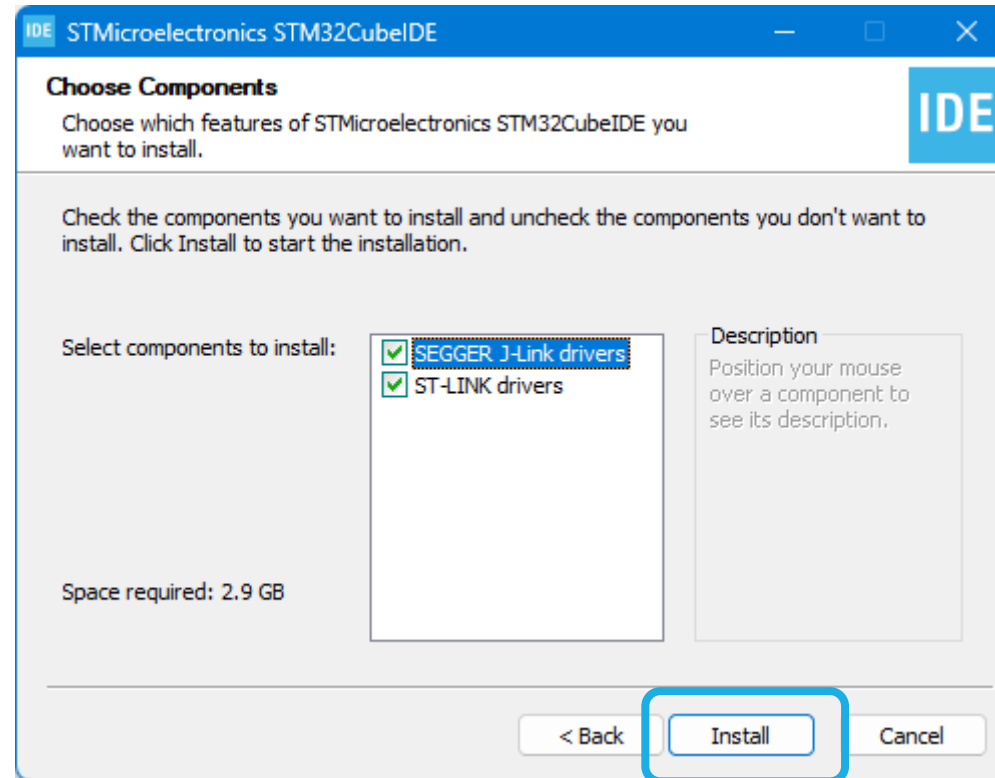
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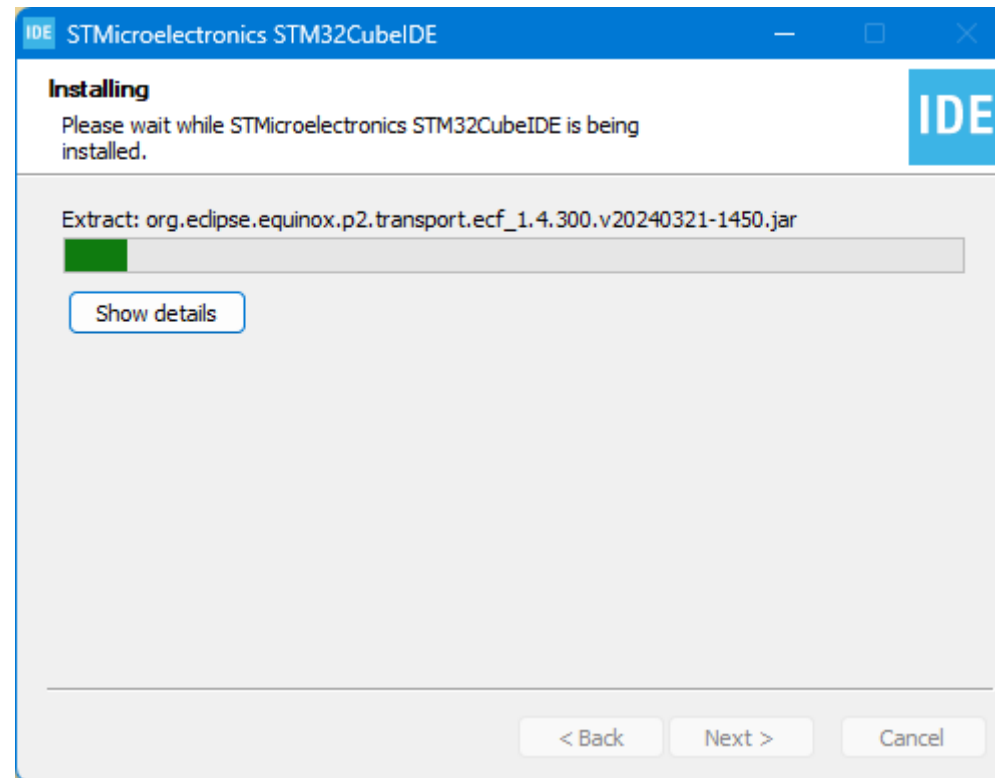
Escogemos la ruta de instalación

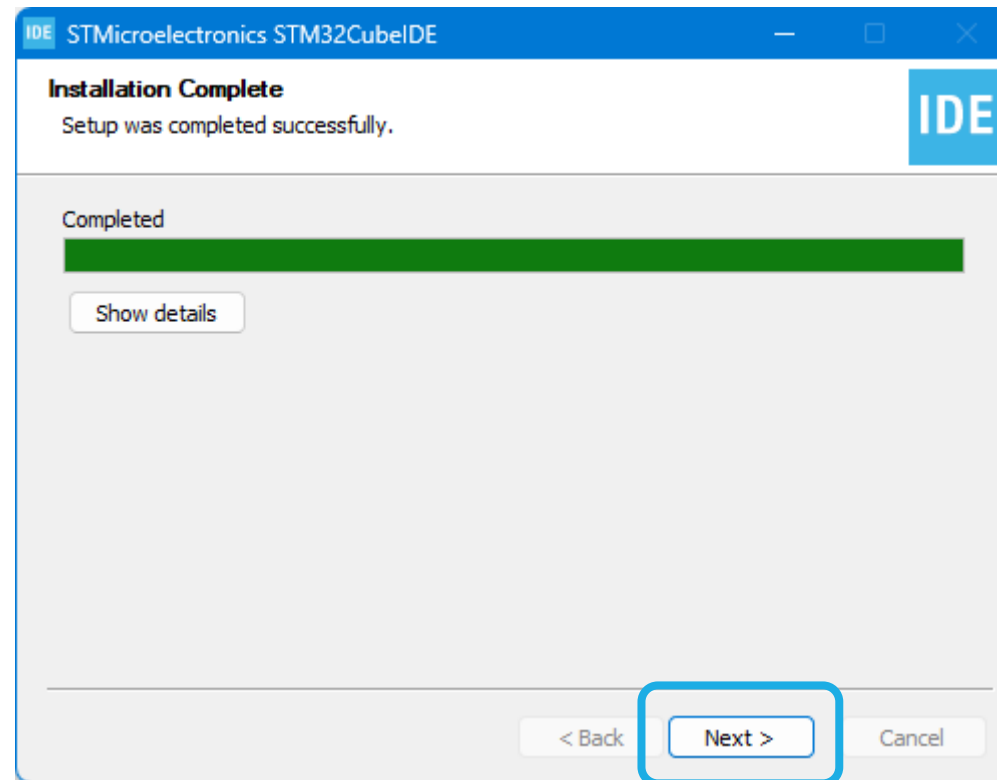


Seleccionamos *Install*



Esperamos a que se termine de instalar



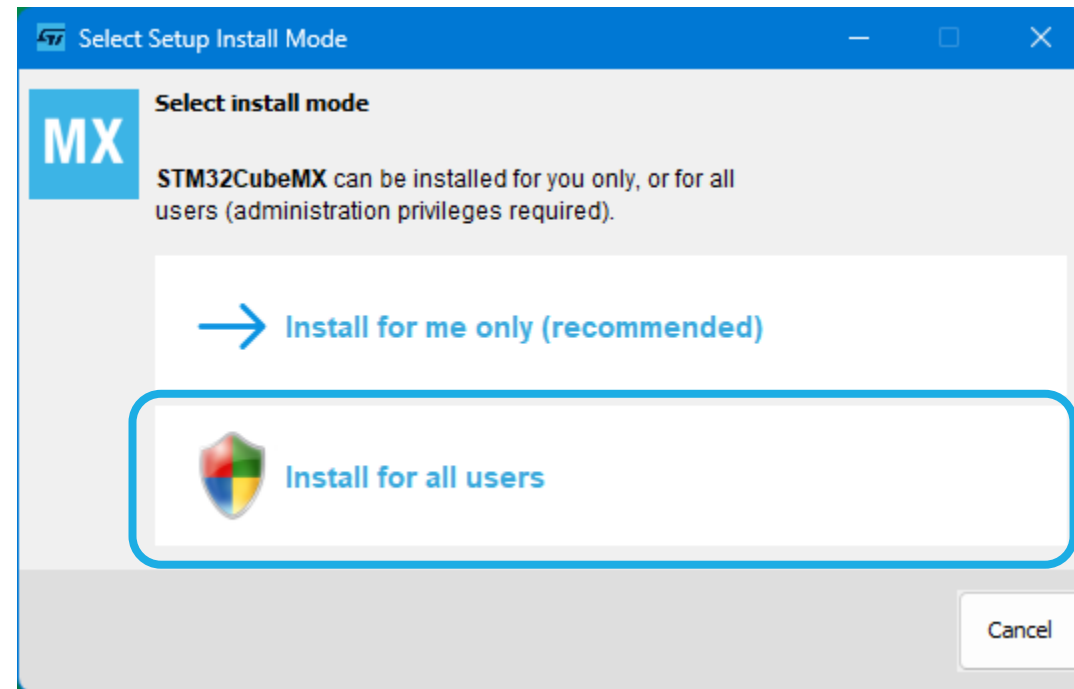
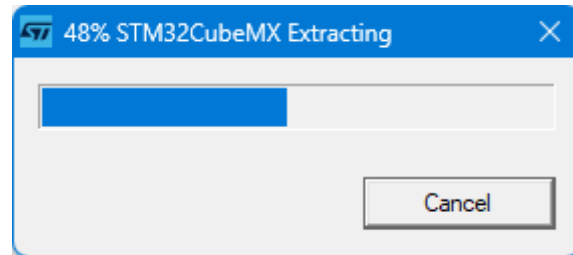


Con esto finalizamos

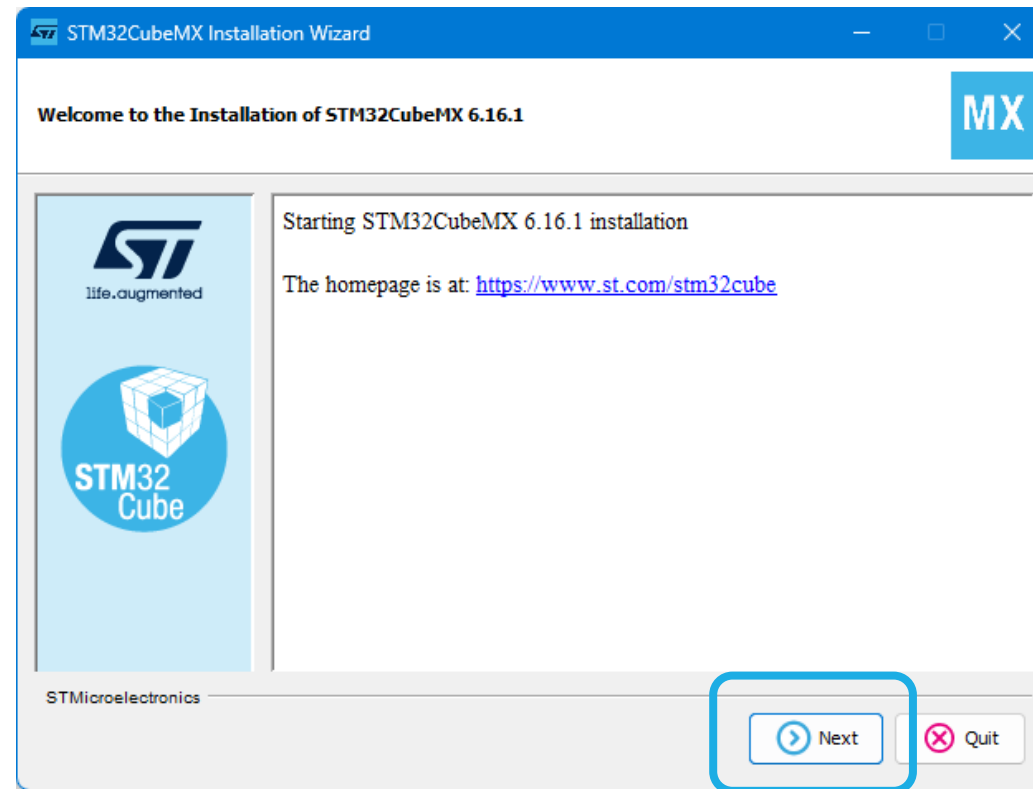


Iniciamos el instalador del STMCubeMX

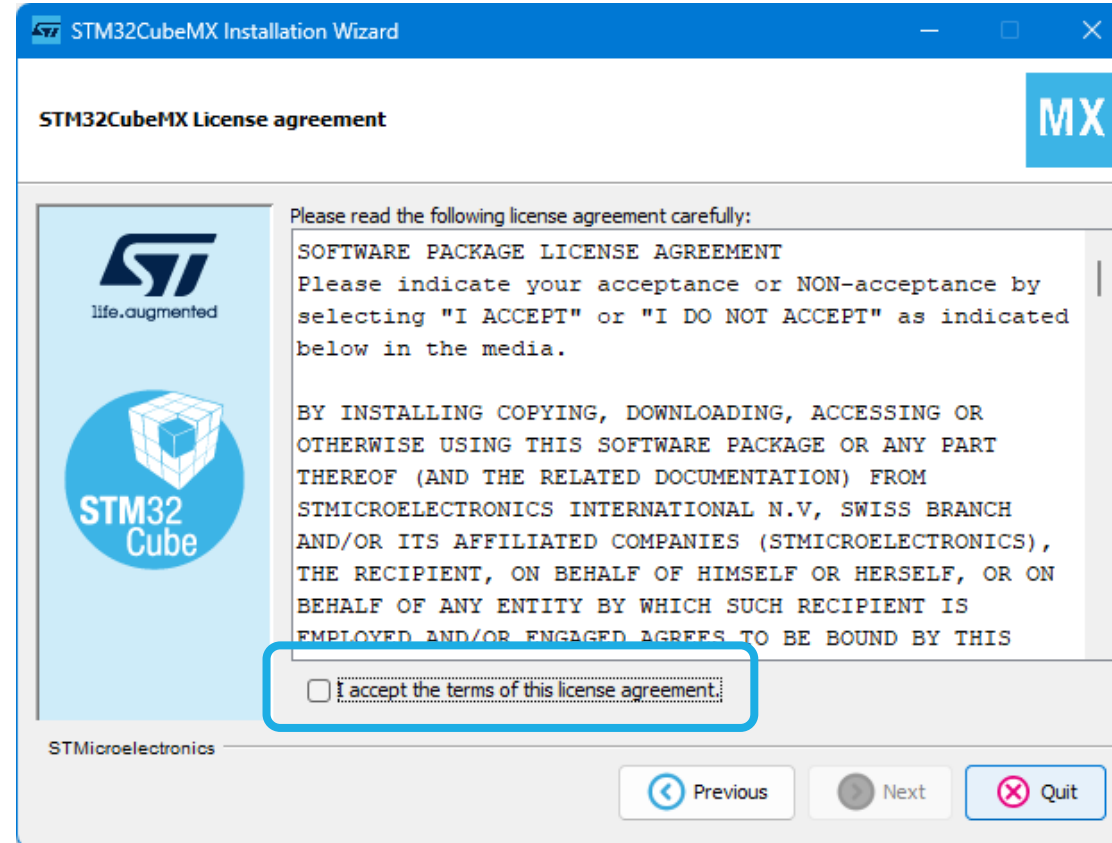
Seleccione la opción para su caso



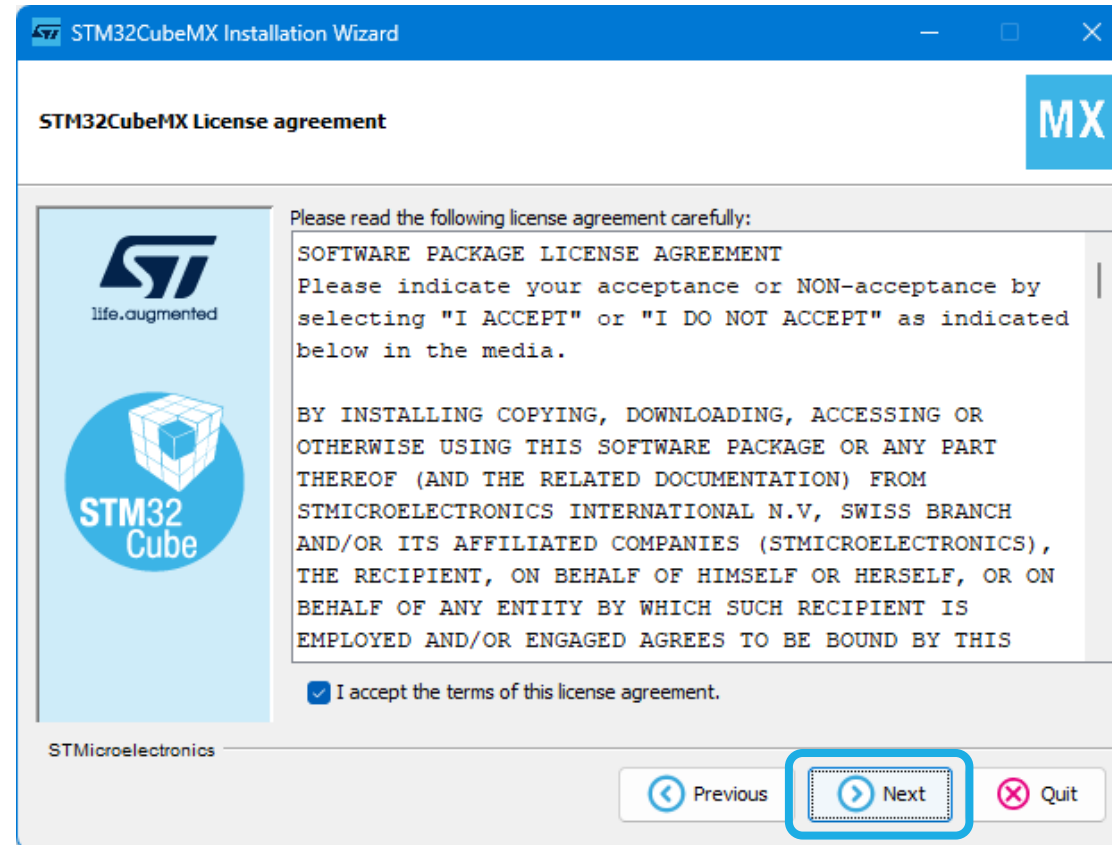
Seleccionamos *Next*



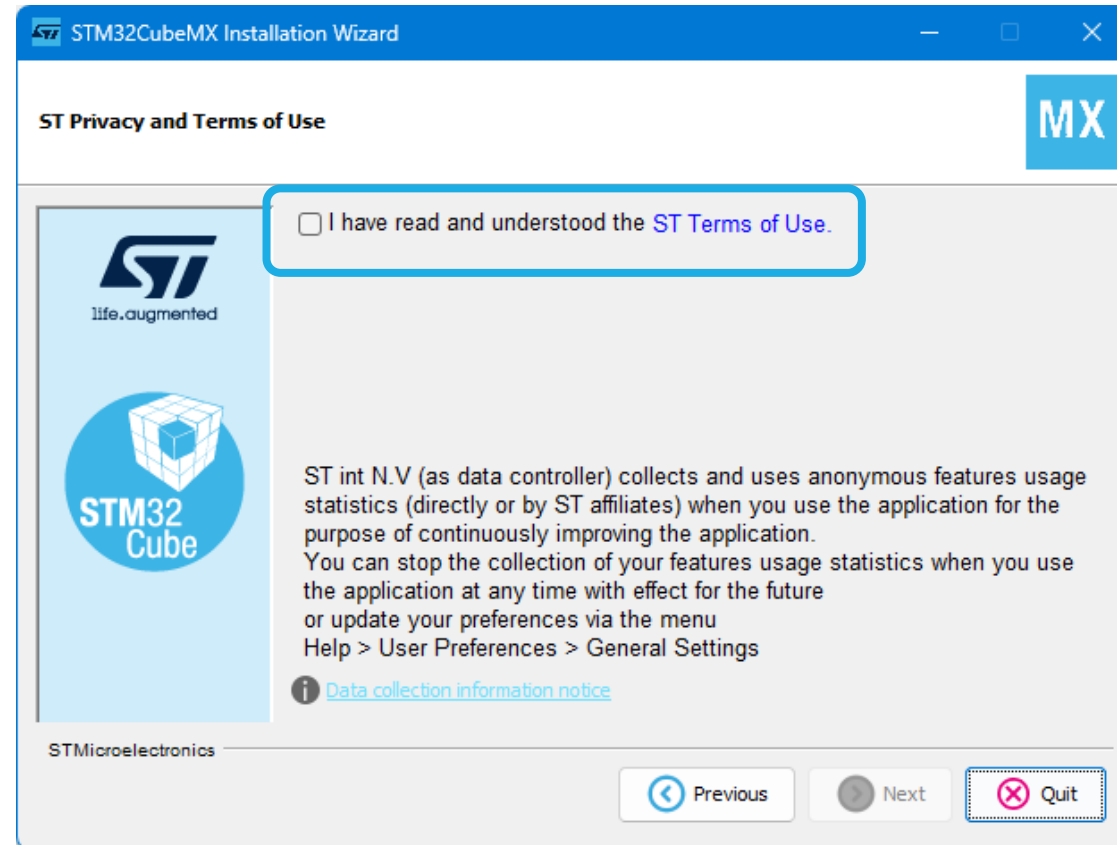
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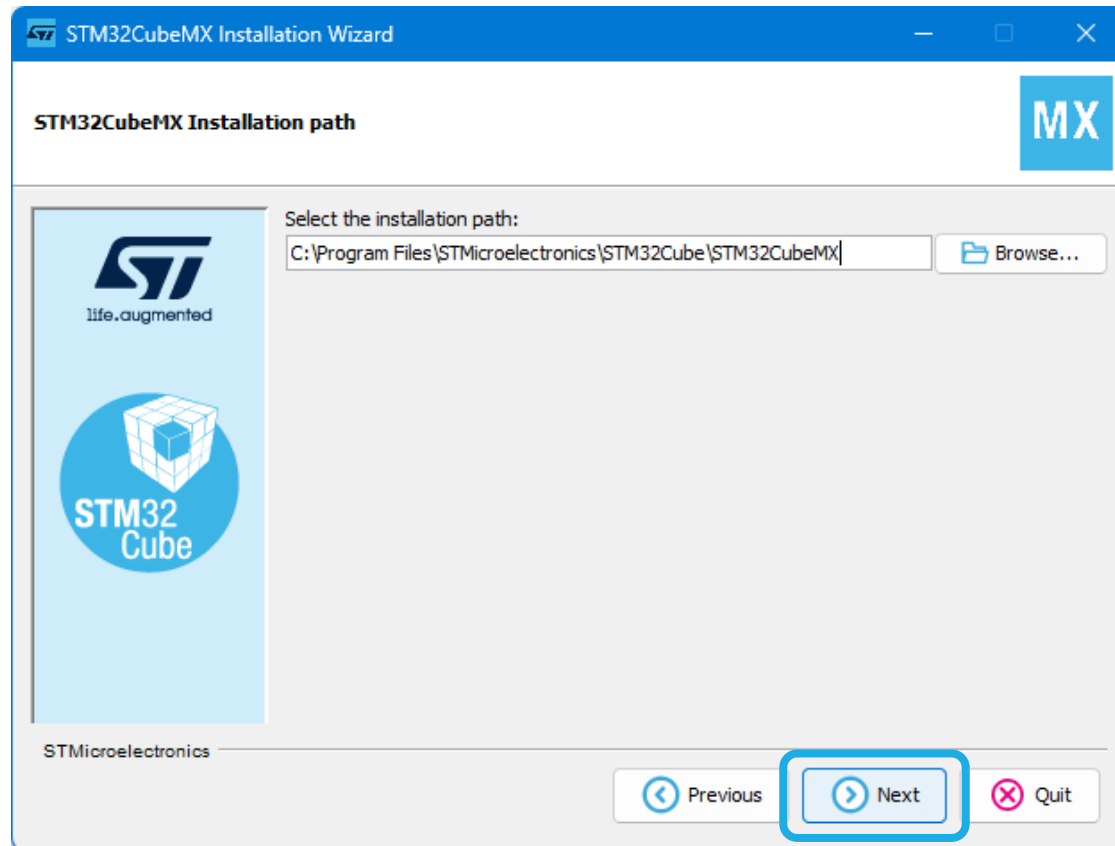
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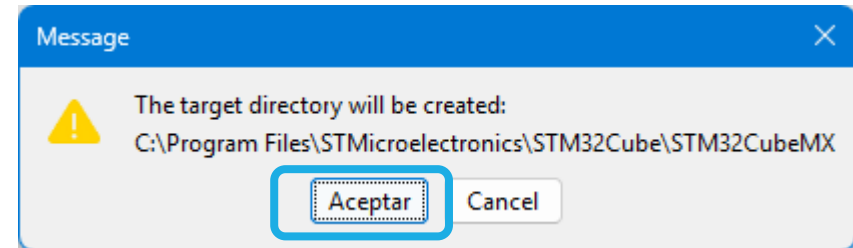
Aceptamos los términos de ST



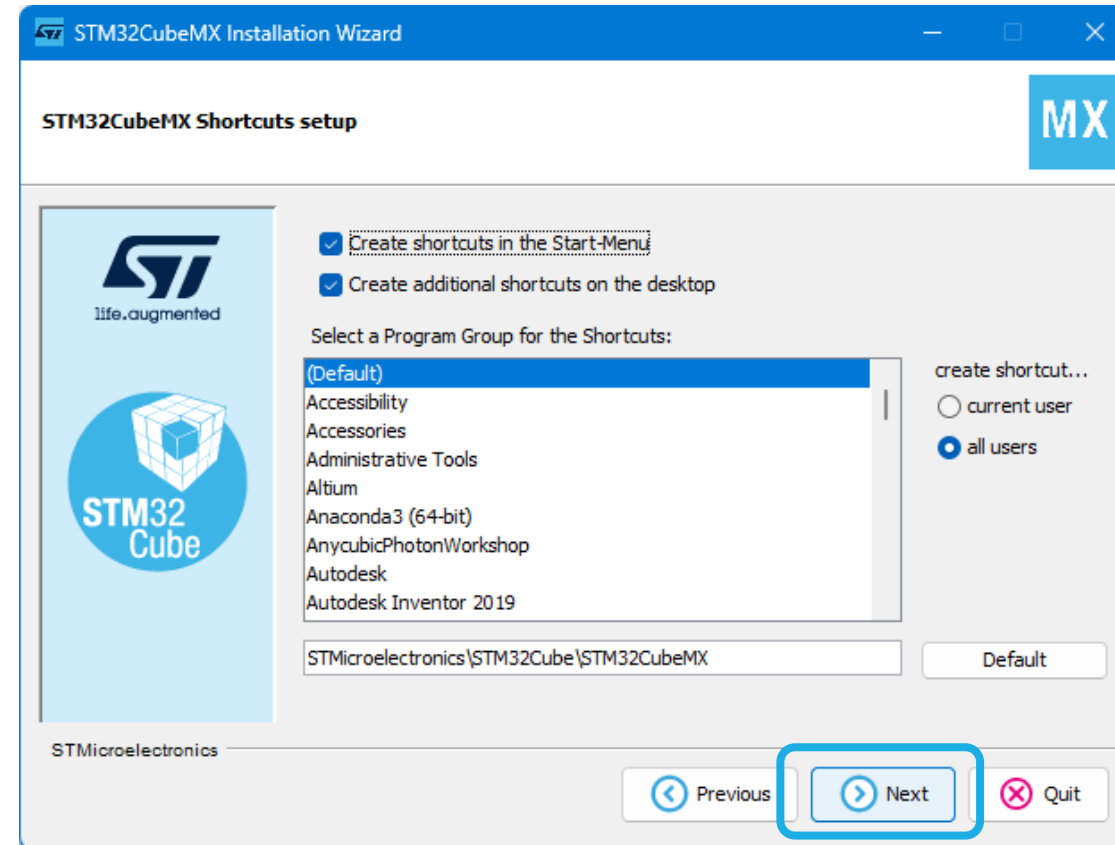
Seleccionamos la ubicación



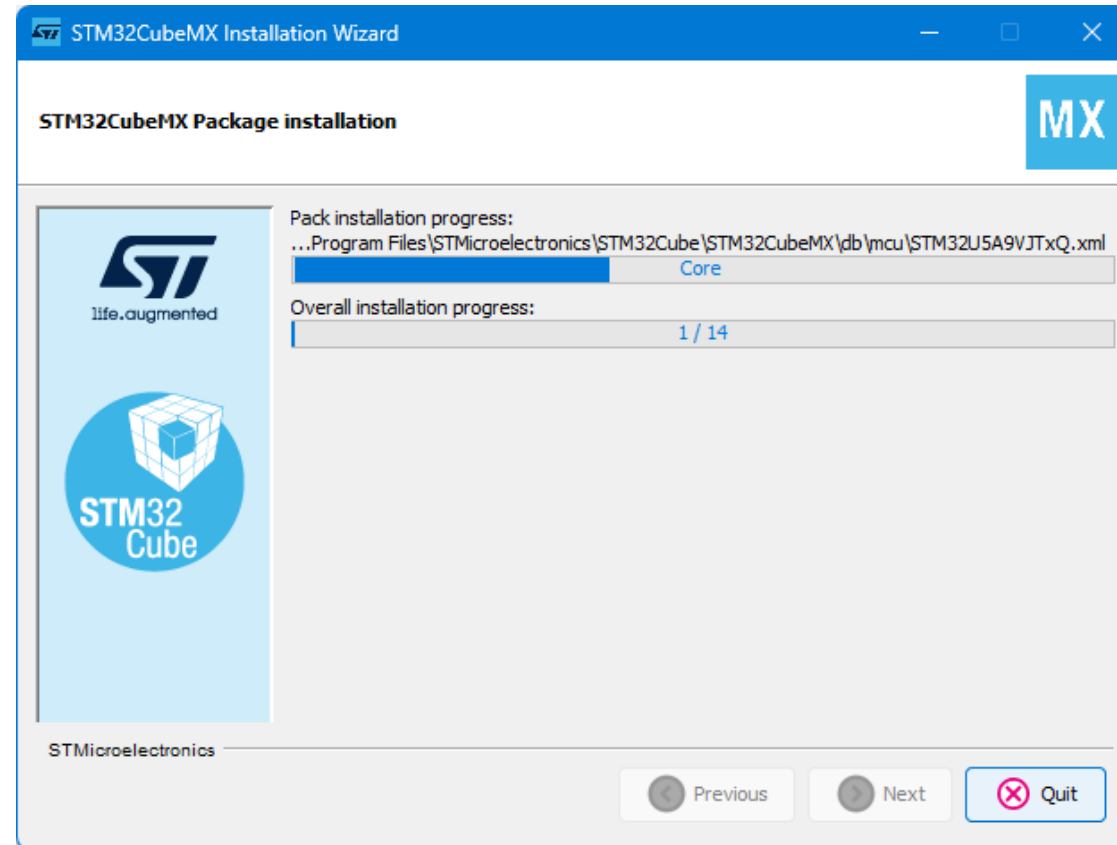
Aparecerá la siguiente ventana, aceptamos



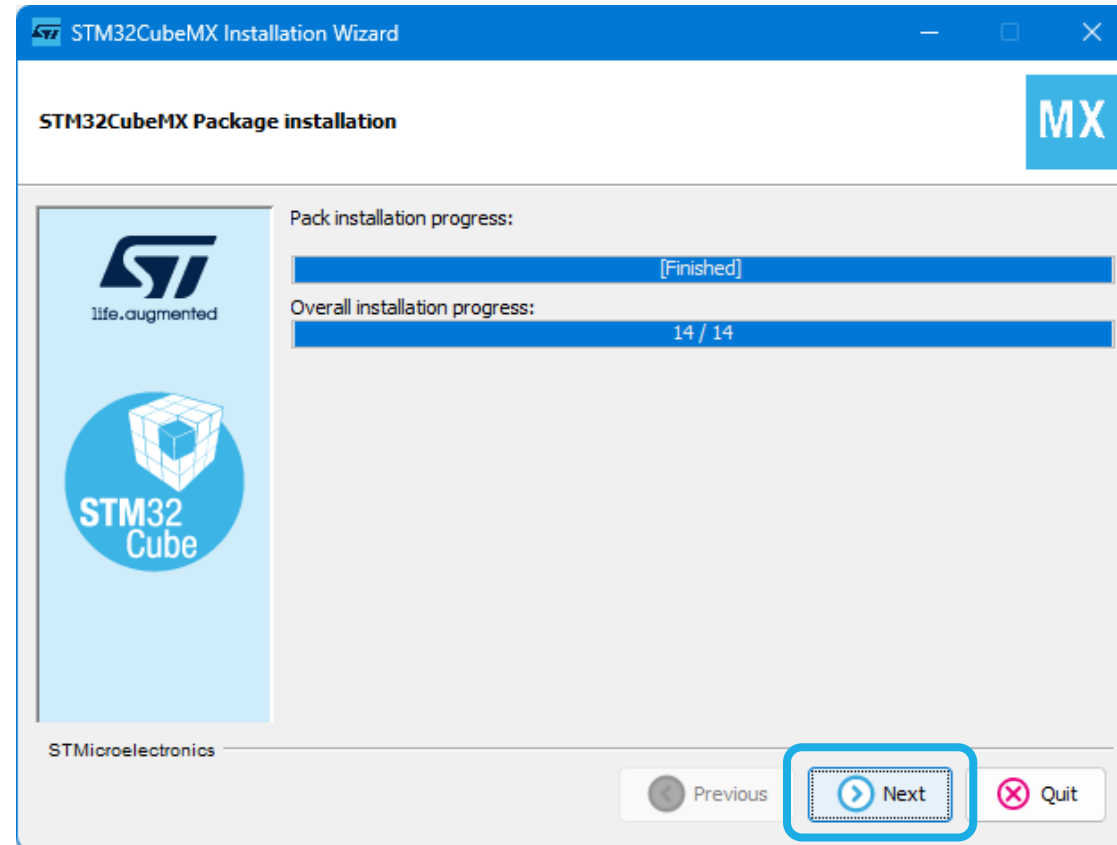
Seleccionamos *Next*



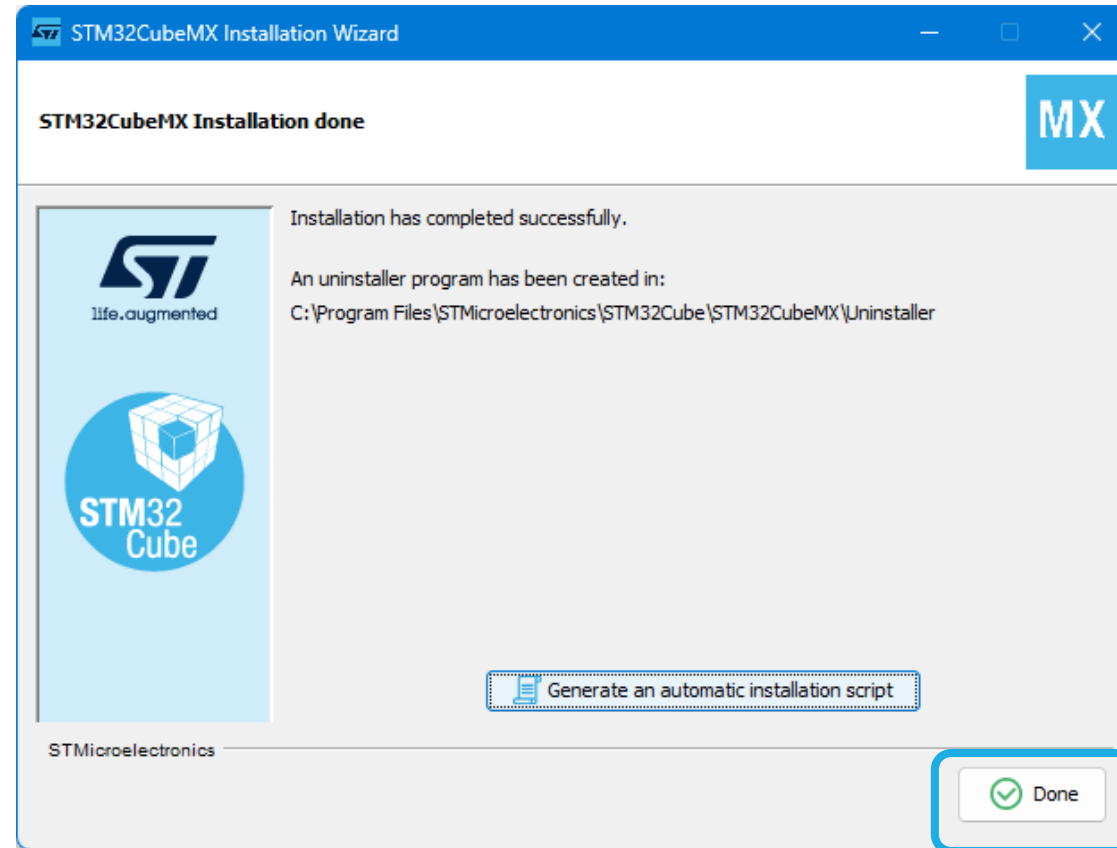
Esperamos a que se termine de instalar



Seleccionamos *Next*



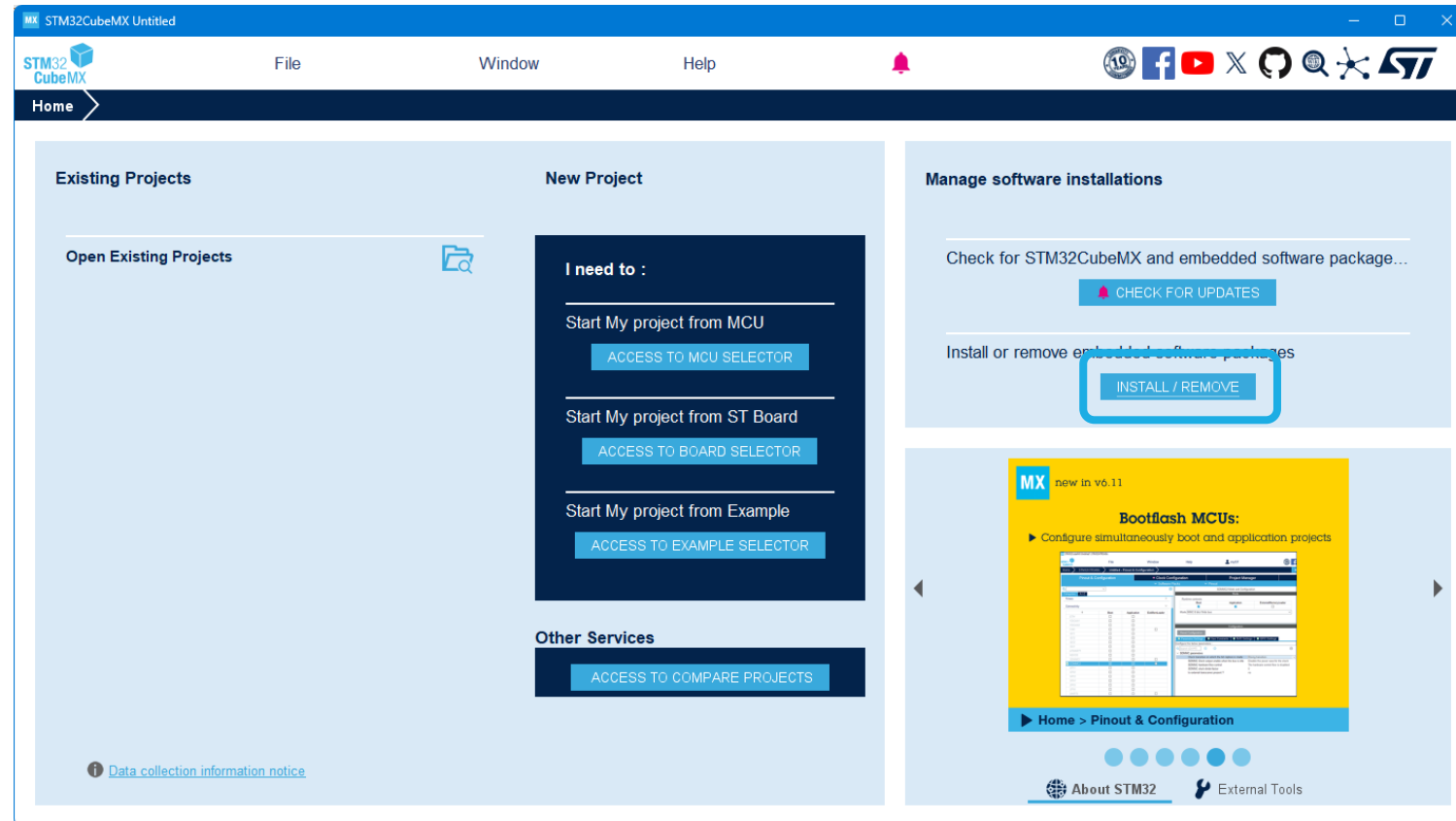
Con esto finalizamos



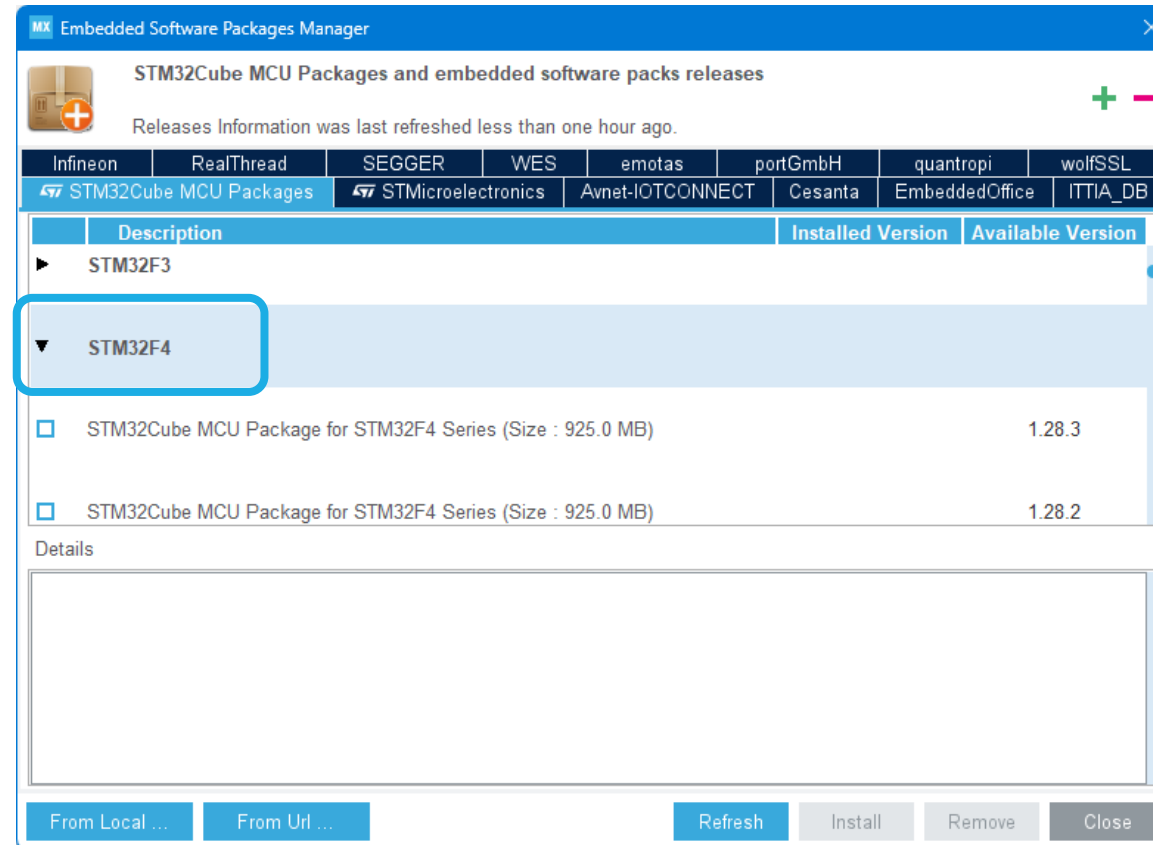
Iniciamos STM32 CubeMX desde el ícono del escritorio



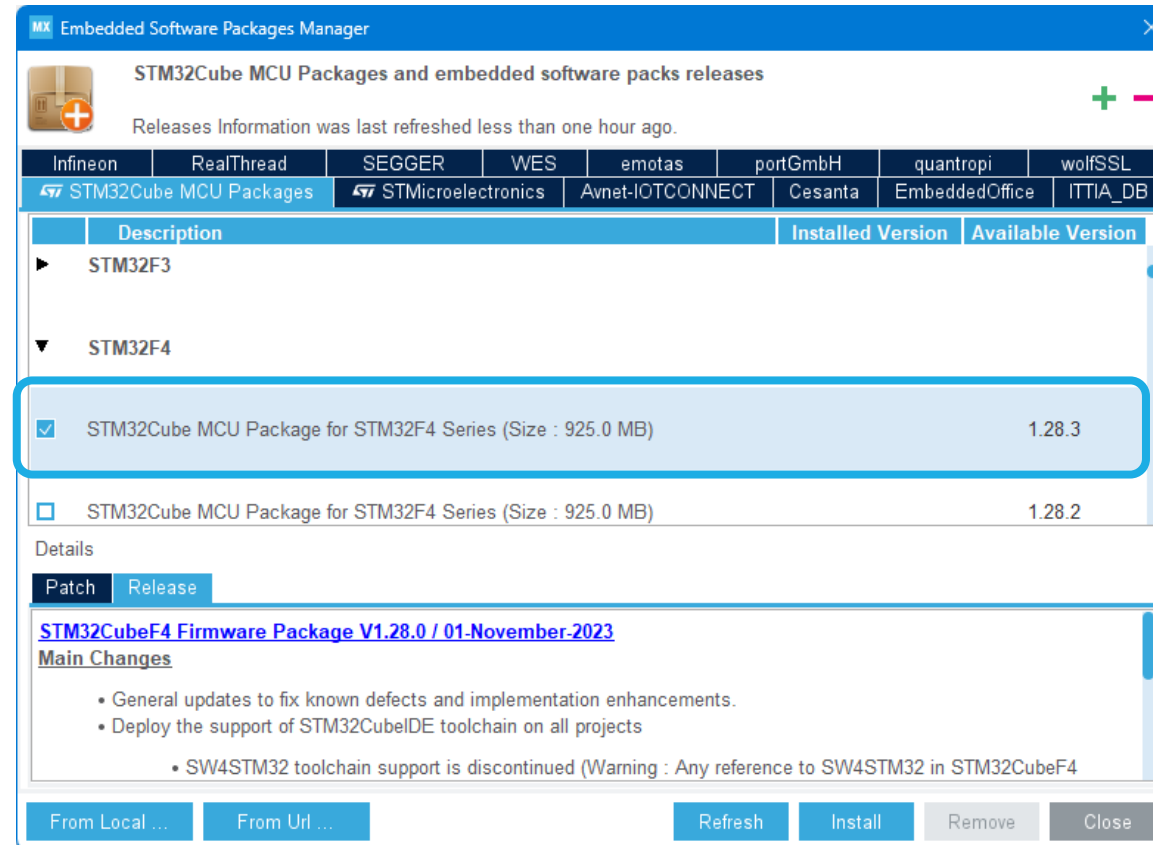
Seleccionamos *Install embedded software packages*



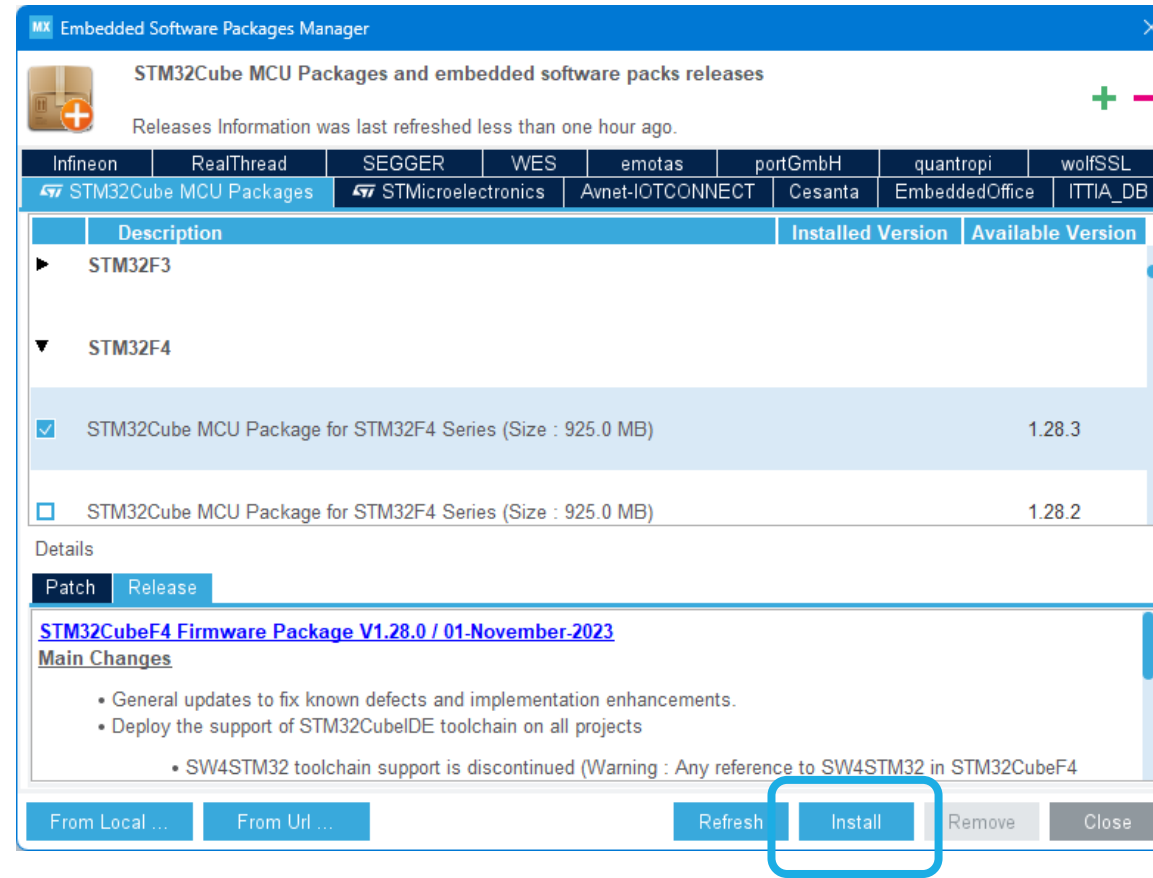
Buscamos el apartado *STM32F4*



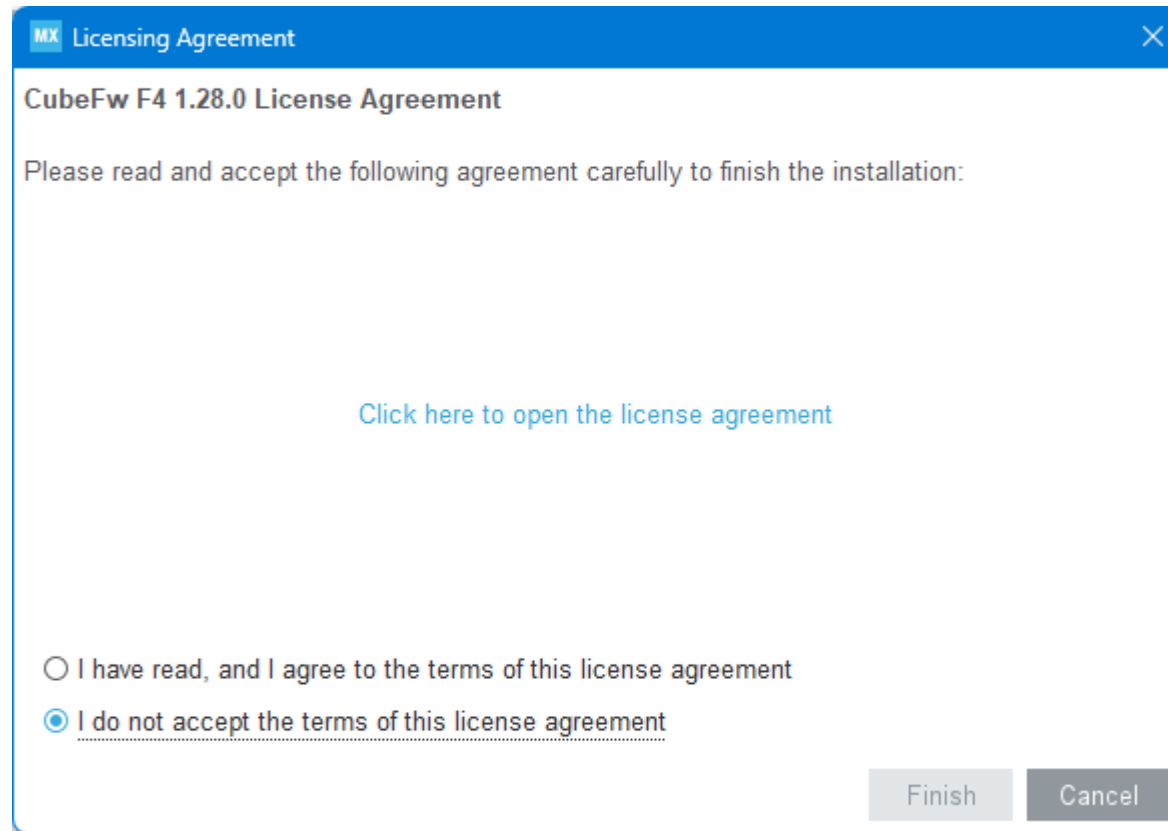
Seleccionamos la última versión



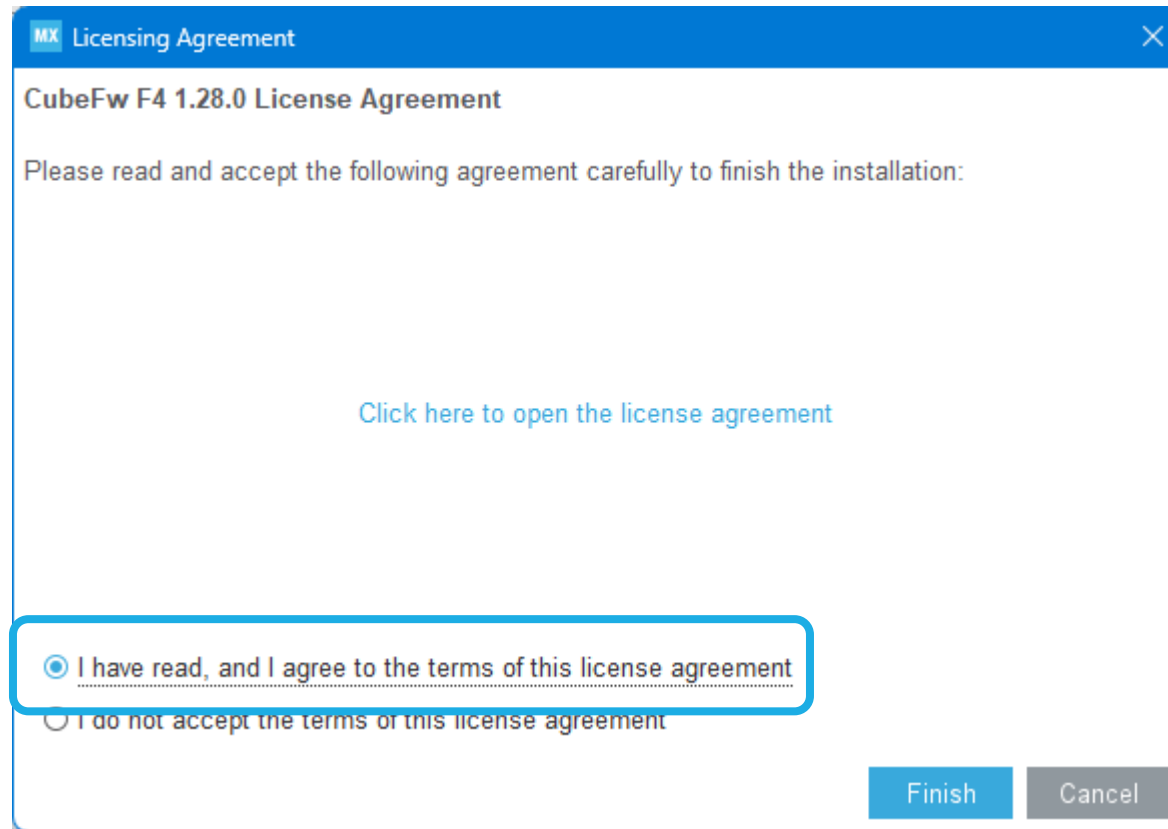
Seleccionamos *Install*



Nos aparecerá la siguiente ventana



Aceptamos los términos



MX Licensing Agreement [X]

CubeFw F4 1.28.0 License Agreement

Please read and accept the following agreement carefully to finish the installation:

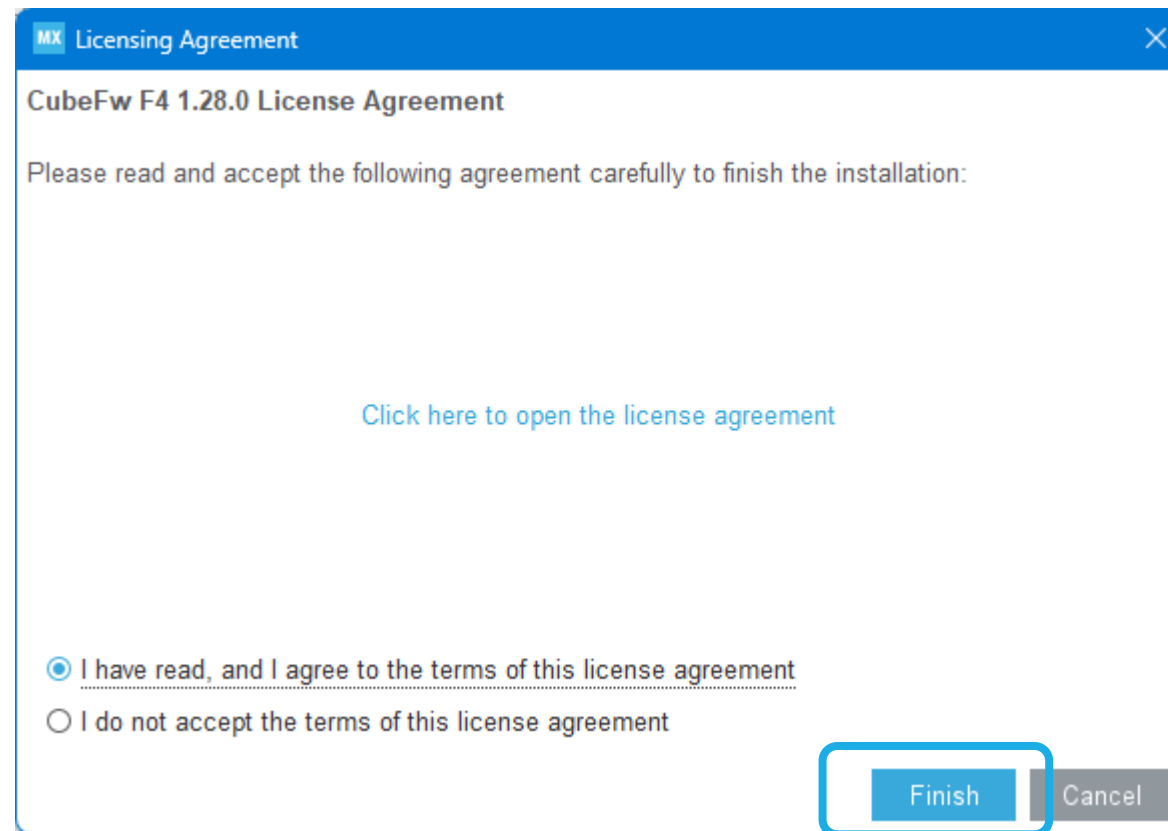
[Click here to open the license agreement](#)

☒ I have read, and I agree to the terms of this license agreement

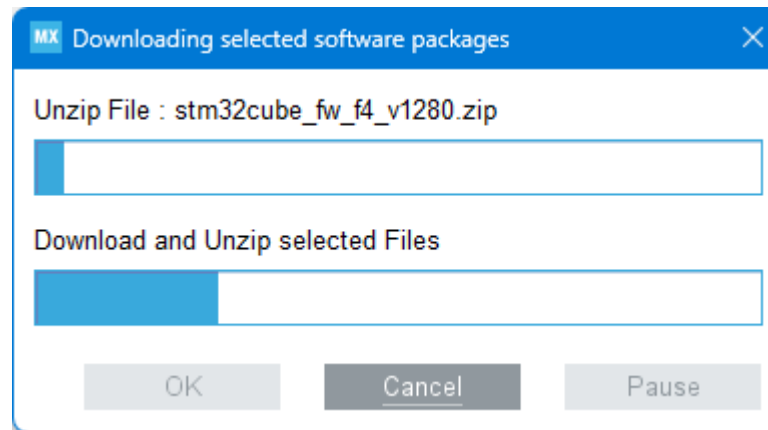
☐ I do not accept the terms of this license agreement

Finish Cancel

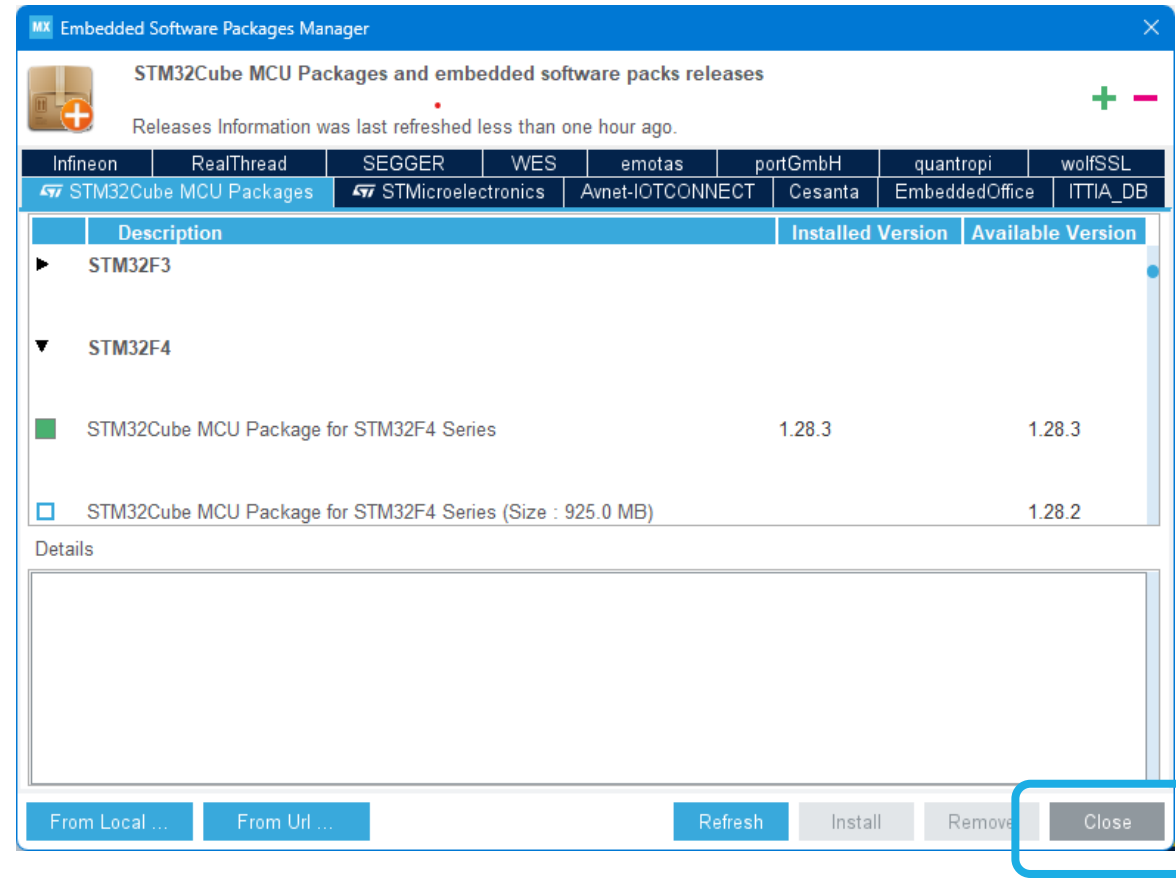
Seleccionamos *Finish*



Esperamos a que se termine de descargar e instalar



Una vez finalizado le damos a *Close*



A partir de este momento
estamos listos para iniciar
nuestros proyectos

Descarga de
Documentos necesarios

Documentación necesaria

Hoja de datos STM32F446RE (Datasheet)

<https://www.st.com/resource/en/datasheet/stm32f446mc.pdf>

Manual de referencia del microcontrolador STM32F446RE (Reference Manual)

https://www.st.com/resource/en/reference_manual/dm00135183-stm32f446xx-advanced-arm-based-32-bit-mcus-stmicroelectronics.pdf

Manual de usuario Nucleo-64 (User Manual)

https://www.st.com/resource/en/user_manual/um1724-stm32-nucleo64-boards-mb1136-stmicroelectronics.pdf

Documentación de STM32F4 HAL y LL

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